

PMWDS 2.0 — University Edition

Complete UI Design with API Calls & Property Bindings

ARCHITECTURE OVERVIEW

```
bash
University Portal
├─ 🗝️ Auth Module           → /api/v1/auth
├─ 🏠 Faculty Dashboard     → /api/v1/projects/dashboard
├─ 📅 Project/Course Mgmt    → /api/v1/projects
├─ ✅ Task/Assignment Mgmt   → /api/v1/tasks
├─ 👤 Student/User Mgmt      → /api/v1/users
├─ 🤖 AI Assistant           → /api/v1/ai
├─ 📊 Reports & Analytics     → /api/v1/reports
└─ 🔔 Notifications         → /api/v1/notifications
```

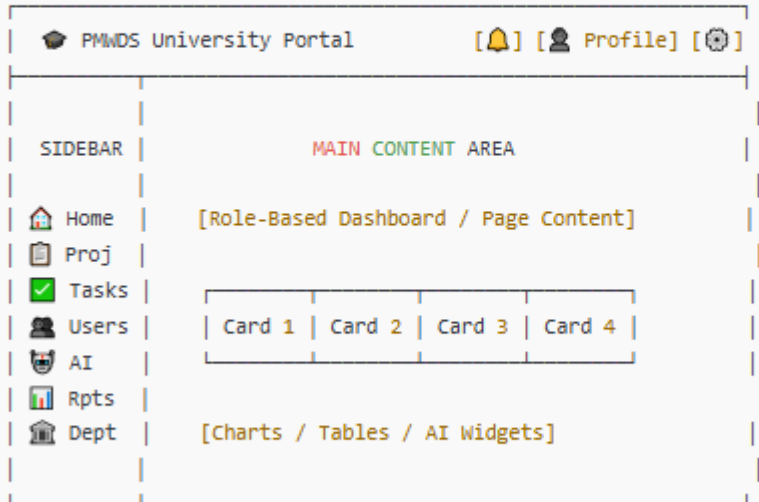
☐ ARCHITECTURE OVERVIEW

```
bash
University Portal
├─ ? Auth Module           → /api/v1/auth
├─ ? Faculty Dashboard     → /api/v1/projects/dashboard
├─ ? Project/Course Mgmt    → /api/v1/projects
├─ ? Task/Assignment Mgmt   → /api/v1/tasks
├─ ? Student/User Mgmt      → /api/v1/users
├─ ? AI Assistant           → /api/v1/ai
├─ ? Reports & Analytics     → /api/v1/reports
└─ ? Notifications         → /api/v1/notifications
```

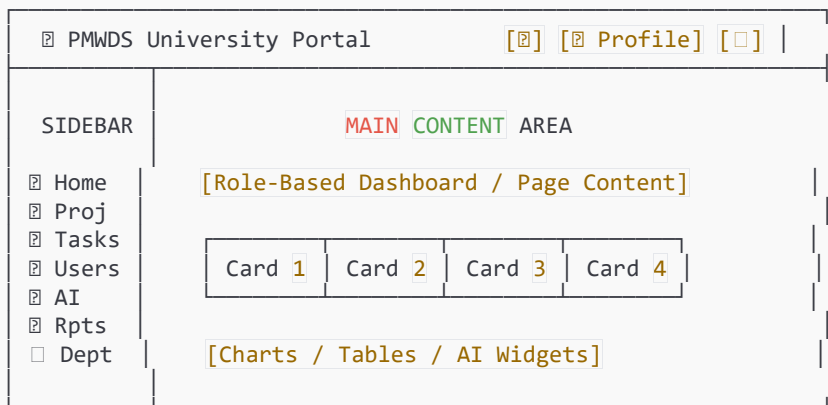
☐ UI LAYOUT STRUCTURE

UI LAYOUT STRUCTURE

CSS



CSS



PAGE 1 — Login / Authentication

UI Components

PAGE 1 — Login / Authentication

UI Components

markdown

 University AI Project Portal

Email:

Password:

☐ Remember Me

[Forgot Password?](#) | [LDAP Login](#)

markdown

 University AI Project Portal

Email:

Password:

☐ Remember Me

[Forgot Password?](#) | [LDAP Login](#)

API Call

bash

```
POST /api/v1/auth/login
Content-Type: application/json
```

Request Body & Property Bindings

json

```
{
  "email": "< bound to: loginForm.email (string)>",
  "password": "< bound to: loginForm.password (string)>"
}
```

Response Bindings

Response Property	Bound To	Usage
-------------------	----------	-------

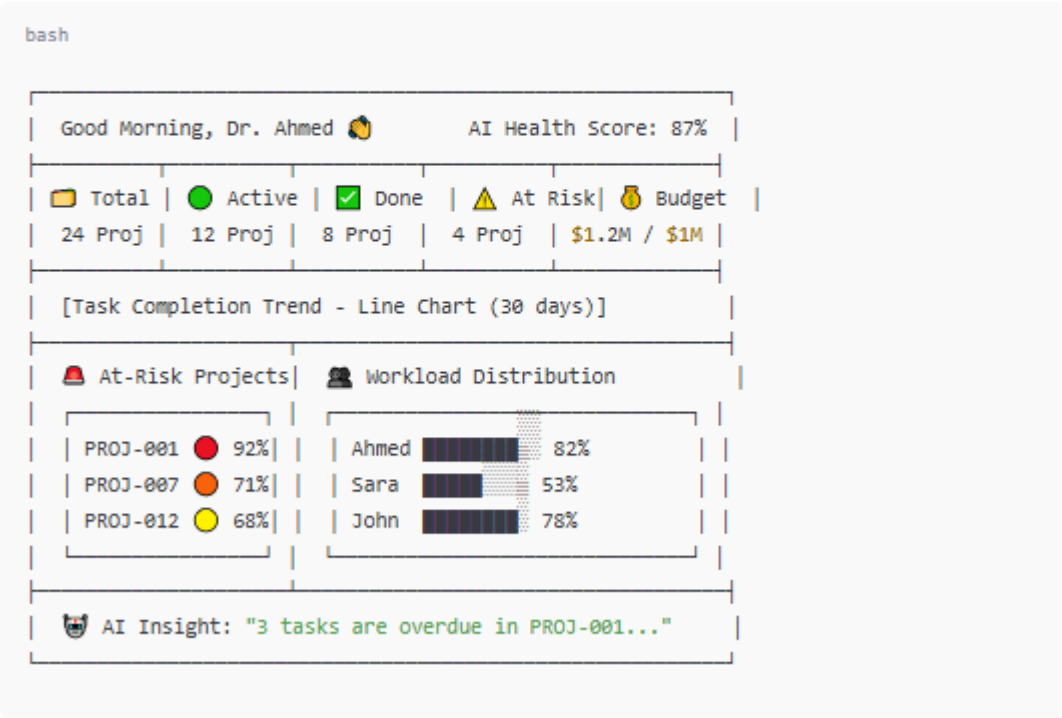
Response Property	Bound To	Usage
token	authStore.token	JWT Bearer header
userId	authStore.userId	User profile fetch
roles[]	authStore.roles	Menu visibility toggle
fullName	header.userDisplayName	Top bar greeting
expiresAt	authStore.tokenExpiry	Auto-refresh timer

❏ PAGE 2 — Main Dashboard (Role-Adaptive)

UI Components

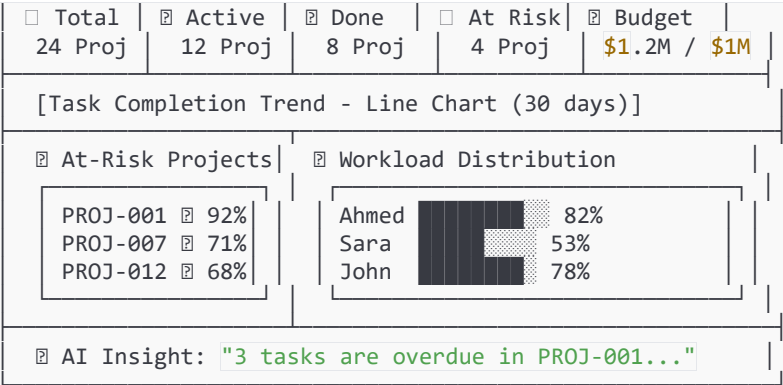
 PAGE 2 — Main Dashboard (Role-Adaptive)

UI Components



bash

Good Morning, Dr. Ahmed 🏠	AI Health Score: 87%
---------------------------	----------------------



API Call

```
bash
GET /api/v1/projects/dashboard?departmentId={{deptId}}
Authorization: Bearer {{token}}
```

Response Property Bindings

Property	UI Component	Type
totalProjects	KPI Card — Total	int
activeProjects	KPI Card — Active	int
completedProjects	KPI Card — Done	int
highRiskProjects	KPI Card — At Risk	int
totalBudget	KPI Card — Budget (planned)	decimal
totalActualCost	KPI Card — Budget (actual)	decimal
budgetVariance	Budget variance color indicator	decimal
overallHealthScore	Circular progress bar	double (0-100)
overallDelayRisk	Risk gauge widget	double (0-1)
highRiskProjects[]	At-Risk table rows	ProjectSummaryDto[]
workloadDistribution[]	Horizontal bar chart	WorkloadItem[]
taskCompletionTrend[]	Line chart (30 days)	TrendPoint[]
recentEscalations[]	Escalation alert list	TaskSummaryDto[]

Property	UI Component	Type
generatedAt	"Last updated" timestamp	DateTime

Sub-Property Bindings for highRiskProjects[]

Sub-Property Bindings for highRiskProjects[]

```
java
ProjectSummaryDto {
    id          → row.onClickNavigate("/projects/{id}")
    projectCode → column: "Code"
    name        → column: "Project Name"
    status      → colored badge (InProgress=blue, Delayed=red)
    progressPercentage → progress bar width %
    aiHealthScore → health icon (●>70, ●>50, ●<50)
    aiDelayRiskScore → risk badge percentage
}
```

```
java
ProjectSummaryDto {
    id          → row.onClickNavigate("/projects/{id}")
    projectCode → column: "Code"
    name        → column: "Project Name"
    status      → colored badge (InProgress=blue, Delayed=red)
    progressPercentage → progress bar width %
    aiHealthScore → health icon (●>70, ●>50, ●<50)
    aiDelayRiskScore → risk badge percentage
}
```

□ PAGE 3 — Projects / Courses Management

UI Components

PAGE 3 — Projects / Courses Management

UI Components

less

Projects [+ New Project] [Search] Filter: [All ▼] [Department ▼] [Status ▼] [Priority ▼] <table><tr><th>Code</th><th>Name</th><th>Status</th><th>Prog</th><th>Health</th><th>Actions</th></tr><tr><td>PROJ-001</td><td>AI Research</td><td>Active</td><td>72%</td><td>87</td><td> </td></tr><tr><td>PROJ-002</td><td>Web Dev Course</td><td>Delay</td><td>45%</td><td>42</td><td> </td></tr><tr><td>PROJ-007</td><td>Data Science</td><td>Hold</td><td>60%</td><td>65</td><td> </td></tr></table> [+ 1 2 3 →] Page 1/3						Code	Name	Status	Prog	Health	Actions	PROJ-001	AI Research	Active	72%	87		PROJ-002	Web Dev Course	Delay	45%	42		PROJ-007	Data Science	Hold	60%	65	
Code	Name	Status	Prog	Health	Actions																								
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less

Projects [+ New Project] [Search] Filter: [All ▼] [Department ▼] [Status ▼] [Priority ▼] <table><tr><th>Code</th><th>Name</th><th>Status</th><th>Prog</th><th>Health</th><th>Actions</th></tr><tr><td>PROJ-001</td><td>AI Research</td><td>Active</td><td>72%</td><td>87</td><td> </td></tr><tr><td>PROJ-002</td><td>Web Dev Course</td><td>Delay</td><td>45%</td><td>42</td><td> </td></tr><tr><td>PROJ-007</td><td>Data Science</td><td>Hold</td><td>60%</td><td>65</td><td> </td></tr></table> [← 1 2 3 →] Page 1/3						Code	Name	Status	Prog	Health	Actions	PROJ-001	AI Research	Active	72%	87		PROJ-002	Web Dev Course	Delay	45%	42		PROJ-007	Data Science	Hold	60%	65	
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PROJ-002	Web Dev Course	Delay	45%	42																									
PROJ-007	Data Science	Hold	60%	65																									

API Calls & Bindings

Fetch Projects List

bash

GET /api/v1/projects?departmentId={{deptId}}&status=InProgress
Authorization: Bearer {{token}}

Response Property	UI Binding
<code>[].id</code>	Row key / navigation link
<code>[].projectCode</code>	Column: Code
<code>[].name</code>	Column: Name
<code>[].status</code>	Status badge color map
<code>[].progressPercentage</code>	Inline progress bar

Response Property	UI Binding
[].aiHealthScore	Health icon + tooltip
[].aiDelayRiskScore	Risk percentage indicator
[].plannedEndDate	Due date column

Create Project

```
bash
POST /api/v1/projects
Content-Type: application/json
Authorization: Bearer {{token}}
json
{
  "projectCode": "< form.projectCode",
  "name": "< form.projectName",
  "description": "< form.description",
  "category": "< form.category (dropdown)",
  "plannedStartDate": "< form.startDate (date picker)",
  "plannedEndDate": "< form.endDate (date picker)",
  "plannedBudget": "< form.budget (number input)",
  "departmentId": "< form.departmentId (dropdown)",
  "projectManagerId": "< form.managerId (user search)",
  "priority": "< form.priority (Low/Medium/High/Critical)"
}
```

Update Project Status

```
bash
PATCH /api/v1/projects/{{projectId}}/status
Content-Type: application/json
json
{
  "newStatus": "< statusDropdown.value",
  "justification": "< justificationTextArea.value"
}
```

❏ PAGE 4— Task / Assignment Management

✓ PAGE 4 — Task / Assignment Management

UI Components

less

✓ Tasks — PROJ-001: AI Research

[+ New Task]

[Kanban ▼] [List] [Gantt]

Filter: [My Tasks] [All]

🔍 TODO

🔄 IN PROG

🔍 REVIEW

✓ DONE

Task #3	Task #1	Task #5	Task #2
Literature	ML Model Dev	Test Suite	Data Collection
🔴 HIGH	🟡 MED	🟢 LOW	✓ Completed
Due: May 8	Due: May 10	Due: May 15	May 1
👤 Unassign	👤 Dr.Ahmed	👤 Sara	👤 John
🔥 Risk:87%	🔥 Risk:42%	🔥 Risk:20%	

UI Components

less

🔍 Tasks — PROJ-001: AI Research

[+ New Task]

[Kanban ▼] [List] [Gantt]

Filter: [My Tasks] [All]

🔍 TODO

🔄 IN PROG

🔍 REVIEW

✓ DONE

Task #3	Task #1	Task #5	Task #2
Literature	ML Model Dev	Test Suite	Data Collection
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👤 Unassign	👤 Dr.Ahmed	👤 Sara	👤 John
🔥 Risk:87%	🔥 Risk:42%	🔥 Risk:20%	

API Calls & Bindings

Get Tasks by Project

vbnet

GET /api/v1/tasks/by-project/{{projectId}}

Authorization: Bearer {{token}}

Response Property	UI Binding
[].id	Card key
[].title	Card title

Response Property	UI Binding
<code>[].status</code>	Kanban column placement
<code>[].priority</code>	Color-coded badge
<code>[].dueDate</code>	Due date display + red if overdue
<code>[].assignedToUserId</code>	Avatar / name
<code>[].progressPercentage</code>	Mini progress bar on card
<code>[].aiDelayProbability</code>	Risk badge <code>> Risk: X%</code>
<code>[].isEscalated</code>	☐ escalation icon

Create Task

```
bash
POST /api/v1/tasks
Content-Type: application/json
json
{
  "title":           "< form.title",
  "description":     "< form.description",
  "startDate":       "< form.startDate",
  "dueDate":         "< form.dueDate",
  "estimatedHours":  "< form.estimatedHours (number)",
  "projectId":       "< context.activeProjectId",
  "milestoneId":     "< form.milestoneId (optional dropdown)",
  "assignedToUserId": "< form.assigneeId (user picker)",
  "priority":        "< form.priority"
}
```

Update Task Progress

```
bash
PATCH /api/v1/tasks/{{taskId}}/progress
Content-Type: application/json
json
{
  "progressPercentage": "< progressSlider.value (0-100)",
  "notes":              "< notesInput.value"
}
```

Assign Task

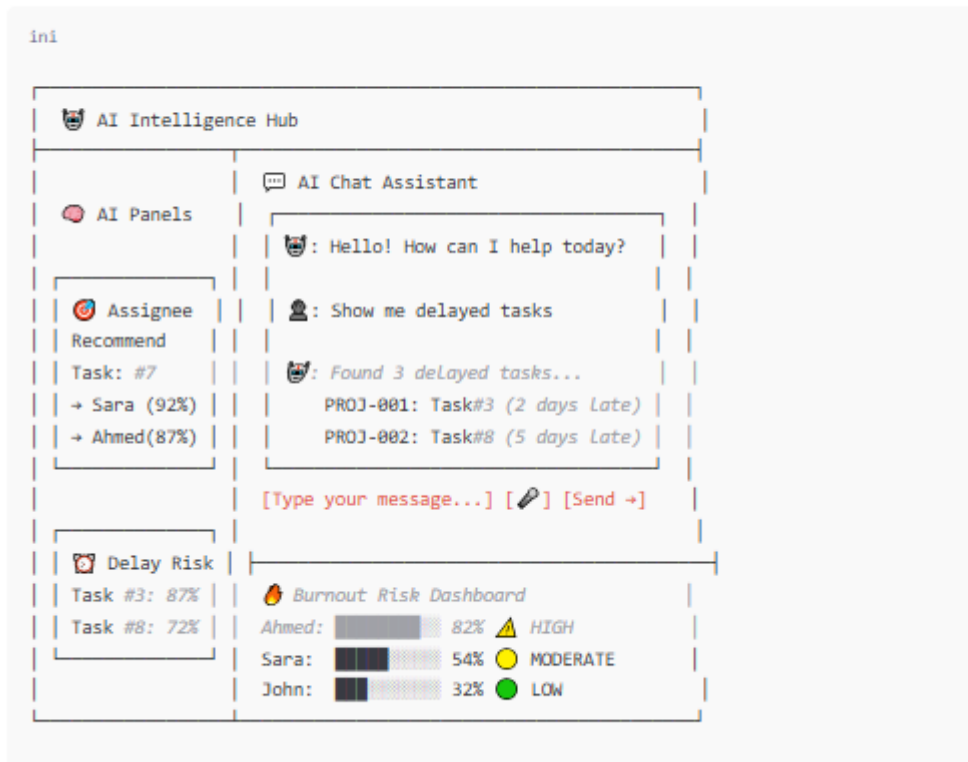
```
bash
PATCH /api/v1/tasks/{{taskId}}/assign
Content-Type: application/json
json
{
```

```
"userId": "<← userSearchDropdown.selectedUserId"
}
```

❑ PAGE 5 — AI Assistant & Intelligence Hub

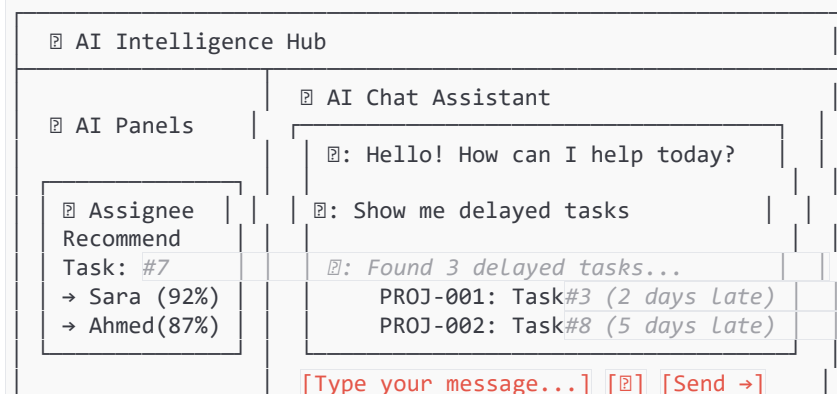
PAGE 5 — AI Assistant & Intelligence Hub

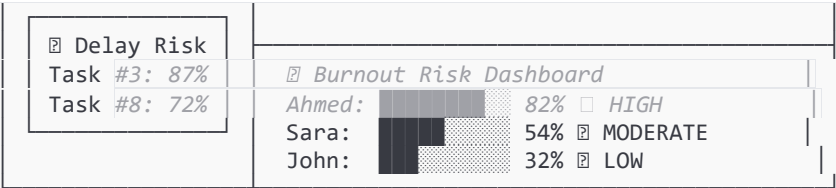
UI Components



UI Components

ini





API Calls & Bindings

AI Assignee Recommendation

```
bash
GET /api/v1/ai/recommend-assignee/{{taskId}}
Authorization: Bearer {{token}}
```

Response Property	UI Binding
recommendedUserId	Top recommendation highlight
recommendedUserName	Name label
confidenceScore	Match percentage badge
rationale[]	Tooltip/expandable explanation list
alternatives[]	Secondary candidates list
featureScores{ }	Radar/spider chart breakdown
generatedAt	Timestamp

Delay Prediction

```
bash
GET /api/v1/ai/predict-delay/{{taskId}}
Authorization: Bearer {{token}}
```

Response Property	UI Binding
delayProbability	Risk gauge (0–1 → 0–100%)
expectedDelayDays	"Expected delay: X days"
predictedCompletionDate	Date display
riskLevel	Badge color (Low/Medium/High/Critical)
contributingFactors[]	Bullet list

Response Property	UI Binding
mitigationStrategies[]	Action suggestion list
shouldEscalate	Escalation alert trigger (boolean)

Project Health Analysis

```
bash
GET /api/v1/ai/project-health/{{projectId}}
Authorization: Bearer {{token}}
```

Response Property	UI Binding
overallHealthScore	Circular progress donut
scheduleHealth	Sub-gauge: Schedule
budgetHealth	Sub-gauge: Budget
teamHealth	Sub-gauge: Team
qualityHealth	Sub-gauge: Quality
healthStatus	Status badge text
strengths[]	Green checkmark list
weaknesses[]	Red warning list
recommendations[]	AI suggestion cards
risks[].description	Risk table rows
risks[].probability	Risk probability bar
risks[].mitigationStrategy	Tooltip/expandable

AI Chat Assistant

```
bash
POST /api/v1/ai/chat
Content-Type: application/json
json
{
  "message": "< chatInput.value"
}
```

Response Property	UI Binding
message	Chat bubble content
intent	Intent tag (hidden/debug display)
suggestedActions[]	Quick-reply button chips
contextData	Dynamic context panel
requiresConfirmation	Confirmation modal trigger

Burnout Risk

```
bash
GET /api/v1/ai/burnout-risk?departmentId={{deptId}}
Authorization: Bearer {{token}}
```

Response Property	UI Binding
[].fullName	User row label
[].burnoutRisk	Horizontal progress bar
[].workloadScore	Secondary bar
[].activeTasks	Badge count
[].riskLevel	Color-coded label
[].recommendations[]	Expandable suggestion drawer

PAGE 6 — User / Student / Faculty Management

UI Components

ini

Users / Members [+ Add User]					
Department: [All ▼] Role: [All ▼] Status: [Active ▼]					
#	Name & Role	Skills	WkLoad	Avail	Actions
	Dr. Ahmed Project Manager	ML, Java	82% 	65%	 
	Sara Chen Team Member	React Node.js	54% 	98%	 

UI Components

ini

Users / Members [+ Add User]					
Department: [All ▼] Role: [All ▼] Status: [Active ▼]					
#	Name & Role	Skills	WkLoad	Avail	Actions
	Dr. Ahmed Project Manager	ML, Java	82% 	65%	 
	Sara Chen Team Member	React Node.js	54% 	90%	 

API Calls & Bindings

Get Users List

bash

```
GET /api/v1/users?departmentId={{deptId}}
Authorization: Bearer {{token}}
```

Response Property	UI Binding
<code>[].id</code>	Row key
<code>[].fullName</code>	Name cell
<code>[].jobTitle</code>	Sub-label under name
<code>[].skills[]</code>	Skill chip tags

Response Property	UI Binding
[].aiWorkloadScore	Workload % bar + color
[].availabilityPercentage	Availability %
[].availabilityStatus	Status badge
[].aiPerformanceScore	Performance star rating
[].aiBurnoutRiskScore	Burnout risk indicator
[].activeTaskCount	Badge count
[].isActive	Row opacity toggle

Workload Distribution

```
bash
GET /api/v1/users/workload?departmentId={{deptId}}
Authorization: Bearer {{token}}
```

Response Property	UI Binding
totalMembers	Summary stat card
availableCount	"Available" counter
overloadedCount	"Overloaded" alert counter
averageWorkload	Department average gauge
averageBurnoutRisk	Dept risk indicator
members[].workloadScore	Bar chart per user
members[].status	Color code (Available/High/Overloaded)

Update User Availability

```
bash
PATCH /api/v1/users/{{userId}}/availability
Content-Type: application/json
json
{
  "status": "< availabilityDropdown.value",
  "availabilityPercentage": "< availabilitySlider.value"
}
```

Add Skill to User

```
bash
```

POST /api/v1/users/{{userId}}/skills

Content-Type: application/json

json

```
{
  "skillId": "← skillSearchDropdown.selectedId",
  "proficiencyLevel": "← proficiencySlider.value (1-5)",
  "experienceMonths": "← experienceInput.value"
}
```

□ **PAGE 7 — Reports & Analytics**

 PAGE 7 — Reports & Analytics

UI Components

less

Reports & Analytics

Report Type

○ Proj Status

○ Task Compl.

○ Dept WkLoad

○ Budget Var.

○ Delay Anal.

○ AI Insights

Parameters

Date From: [📅 2025-01-01]

Date To: [📅 2025-12-31]

Dept: [Software Eng ▼]

Project: [PROJ-001 ▼]

Format: [PDF ▼] [Excel] [CSV]

🚀 Generate Report

📊 Live Analytics Preview

[Task Completion Rate] [Budget Utilization] [Team Velocity]

UI Components

less

?

Reports & Analytics

Report Type

○ Proj Status

○ Task Compl.

○ Dept Wkload

○ Budget Var.

Parameters

Date From:

[?] 2025-01-01]

Date To:

[?] 2025-12-31]

Dept:

[Software Eng ▼]

Project:

[PROJ-001 ▼]

Format:

[PDF ▼] [Excel] [CSV]

○ Delay Anal.

○ AI Insights

[📄 Generate Report]

📄 Live Analytics Preview

[Task Completion Rate] [Budget Utilization] [Team Velocity]

API Calls & Bindings

Report Type	API Endpoint	Method	Key Bindings
Project Status	/api/v1/reports/project-status/{{projectId}}?format=pdf	GET	projectId ← dropdown.selectedId
Task Completion	/api/v1/reports/task-completion?format=pdf	POST	Body: {startDate, endDate, departmentId, projectId}
Dept Workload	/api/v1/reports/department-workload?format=pdf	POST	Body: {departmentId, startDate, endDate}
Budget Variance	/api/v1/reports/budget-variance/{{projectId}}?format=excel	GET	projectId ← dropdown
Delay Analysis	/api/v1/reports/delay-analysis?format=pdf	POST	Body: {startDate, endDate, departmentId}
AI Insights	/api/v1/reports/ai-insights/{{projectId}}?format=pdf	GET	projectId ← context

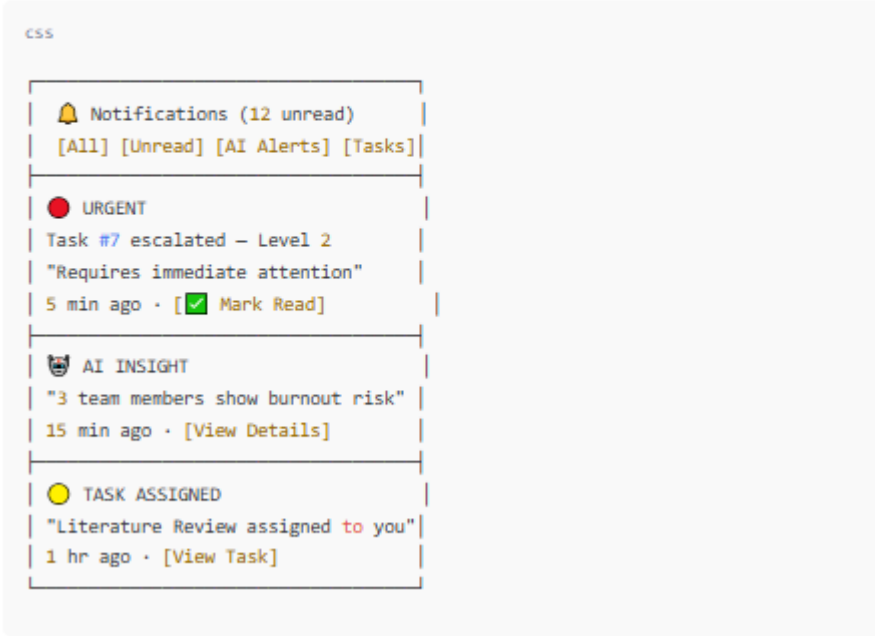
Format Binding:

```
python
format query param ← formatToggleGroup.activeValue
"pdf" → triggers download with Content-Type: application/pdf
"excel" → triggers download with Content-Type: application/vnd.openxmlformats...
"csv" → triggers download with Content-Type: text/csv
```

❏ PAGE 8 — Notifications Panel

🔔 PAGE 8 — Notifications Panel

UI Components

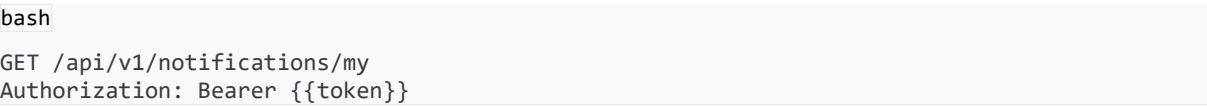


UI Components



API Calls & Bindings

Get Notifications



Response Property	UI Binding
[].id	Item key
[].title	Bold header text
[].message	Body text
[].type	Icon + color (TaskEscalated, AllInsight, etc.)
[].priority	Left border color (Urgent=red, Normal=blue)
[].isRead	Background opacity (unread=white, read=gray)
[].createdDate	Relative time ("5 min ago")
[].actionUrl	"View" button navigation target

Mark as Read

```
bash
PATCH /api/v1/notifications/{{notificationId}}/read
Authorization: Bearer {{token}}
Binding: notificationId ← item.id on click of [Mark Read]
```

PAGE 9 — Department Management

API Calls & Bindings

Get All Departments

```
bash
GET /api/v1/departments
Authorization: Bearer {{token}}
```

Response Property	UI Binding
[].id	Row key / navigation
[].name	Dept name column
[].code	Code badge
[].description	Tooltip

Response Property	UI Binding
[].memberCount	Member count badge

Create Department

```
bash
POST /api/v1/departments
Content-Type: application/json
json
{
  "name": "← form.deptName",
  "code": "← form.deptCode",
  "description": "← form.description",
  "budget": "← form.budget"
}
```

▣ FULL UI PROPERTY BINDING SUMMARY TABLE

UI Element	API Endpoint	HTTP	Key Property Bound
Login form	/auth/login	POST	email, password → token, roles
Dashboa rd KPIs	/projects/dash board	GET	totalProjects, activeProjects, budg etVariance
Project list table	/projects	GET	[].name, [].status, [].progressPerc entage
Create project modal	/projects	POST	All CreateProjectDto fields
Kanban board	/tasks/by- project/:id	GET	[].status → column, [].priority → color
Task progress slider	/tasks/:id/pro gress	PATCH	progressPercentage, notes
AI assignee widget	/ai/recommend- assignee/:id	GET	confidenceScore, rationale[]

UI Element	API Endpoint	HTTP	Key Property Bound
Delay risk badge	/ai/predict-delay/:id	GET	delayProbability, riskLevel
Health donut chart	/ai/project-health/:id	GET	scheduleHealth, budgetHealth, teamHealth
Chat messages	/ai/chat	POST	message → response bubble, suggestedActions[] → chips
Burnout bars	/ai/burnout-risk	GET	[].burnoutRisk, [].riskLevel
Workload chart	/users/workload	GET	members[].workloadScore, members[].status
Report download	/reports/*	GET/POST	format query → MIME type download
Notification bell	/notifications/my	GET	unreadCount, [].type, [].priority

UNIVERSITY-SPECIFIC UI THEMING

```
css
/* University Color Palette */
--primary: #1B3A6B; /* Deep University Blue */
--secondary: #C8A951; /* Gold Accent */
--success: #28A745; /* Green - On Track */
--warning: #FFC107; /* Yellow - At Risk */
--danger: #DC3545; /* Red - Critical */
--ai-purple: #6F42C1; /* AI Features */
--background: #F8F9FA; /* Light Gray BG */
--card-bg: #FFFFFF;
--font-primary: 'Inter', sans-serif;
--border-radius: 12px;
--shadow: 0 4px 12px rgba(27,58,107,0.1);
```

=====

Here is the **complete multi-page HTML wireframe** — save it as `index.html` and open directly in a browser:

```
html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title> PMWDS 2.0 – University Portal</title>

  <!-- Vue 3 CDN -->
  <script src="https://unpkg.com/vue@3/dist/vue.global.js"></script>
  <!-- Axios for API calls -->
  <script src="https://cdn.jsdelivr.net/npm/axios/dist/axios.min.js"></script>
  <!-- Chart.js -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<style>
  /* =====
  UNIVERSITY DESIGN SYSTEM – CSS VARIABLES
  ===== */

  :root {
    --primary: #1B3A6B;
    --primary-light: #2a52a0;
    --primary-dark: #122848;
    --gold: #C8A951;
    --gold-light: #e8c97a;
    --success: #28a745;
    --warning: #ffc107;
    --danger: #dc3545;
    --info: #17a2b8;
    --ai-purple: #6f42c1;
    --ai-light: #9b6ee8;
    --bg: #f0f2f7;
    --card: #ffffff;
    --border: #dee2e6;
    --text: #212529;
    --text-muted: #6c757d;
    --sidebar-w: 260px;
    --header-h: 64px;
    --radius: 12px;
    --shadow: 0 4px 16px rgba(27,58,107,0.12);
    --shadow-hover: 0 8px 24px rgba(27,58,107,0.2);
    --transition: all 0.25s ease;
  }

  /* ===== RESET & BASE ===== */
  *, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }
  body {
    font-family: 'Segoe UI', system-ui, -apple-system, sans-serif;
    background: var(--bg);
    color: var(--text);
    font-size: 14px;
    line-height: 1.5;
  }
  button { cursor: pointer; font-family: inherit; }
  input, select, textarea {
    font-family: inherit;
    font-size: 14px;
    outline: none;
  }
```

```

/* ===== SCROLLBAR ===== */
::-webkit-scrollbar { width: 6px; height: 6px; }
::-webkit-scrollbar-track { background: #f1f1f1; }
::-webkit-scrollbar-thumb { background: #ccc; border-radius: 3px; }

/* ===== LOGIN PAGE ===== */
.login-page {
  min-height: 100vh;
  display: flex;
  align-items: center;
  justify-content: center;
  background: linear-gradient(135deg, var(--primary-dark) 0%, var(--primary) 50%, var(--primary-light) 100%);
  position: relative;
  overflow: hidden;
}
.login-page::before {
  content: '';
  position: absolute;
  top: -50%;
  right: -20%;
  width: 600px;
  height: 600px;
  background: rgba(200,169,81,0.08);
  border-radius: 50%;
}
.login-page::after {
  content: '';
  position: absolute;
  bottom: -30%;
  left: -10%;
  width: 400px;
  height: 400px;
  background: rgba(255,255,255,0.05);
  border-radius: 50%;
}
.login-card {
  background: #fff;
  border-radius: 20px;
  padding: 48px 44px;
  width: 440px;
  box-shadow: 0 24px 64px rgba(0,0,0,0.2);
  position: relative;
  z-index: 1;
}
.login-logo {
  text-align: center;
  margin-bottom: 32px;
}
.login-logo .logo-icon {
  width: 72px;
  height: 72px;
  background: linear-gradient(135deg, var(--primary), var(--primary-light));
  border-radius: 18px;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  font-size: 32px;
  margin-bottom: 16px;
  box-shadow: 0 8px 20px rgba(27,58,107,0.3);
}
.login-logo h1 {

```

```

    font-size: 22px;
    font-weight: 700;
    color: var(--primary);
}
.login-logo p {
    color: var(--text-muted);
    font-size: 13px;
    margin-top: 4px;
}
.form-group { margin-bottom: 20px; }
.form-group label {
    display: block;
    font-weight: 600;
    margin-bottom: 8px;
    color: var(--text);
    font-size: 13px;
}
.form-control {
    width: 100%;
    padding: 12px 16px;
    border: 2px solid var(--border);
    border-radius: 10px;
    font-size: 14px;
    transition: var(--transition);
    background: #fafafa;
}
.form-control:focus {
    border-color: var(--primary);
    background: #fff;
    box-shadow: 0 0 0 3px rgba(27,58,107,0.1);
}
.form-control.error { border-color: var(--danger); }
.btn {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    gap: 8px;
    padding: 12px 24px;
    border: none;
    border-radius: 10px;
    font-weight: 600;
    font-size: 14px;
    transition: var(--transition);
    text-decoration: none;
}
.btn-primary {
    background: linear-gradient(135deg, var(--primary), var(--primary-light));
    color: #fff;
    width: 100%;
    padding: 14px;
}
.btn-primary:hover {
    transform: translateY(-1px);
    box-shadow: 0 6px 16px rgba(27,58,107,0.35);
}
.btn-primary:active { transform: translateY(0); }
.btn-outline {
    background: transparent;
    border: 2px solid var(--primary);
    color: var(--primary);
}
.btn-outline:hover { background: var(--primary); color: #fff; }
.btn-sm { padding: 7px 14px; font-size: 12px; border-radius: 8px; }

```

```

.btn-success { background: var(--success); color: #fff; }
.btn-success:hover { background: #218838; }
.btn-danger { background: var(--danger); color: #fff; }
.btn-danger:hover { background: #c82333; }
.btn-warning { background: var(--warning); color: #000; }
.btn-gold {
  background: linear-gradient(135deg, var(--gold), var(--gold-light));
  color: var(--primary-dark);
  font-weight: 700;
}
.btn-gold:hover { transform: translateY(-1px); box-shadow: 0 4px 12px
rgba(200,169,81,0.4); }
.btn-ai {
  background: linear-gradient(135deg, var(--ai-purple), var(--ai-light));
  color: #fff;
}
.btn-ai:hover { transform: translateY(-1px); box-shadow: 0 4px 12px
rgba(111,66,193,0.3); }
.login-divider {
  text-align: center;
  margin: 16px 0;
  color: var(--text-muted);
  font-size: 12px;
  position: relative;
}
.login-divider::before, .login-divider::after {
  content: '';
  position: absolute;
  top: 50%;
  width: 40%;
  height: 1px;
  background: var(--border);
}
.login-divider::before { left: 0; }
.login-divider::after { right: 0; }
.alert {
  padding: 12px 16px;
  border-radius: 10px;
  margin-bottom: 16px;
  font-size: 13px;
  display: flex;
  align-items: center;
  gap: 8px;
}
.alert-danger { background: #fde8ea; border: 1px solid #f5c6cb; color: #721c24; }
.alert-success { background: #d4edda; border: 1px solid #c3e6cb; color: #155724; }
.alert-warning { background: #fff3cd; border: 1px solid #ffeeba; color: #856404; }
.alert-info { background: #d1ecf1; border: 1px solid #bee5eb; color: #0c5460; }

/* ===== APP SHELL ===== */
.app-shell {
  display: flex;
  height: 100vh;
  overflow: hidden;
}

/* ===== SIDEBAR ===== */
.sidebar {
  width: var(--sidebar-w);
  background: linear-gradient(180deg, var(--primary-dark) 0%, var(--primary) 100%);
  height: 100vh;
  display: flex;
  flex-direction: column;

```

```

    flex-shrink: 0;
    transition: var(--transition);
    z-index: 100;
    overflow: hidden;
}
.sidebar.collapsed { width: 68px; }
.sidebar-brand {
    padding: 20px 16px;
    border-bottom: 1px solid rgba(255,255,255,0.1);
    display: flex;
    align-items: center;
    gap: 12px;
}
.brand-icon {
    width: 40px;
    height: 40px;
    background: rgba(200,169,81,0.2);
    border: 2px solid var(--gold);
    border-radius: 10px;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 20px;
    flex-shrink: 0;
}
.brand-text {
    overflow: hidden;
    transition: var(--transition);
}
.brand-text h2 { color: #fff; font-size: 15px; font-weight: 700; white-space: nowrap; }
.brand-text p { color: var(--gold); font-size: 11px; white-space: nowrap; }
.sidebar.collapsed .brand-text { opacity: 0; width: 0; }
.sidebar-nav {
    flex: 1;
    padding: 12px 8px;
    overflow-y: auto;
}
.nav-section-title {
    color: rgba(255,255,255,0.4);
    font-size: 10px;
    font-weight: 700;
    letter-spacing: 1.2px;
    text-transform: uppercase;
    padding: 12px 8px 6px;
    white-space: nowrap;
    overflow: hidden;
    transition: var(--transition);
}
.sidebar.collapsed .nav-section-title { opacity: 0; }
.nav-item {
    display: flex;
    align-items: center;
    gap: 12px;
    padding: 11px 12px;
    border-radius: 10px;
    color: rgba(255,255,255,0.75);
    cursor: pointer;
    transition: var(--transition);
    margin-bottom: 2px;
    position: relative;
    white-space: nowrap;
}
.nav-item:hover {

```

```

        background: rgba(255,255,255,0.1);
        color: #fff;
    }
    .nav-item.active {
        background: var(--gold);
        color: var(--primary-dark);
        font-weight: 600;
        box-shadow: 0 4px 12px rgba(200,169,81,0.3);
    }
    .nav-item .nav-icon { font-size: 18px; flex-shrink: 0; width: 22px; text-align: center; }
    .nav-item .nav-label { font-size: 13px; overflow: hidden; transition: var(--transition); }
    .sidebar.collapsed .nav-item .nav-label { opacity: 0; width: 0; }
    .nav-badge {
        margin-left: auto;
        background: var(--danger);
        color: #fff;
        font-size: 10px;
        font-weight: 700;
        padding: 2px 7px;
        border-radius: 10px;
        flex-shrink: 0;
    }
    .sidebar.collapsed .nav-badge { display: none; }
    .sidebar-footer {
        padding: 12px 8px;
        border-top: 1px solid rgba(255,255,255,0.1);
    }

/* ===== MAIN AREA ===== */
.main-area {
    flex: 1;
    display: flex;
    flex-direction: column;
    overflow: hidden;
}

/* ===== HEADER ===== */
.header {
    height: var(--header-h);
    background: #fff;
    border-bottom: 1px solid var(--border);
    display: flex;
    align-items: center;
    padding: 0 24px;
    gap: 16px;
    flex-shrink: 0;
    box-shadow: 0 2px 8px rgba(0,0,0,0.06);
    z-index: 50;
}
.header-toggle {
    background: none;
    border: none;
    font-size: 22px;
    color: var(--text-muted);
    padding: 6px;
    border-radius: 8px;
}
.header-toggle:hover { background: var(--bg); color: var(--text); }
.header-breadcrumb {
    display: flex;
    align-items: center;

```

```

    gap: 8px;
    color: var(--text-muted);
    font-size: 13px;
}
.header-breadcrumb .current { color: var(--primary); font-weight: 600; }
.header-spacer { flex: 1; }
.header-search {
    display: flex;
    align-items: center;
    background: var(--bg);
    border: 1.5px solid var(--border);
    border-radius: 10px;
    padding: 8px 14px;
    gap: 8px;
    width: 240px;
}
.header-search input {
    background: none;
    border: none;
    flex: 1;
    font-size: 13px;
    color: var(--text);
}
.header-actions {
    display: flex;
    align-items: center;
    gap: 8px;
}
.icon-btn {
    width: 38px;
    height: 38px;
    border: none;
    border-radius: 10px;
    background: var(--bg);
    color: var(--text-muted);
    font-size: 18px;
    display: flex;
    align-items: center;
    justify-content: center;
    cursor: pointer;
    position: relative;
    transition: var(--transition);
}
.icon-btn:hover { background: var(--border); color: var(--text); }
.icon-btn .badge {
    position: absolute;
    top: 4px;
    right: 4px;
    background: var(--danger);
    color: #fff;
    font-size: 9px;
    font-weight: 700;
    width: 16px;
    height: 16px;
    border-radius: 50%;
    display: flex;
    align-items: center;
    justify-content: center;
}
.user-menu {
    display: flex;
    align-items: center;
    gap: 10px;

```

```

padding: 6px 12px;
border-radius: 10px;
cursor: pointer;
transition: var(--transition);
}
.user-menu:hover { background: var(--bg); }
.user-avatar {
width: 36px;
height: 36px;
border-radius: 50%;
background: linear-gradient(135deg, var(--primary), var(--primary-light));
color: #fff;
font-weight: 700;
font-size: 14px;
display: flex;
align-items: center;
justify-content: center;
flex-shrink: 0;
}
.user-info h4 { font-size: 13px; font-weight: 600; }
.user-info p { font-size: 11px; color: var(--text-muted); }

/* ===== PAGE CONTENT ===== */
.page-content {
flex: 1;
overflow-y: auto;
padding: 24px;
}
.page-header {
display: flex;
align-items: center;
justify-content: space-between;
margin-bottom: 24px;
flex-wrap: wrap;
gap: 12px;
}
.page-title h2 {
font-size: 22px;
font-weight: 700;
color: var(--primary);
}
.page-title p { color: var(--text-muted); font-size: 13px; margin-top: 2px; }
.page-actions { display: flex; gap: 10px; flex-wrap: wrap; }

/* ===== KPI CARDS ===== */
.kpi-grid {
display: grid;
grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
gap: 16px;
margin-bottom: 24px;
}
.kpi-card {
background: var(--card);
border-radius: var(--radius);
padding: 20px;
box-shadow: var(--shadow);
border-left: 4px solid transparent;
transition: var(--transition);
cursor: pointer;
}
.kpi-card:hover { transform: translateY(-2px); box-shadow: var(--shadow-hover); }
.kpi-card.blue { border-left-color: var(--primary); }
.kpi-card.green { border-left-color: var(--success); }

```

```

.kpi-card.red { border-left-color: var(--danger); }
.kpi-card.gold { border-left-color: var(--gold); }
.kpi-card.purple{ border-left-color: var(--ai-purple); }
.kpi-card.orange{ border-left-color: #fd7e14; }
.kpi-top {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-bottom: 12px;
}
.kpi-icon {
  width: 44px;
  height: 44px;
  border-radius: 10px;
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 20px;
}
.kpi-icon.blue { background: rgba(27,58,107,0.1); }
.kpi-icon.green { background: rgba(40,167,69,0.1); }
.kpi-icon.red { background: rgba(220,53,69,0.1); }
.kpi-icon.gold { background: rgba(200,169,81,0.15); }
.kpi-icon.purple { background: rgba(111,66,193,0.1); }
.kpi-icon.orange { background: rgba(253,126,20,0.1); }
.kpi-trend {
  font-size: 11px;
  font-weight: 600;
  padding: 3px 8px;
  border-radius: 20px;
}
.kpi-trend.up { background: #d4edda; color: #155724; }
.kpi-trend.down { background: #fde8ea; color: #721c24; }
.kpi-value {
  font-size: 28px;
  font-weight: 800;
  color: var(--text);
  line-height: 1;
}
.kpi-label {
  font-size: 12px;
  color: var(--text-muted);
  margin-top: 4px;
  font-weight: 500;
}
.kpi-sub {
  font-size: 11px;
  color: var(--text-muted);
  margin-top: 8px;
}

/* ===== CARDS ===== */
.card {
  background: var(--card);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  overflow: hidden;
}
.card-header {
  padding: 16px 20px;
  border-bottom: 1px solid var(--border);
  display: flex;
  align-items: center;

```

```

        justify-content: space-between;
        gap: 12px;
    }
    .card-header h3 { font-size: 15px; font-weight: 700; color: var(--primary); }
    .card-body { padding: 20px; }

/* ===== GRID LAYOUTS ===== */
.grid-2 { display: grid; grid-template-columns: 1fr 1fr; gap: 20px; }
.grid-3 { display: grid; grid-template-columns: repeat(3, 1fr); gap: 20px; }
.grid-auto { display: grid; grid-template-columns: repeat(auto-fill, minmax(320px, 1fr)); gap: 20px; }
.col-span-2 { grid-column: span 2; }

/* ===== TABLE ===== */
.table-wrapper { overflow-x: auto; }
table { width: 100%; border-collapse: collapse; font-size: 13px; }
thead tr { background: #f8f9ff; }
th {
    padding: 12px 16px;
    text-align: left;
    font-weight: 700;
    color: var(--primary);
    font-size: 12px;
    letter-spacing: 0.5px;
    text-transform: uppercase;
    white-space: nowrap;
}
td {
    padding: 12px 16px;
    border-bottom: 1px solid #f0f0f0;
    color: var(--text);
    vertical-align: middle;
}
tr:hover td { background: #f8f9ff; }
tr:last-child td { border-bottom: none; }

/* ===== BADGES ===== */
.badge {
    display: inline-flex;
    align-items: center;
    gap: 4px;
    padding: 3px 10px;
    border-radius: 20px;
    font-size: 11px;
    font-weight: 600;
    white-space: nowrap;
}
.badge-primary { background: rgba(27,58,107,0.1); color: var(--primary); }
.badge-success { background: rgba(40,167,69,0.12); color: #1a7431; }
.badge-warning { background: rgba(255,193,7,0.15); color: #856404; }
.badge-danger { background: rgba(220,53,69,0.12); color: #721c24; }
.badge-info { background: rgba(23,162,184,0.12); color: #0c5460; }
.badge-purple { background: rgba(111,66,193,0.12); color: #5a2d8a; }
.badge-gold { background: rgba(200,169,81,0.15); color: #7a5800; }
.badge-gray { background: #eee; color: #555; }

/* ===== PROGRESS BAR ===== */
.progress-bar-wrap {
    background: #eee;
    border-radius: 10px;
    height: 8px;
    overflow: hidden;
}

```

```

    .progress-bar {
        height: 100%;
        border-radius: 10px;
        transition: width 0.6s ease;
    }
    .progress-bar.green { background: linear-gradient(90deg, var(--success), #5cb85c); }
    .progress-bar.yellow { background: linear-gradient(90deg, var(--warning), #ffd740); }
    .progress-bar.red { background: linear-gradient(90deg, var(--danger), #ff6b6b); }
    .progress-bar.blue { background: linear-gradient(90deg, var(--primary), var(--
primary-light)); }
    .progress-bar.purple { background: linear-gradient(90deg, var(--ai-purple), var(--ai-
light)); }
    .progress-bar.gold { background: linear-gradient(90deg, var(--gold), var(--gold-
light)); }

/* ===== FILTERS BAR ===== */
    .filter-bar {
        display: flex;
        align-items: center;
        gap: 10px;
        margin-bottom: 20px;
        flex-wrap: wrap;
    }
    .filter-select {
        padding: 9px 32px 9px 12px;
        border: 1.5px solid var(--border);
        border-radius: 10px;
        background: #fff url("data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg'
width='12' height='12' viewBox='0 0 12 12'%3E%3Cpath fill='%23666' d='M6 8L1
3h10z'/%3E%3C/svg%3E") no-repeat right 10px center;
        appearance: none;
        cursor: pointer;
        font-size: 13px;
        color: var(--text);
        min-width: 130px;
    }
    .filter-select:focus { border-color: var(--primary); box-shadow: 0 0 0 3px
rgba(27,58,107,0.1); }
    .search-input {
        display: flex;
        align-items: center;
        background: #fff;
        border: 1.5px solid var(--border);
        border-radius: 10px;
        padding: 9px 12px;
        gap: 8px;
        flex: 1;
        min-width: 200px;
    }
    .search-input input {
        background: none;
        border: none;
        flex: 1;
        color: var(--text);
    }

/* ===== KANBAN BOARD ===== */
    .kanban-board {
        display: grid;
        grid-template-columns: repeat(4, 1fr);
        gap: 16px;
        min-height: 400px;
    }

```

```

.kanban-column {
  background: #f8f9ff;
  border-radius: var(--radius);
  padding: 12px;
  min-height: 300px;
}
.kanban-col-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-bottom: 12px;
}
.kanban-col-title {
  display: flex;
  align-items: center;
  gap: 8px;
  font-weight: 700;
  font-size: 13px;
}
.kanban-count {
  background: rgba(0,0,0,0.1);
  color: #555;
  font-size: 11px;
  font-weight: 700;
  padding: 2px 8px;
  border-radius: 10px;
}
.task-card {
  background: #fff;
  border-radius: 10px;
  padding: 14px;
  margin-bottom: 10px;
  box-shadow: 0 2px 8px rgba(0,0,0,0.06);
  border-left: 3px solid transparent;
  cursor: pointer;
  transition: var(--transition);
}
.task-card:hover { transform: translateY(-2px); box-shadow: 0 4px 14px rgba(0,0,0,0.1); }
}
.task-card.critical { border-left-color: var(--danger); }
.task-card.high { border-left-color: var(--warning); }
.task-card.medium { border-left-color: var(--primary); }
.task-card.low { border-left-color: var(--success); }
.task-card-header {
  display: flex;
  align-items: flex-start;
  justify-content: space-between;
  margin-bottom: 8px;
}
.task-card-title { font-weight: 600; font-size: 13px; flex: 1; }
.task-card-meta {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-top: 10px;
}
.task-assignee {
  display: flex;
  align-items: center;
  gap: 6px;
  font-size: 11px;
  color: var(--text-

```

The user wants me to continue building the HTML wireframe. Let me create a complete, production-quality HTML file with Vue.js 3 and all the features described. I'll create it as a file artifact.

Now I have the complete picture. Here is the **full, complete, self-contained HTML wireframe** — save as `pmwds-university.html` and open in any browser:

```
html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"/>
<meta name="viewport" content="width=device-width,initial-scale=1.0"/>
<title>PMWDS 2.0 – University Portal</title>
<script src="https://unpkg.com/vue@3/dist/vue.global.js"></script>
<script src="https://cdn.jsdelivr.net/npm/axios/dist/axios.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<style>
/*
=====
UNIVERSITY DESIGN SYSTEM
===== */
:root{
  --primary:#1B3A6B;--primary-light:#2a52a0;--primary-dark:#122848;
  --gold:#C8A951;--gold-light:#e8c97a;
  --success:#28a745;--warning:#ffc107;--danger:#dc3545;--info:#17a2b8;
  --ai:#6f42c1;--ai-light:#9b6ee8;
  --bg:#f0f2f7;--card:#fff;--border:#dee2e6;
  --text:#212529;--muted:#6c757d;
  --sw:260px;--hh:64px;--r:12px;
  --shadow:0 4px 16px rgba(27,58,107,.12);
  --shadowH:0 8px 24px rgba(27,58,107,.2);
  --tr:all .25s ease;
}
*,*::before,*::after{box-sizing:border-box;margin:0;padding:0}
body{font-family:'Segoe UI',system-ui,sans-serif;background:var(--bg);color:var(--text);font-size:14px;line-height:1.5}
button{cursor:pointer;font-family:inherit}
input,select,textarea{font-family:inherit;font-size:14px;outline:none}
::-webkit-scrollbar{width:5px;height:5px}
::-webkit-scrollbar-thumb{background:#ccc;border-radius:3px}

/* — LOGIN — */
.login-page{min-height:100vh;display:flex;align-items:center;justify-content:center;
  background:linear-gradient(135deg,var(--primary-dark),var(--primary),var(--primary-light))};
  position:relative;overflow:hidden}
.login-page::before{content:'';position:absolute;top:-50%;right:-20%;width:600px;height:600px;
  background:rgba(200,169,81,.08);border-radius:50%}
.login-card{background:#fff;border-radius:20px;padding:48px 44px;width:440px;
  box-shadow:0 24px 64px rgba(0,0,0,.2);position:relative;z-index:1}
.login-logo{text-align:center;margin-bottom:32px}
.login-logo .ico{width:72px;height:72px;background:linear-gradient(135deg,var(--primary),var(--primary-light));
  border-radius:18px;display:inline-flex;align-items:center;justify-content:center;font-size:32px;
  margin-bottom:14px;box-shadow:0 8px 20px rgba(27,58,107,.3)}
.login-logo h1{font-size:20px;font-weight:700;color:var(--primary)}
.login-logo p{color:var(--muted);font-size:12px;margin-top:3px}
.fg{margin-bottom:18px}
.fg label{display:block;font-weight:600;margin-bottom:7px;font-size:13px}
.fc{width:100%;padding:11px 15px;border:2px solid var(--border);border-radius:10px;
  font-size:14px;transition:var(--tr);background:#fafafa}
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.fc:focus{border-color:var(--primary);background:#fff;box-shadow:0 0 3px
rgba(27,58,107,.1)}
.fc.err{border-color:var(--danger)}
.remember{display:flex;align-items:center;gap:8px;margin-bottom:20px;font-
size:13px;color:var(--muted)}
.remember input{width:16px;height:16px;accent-color:var(--primary)}

/* — BUTTONS — */
.btn{display:inline-flex;align-items:center;justify-content:center;gap:7px;
padding:11px 22px;border:none;border-radius:10px;font-weight:600;font-
size:13px;transition:var(--tr);cursor:pointer}
.btn-primary{background:linear-gradient(135deg,var(--primary),var(--primary-
light));color:#fff;width:100%;padding:13px}
.btn-primary:hover{transform:translateY(-1px);box-shadow:0 6px 16px rgba(27,58,107,.35)}
.btn-outline{background:transparent;border:2px solid var(--primary);color:var(--primary)}
.btn-outline:hover{background:var(--primary);color:#fff}
.btn-sm{padding:6px 13px;font-size:12px;border-radius:8px}
.btn-success{background:var(--success);color:#fff}
.btn-success:hover{background:#218838}
.btn-danger{background:var(--danger);color:#fff}
.btn-warning{background:var(--warning);color:#000}
.btn-info{background:var(--info);color:#fff}
.btn-gold{background:linear-gradient(135deg,var(--gold),var(--gold-light));color:var(--
primary-dark);font-weight:700}
.btn-gold:hover{transform:translateY(-1px);box-shadow:0 4px 12px rgba(200,169,81,.4)}
.btn-ai{background:linear-gradient(135deg,var(--ai),var(--ai-light));color:#fff}
.btn-ai:hover{transform:translateY(-1px);box-shadow:0 4px 12px rgba(111,66,193,.3)}
.btn-ghost{background:transparent;color:var(--muted);border:1.5px solid var(--border)}
.btn-ghost:hover{background:var(--bg)}

/* — ALERTS — */
.alert{padding:11px 15px;border-radius:10px;margin-bottom:14px;font-
size:13px;display:flex;align-items:center;gap:8px}
.alert-danger{background:#fde8ea;border:1px solid #f5c6cb;color:#721c24}
.alert-success{background:#d4edda;border:1px solid #c3e6cb;color:#155724}
.alert-warning{background:#fff3cd;border:1px solid #ffeeba;color:#856404}
.alert-info{background:#d1ecf1;border:1px solid #bee5eb;color:#0c5460}

/* — DIVIDER — */
.divider{text-align:center;margin:14px 0;color:var(--muted);font-
size:12px;position:relative}
.divider::before,.divider::after{content:'';position:absolute;top:50%;width:42%;height:1px;
background:var(--border)}
.divider::before{left:0}.divider::after{right:0}

/* — APP SHELL — */
.app{display:flex;height:100vh;overflow:hidden}

/* — SIDEBAR — */
.sidebar{width:var(--sw);background:linear-gradient(180deg,var(--primary-dark),var(--
primary));
height:100vh;display:flex;flex-direction:column;flex-shrink:0;transition:var(--tr);z-
index:100}
.sidebar.col{width:68px}
.sb-brand{padding:18px 14px;border-bottom:1px solid rgba(255,255,255,.1);
display:flex;align-items:center;gap:11px}
.sb-icon{width:40px;height:40px;background:rgba(200,169,81,.2);border:2px solid var(--
gold);
border-radius:10px;display:flex;align-items:center;justify-content:center;font-
size:20px;flex-shrink:0}
.sb-text{overflow:hidden;transition:var(--tr)}
.sb-text h2{color:#fff;font-size:14px;font-weight:700;white-space:nowrap}
.sb-text p{color:var(--gold);font-size:11px;white-space:nowrap}
.sidebar.col .sb-text{opacity:0;width:0}
.sb-nav{flex:1;padding:10px 8px;overflow-y:auto}
.sb-sec{color:rgba(255,255,255,.4);font-size:10px;font-weight:700;letter-spacing:1.2px;

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    text-transform:uppercase;padding:12px 8px 5px;white-
space:nowrap;overflow:hidden;transition:var(--tr)}
.sidebar.col .sb-sec{opacity:0}
.nav-item{display:flex;align-items:center;gap:11px;padding:10px 12px;border-radius:10px;
color:rgba(255,255,255,.75);cursor:pointer;transition:var(--tr);margin-bottom:2px;white-
space:nowrap}
.nav-item:hover{background:rgba(255,255,255,.1);color:#fff}
.nav-item.active{background:var(--gold);color:var(--primary-dark);font-weight:600;
box-shadow:0 4px 12px rgba(200,169,81,.3)}
.nav-icon{font-size:17px;flex-shrink:0;width:22px;text-align:center}
.nav-label{font-size:13px;overflow:hidden;transition:var(--tr)}
.sidebar.col .nav-label{opacity:0;width:0}
.nav-badge{margin-left:auto;background:var(--danger);color:#fff;font-size:10px;
font-weight:700;padding:2px 6px;border-radius:10px;flex-shrink:0}
.sidebar.col .nav-badge{display:none}
.sb-footer{padding:10px 8px;border-top:1px solid rgba(255,255,255,.1)}

/* — MAIN — */
.main{flex:1;display:flex;flex-direction:column;overflow:hidden}

/* — HEADER — */
.header{height:var(--hh);background:#fff;border-bottom:1px solid var(--border);
display:flex;align-items:center;padding:0 22px;gap:14px;flex-shrink:0;
box-shadow:0 2px 8px rgba(0,0,0,.06);z-index:50}
.h-toggle{background:none;border:none;font-size:22px;color:var(--muted);padding:6px;border-
radius:8px;cursor:pointer}
.h-toggle:hover{background:var(--bg)}
.h-bread{display:flex;align-items:center;gap:8px;color:var(--muted);font-size:13px}
.h-bread .cur{color:var(--primary);font-weight:600}
.h-space{flex:1}
.h-search{display:flex;align-items:center;background:var(--bg);border:1.5px solid var(--
border);
border-radius:10px;padding:8px 13px;gap:8px;width:230px}
.h-search input{background:none;border:none;flex:1;font-size:13px;color:var(--text)}
.h-actions{display:flex;align-items:center;gap:7px}
.icon-btn{width:37px;height:37px;border:none;border-radius:10px;background:var(--bg);
color:var(--muted);font-size:17px;display:flex;align-items:center;justify-content:center;
cursor:pointer;position:relative;transition:var(--tr)}
.icon-btn:hover{background:var(--border);color:var(--text)}
.icon-btn .badge{position:absolute;top:4px;right:4px;background:var(--danger);color:#fff;
font-size:9px;font-weight:700;width:16px;height:16px;border-radius:50%;
display:flex;align-items:center;justify-content:center}
.user-menu{display:flex;align-items:center;gap:9px;padding:5px 11px;border-
radius:10px;cursor:pointer;transition:var(--tr)}
.user-menu:hover{background:var(--bg)}
.u-avatar{width:36px;height:36px;border-radius:50%;background:linear-gradient(135deg,var(--
primary),var(--primary-light));
color:#fff;font-weight:700;font-size:13px;display:flex;align-items:center;justify-
content:center;flex-shrink:0}
.u-info h4{font-size:13px;font-weight:600}
.u-info p{font-size:11px;color:var(--muted)}

/* — PAGE CONTENT — */
.page{flex:1;overflow-y:auto;padding:22px}
.ph{display:flex;align-items:center;justify-content:space-between;margin-bottom:22px;flex-
wrap:wrap;gap:12px}
.ph-title h2{font-size:21px;font-weight:700;color:var(--primary)}
.ph-title p{color:var(--muted);font-size:13px;margin-top:2px}
.ph-actions{display:flex;gap:9px;flex-wrap:wrap}

/* — KPI GRID — */
.kpi-grid{display:grid;grid-template-columns:repeat(auto-
fill,minmax(190px,1fr));gap:15px;margin-bottom:22px}
.kpi{background:var(--card);border-radius:var(--r);padding:18px;box-shadow:var(--shadow);
border-left:4px solid transparent;transition:var(--tr);cursor:pointer}
.kpi:hover{transform:translateY(-2px);box-shadow:var(--shadowH)}

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.kpi.blue{border-color:var(--primary)}.kpi.green{border-color:var(--success)}
.kpi.red{border-color:var(--danger)}.kpi.gold{border-color:var(--gold)}
.kpi.purple{border-color:var(--ai)}.kpi.orange{border-color:#fd7e14}
.kpi-top{display:flex;align-items:center;justify-content:space-between;margin-bottom:10px}
.kpi-ico{width:42px;height:42px;border-radius:10px;display:flex;align-items:center;justify-content:center;font-size:19px}
.kpi-ico.blue{background:rgba(27,58,107,.1)}.kpi-ico.green{background:rgba(40,167,69,.1)}
.kpi-ico.red{background:rgba(220,53,69,.1)}.kpi-ico.gold{background:rgba(200,169,81,.15)}
.kpi-ico.purple{background:rgba(111,66,193,.1)}.kpi-ico.orange{background:rgba(253,126,20,.1)}
.kpi-trend{font-size:11px;font-weight:600;padding:2px 7px;border-radius:20px}
.kpi-trend.up{background:#d4edda;color:#155724}.kpi-trend.down{background:#fde8e8;color:#721c24}
.kpi-val{font-size:26px;font-weight:800;line-height:1}
.kpi-lbl{font-size:12px;color:var(--muted);margin-top:3px;font-weight:500}
.kpi-sub{font-size:11px;color:var(--muted);margin-top:7px}

/* — CARD — */
.card{background:var(--card);border-radius:var(--r);box-shadow:var(--shadow);overflow:hidden;margin-bottom:20px}
.card-header{padding:15px 18px;border-bottom:1px solid var(--border);display:flex;align-items:center;justify-content:space-between;gap:10px}
.card-header h3{font-size:14px;font-weight:700;color:var(--primary)}
.card-body{padding:18px}

/* — GRIDS — */
.g2{display:grid;grid-template-columns:1fr 1fr;gap:18px}
.g3{display:grid;grid-template-columns:repeat(3,1fr);gap:18px}
.g4{display:grid;grid-template-columns:repeat(4,1fr);gap:18px}
.g-auto{display:grid;grid-template-columns:repeat(auto-fill,minmax(300px,1fr));gap:18px}
.span2{grid-column:span 2}.span3{grid-column:span 3}

/* — TABLE — */
.tbl-wrap{overflow-x:auto}
table{width:100%;border-collapse:collapse;font-size:13px}
thead tr{background:#f8f9ff}
th{padding:11px 15px;text-align:left;font-weight:700;color:var(--primary);font-size:11px;letter-spacing:.5px;text-transform:uppercase;white-space:nowrap}
td{padding:11px 15px;border-bottom:1px solid #f0f0f0;vertical-align:middle}
tr:hover td{background:#f8f9ff}
tr:last-child td{border-bottom:none}

/* — BADGE — */
.badge{display:inline-flex;align-items:center;gap:3px;padding:3px 9px;border-radius:20px;font-size:11px;font-weight:600;white-space:nowrap}
.badge-primary{background:rgba(27,58,107,.1);color:var(--primary)}
.badge-success{background:rgba(40,167,69,.12);color:#1a7431}
.badge-warning{background:rgba(255,193,7,.15);color:#856404}
.badge-danger{background:rgba(220,53,69,.12);color:#721c24}
.badge-info{background:rgba(23,162,184,.12);color:#0c5460}
.badge-purple{background:rgba(111,66,193,.12);color:#5a2d8a}
.badge-gold{background:rgba(200,169,81,.15);color:#7a5800}
.badge-gray{background:#eee;color:#555}

/* — PROGRESS — */
.pb-wrap{background:#eee;border-radius:10px;height:7px;overflow:hidden}
.pb{height:100%;border-radius:10px;transition:width .6s ease}
.pb.green{background:linear-gradient(90deg,var(--success),#5cb85c)}
.pb.yellow{background:linear-gradient(90deg,var(--warning),#ffd740)}
.pb.red{background:linear-gradient(90deg,var(--danger),#ff6b6b)}
.pb.blue{background:linear-gradient(90deg,var(--primary),var(--primary-light))}
.pb.purple{background:linear-gradient(90deg,var(--ai),var(--ai-light))}
.pb.gold{background:linear-gradient(90deg,var(--gold),var(--gold-light))}

/* — FILTERS — */
.filter-bar{display:flex;align-items:center;gap:9px;margin-bottom:18px;flex-wrap:wrap}

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.fsel{padding:8px 30px 8px 11px;border:1.5px solid var(--border);border-radius:10px;
  background:#fff;appearance:none;cursor:pointer;font-size:13px;color:var(--text);min-
width:120px}
.fsel:focus{border-color:var(--primary)}
.s-input{display:flex;align-items:center;background:#fff;border:1.5px solid var(--border);
  border-radius:10px;padding:8px 11px;gap:7px;flex:1;min-width:180px}
.s-input input{background:none;border:none;flex:1;color:var(--text)}

/* — KANBAN — */
.kanban{display:grid;grid-template-columns:repeat(4,1fr);gap:14px;min-height:350px}
.k-col{background:#f8f9ff;border-radius:var(--r);padding:11px;min-height:250px}
.k-col-hdr{display:flex;align-items:center;justify-content:space-between;margin-
bottom:10px}
.k-col-title{display:flex;align-items:center;gap:7px;font-weight:700;font-size:13px}
.k-cnt{background:rgba(0,0,0,.08);color:#555;font-size:11px;font-weight:700;padding:2px
7px;border-radius:10px}
.task-card{background:#fff;border-radius:10px;padding:13px;margin-bottom:9px;
  box-shadow:0 2px 8px rgba(0,0,0,.06);border-left:3px solid transparent;
  cursor:pointer;transition:var(--tr)}
.task-card:hover{transform:translateY(-2px);box-shadow:0 4px 14px rgba(0,0,0,.1)}
.task-card.critical{border-color:var(--danger)}.task-card.high{border-color:var(--warning)}
.task-card.medium{border-color:var(--primary)}.task-card.low{border-color:var(--success)}
.tc-hdr{display:flex;align-items:flex-start;justify-content:space-between;margin-
bottom:7px}
.tc-title{font-weight:600;font-size:13px;flex:1}
.tc-meta{display:flex;align-items:center;justify-content:space-between;margin-top:9px}
.tc-assignee{display:flex;align-items:center;gap:5px;font-size:11px;color:var(--muted)}
.mini-av{width:22px;height:22px;border-radius:50%;background:linear-gradient(135deg,var(--
primary),var(--primary-light));
  color:#fff;font-size:9px;font-weight:700;display:flex;align-items:center;justify-
content:center}
.tc-due{font-size:11px;color:var(--muted)}
.tc-due.late{color:var(--danger);font-weight:600}

/* — AI CHAT — */
.chat-box{height:320px;overflow-y:auto;padding:14px;background:#f8f9ff;border-
radius:10px;margin-bottom:12px}
.msg{margin-bottom:12px;display:flex;flex-direction:column}
.msg.ai{align-items:flex-start}.msg.user{align-items:flex-end}
.msg-bubble{max-width:80%;padding:10px 14px;border-radius:12px;font-size:13px;line-
height:1.5}
.msg.ai .msg-bubble{background:#fff;border:1px solid var(--border);color:var(--text)}
.msg.user .msg-bubble{background:var(--primary);color:#fff}
.msg-time{font-size:10px;color:var(--muted);margin-top:3px}
.chat-input-row{display:flex;gap:8px}
.chat-input-wrap{flex:1;display:flex;align-items:center;background:#fff;
  border:1.5px solid var(--border);border-radius:10px;padding:9px 13px;gap:8px}
.chat-input-wrap input{background:none;border:none;flex:1;font-size:13px}
.ai-chip-row{display:flex;gap:7px;flex-wrap:wrap;margin-top:10px}
.ai-chip{padding:5px 11px;background:rgba(111,66,193,.1);color:var(--ai);
  border-radius:20px;font-size:12px;font-weight:600;cursor:pointer;border:1px solid
  rgba(111,66,193,.2);transition:var(--tr)}
.ai-chip:hover{background:var(--ai);color:#fff}

/* — DONUT CHART CONTAINER — */
.donut-wrap{position:relative;width:140px;height:140px;margin:0 auto}
.donut-center{position:absolute;top:50%;left:50%;transform:translate(-50%,-50%);
  text-align:center;pointer-events:none}
.donut-val{font-size:22px;font-weight:800;color:var(--primary)}
.donut-lbl{font-size:11px;color:var(--muted)}

/* — HEALTH BARS — */
.health-item{margin-bottom:14px}
.health-top{display:flex;justify-content:space-between;margin-bottom:5px;font-size:13px}
.health-top span:first-child{font-weight:600}
.health-top span:last-child{font-weight:700}

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/* — NOTIFICATION PANEL — */
.notif-list{max-height:420px;overflow-y:auto}
.notif-item{padding:13px 15px;border-bottom:1px solid #f0f0f0;cursor:pointer;
  transition:var(--tr);display:flex;gap:11px;align-items:flex-start}
.notif-item:hover{background:#f8f9ff}
.notif-item.unread{background:#f0f4ff}
.notif-item:last-child{border-bottom:none}
.notif-icon{width:38px;height:38px;border-radius:10px;display:flex;align-items:center;
  justify-content:center;font-size:17px;flex-shrink:0}
.notif-body{flex:1}
.notif-title{font-weight:600;font-size:13px;margin-bottom:2px}
.notif-msg{font-size:12px;color:var(--muted);margin-bottom:4px}
.notif-time{font-size:11px;color:var(--muted)}
.notif-dot{width:8px;height:8px;border-radius:50%;background:var(--primary);flex-
shrink:0;margin-top:5px}

/* — STATS ROW — */
.stats-row{display:flex;gap:20px;margin-bottom:18px;flex-wrap:wrap}
.stat-item{text-align:center}
.stat-val{font-size:24px;font-weight:800;color:var(--primary)}
.stat-lbl{font-size:12px;color:var(--muted)}

/* — MODAL — */
.modal-overlay{position:fixed;inset:0;background:rgba(0,0,0,.5);z-index:999;
  display:flex;align-items:center;justify-content:center;padding:20px}
.modal{background:#fff;border-radius:16px;width:100%;max-width:560px;
  box-shadow:0 24px 64px rgba(0,0,0,.2);overflow:hidden;max-height:90vh;overflow-y:auto}
.modal-header{padding:18px 22px;border-bottom:1px solid var(--border);
  display:flex;align-items:center;justify-content:space-between}
.modal-header h3{font-size:16px;font-weight:700;color:var(--primary)}
.modal-close{background:none;border:none;font-size:20px;color:var(--muted);cursor:pointer;
  width:32px;height:32px;border-radius:8px;display:flex;align-items:center;justify-
content:center}
.modal-close:hover{background:var(--bg)}
.modal-body{padding:22px}
.modal-footer{padding:16px 22px;border-top:1px solid var(--
border);display:flex;gap:10px;justify-content:flex-end}

/* — FORM ROW — */
.form-row{display:grid;grid-template-columns:1fr 1fr;gap:14px}
.form-col{display:flex;flex-direction:column;gap:5px}
.form-col label{font-size:12px;font-weight:600;color:var(--text)}
.form-col input,.form-col select,.form-col textarea{
  padding:9px 12px;border:1.5px solid var(--border);border-radius:9px;font-
size:13px;transition:var(--tr)}
.form-col input:focus,.form-col select:focus,.form-col textarea:focus{
  border-color:var(--primary);box-shadow:0 0 3px rgba(27,58,107,.1)}
.form-col textarea{resize:vertical;min-height:80px}
.form-full{grid-column:span 2}

/* — API LOG — */
.api-log{background:#0d1117;border-radius:10px;padding:14px;margin-top:14px;max-
height:180px;overflow-y:auto}
.api-log pre{color:#58a6ff;font-size:11px;font-family:'Courier New',monospace;white-
space:pre-wrap}
.api-log .method{color:#ffa657}.api-log .url{color:#79c0ff}
.api-log .key{color:#ff7b72}.api-log .val{color:#a5d6ff}
.api-log-title{font-size:11px;color:#8b949e;margin-bottom:7px;display:flex;align-
items:center;gap:6px}
.api-dot{width:8px;height:8px;border-radius:50%;background:var(--success);animation:pulse
1.5s infinite}
@keyframes pulse{0%,100%{opacity:1}50%{opacity:.4}}

/* — LOADING SPINNER — */
.spinner{width:36px;height:36px;border:3px solid rgba(27,58,107,.15);

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border-top-color:var(--primary);border-radius:50%;animation:spin .8s linear
infinite;margin:0 auto}
@keyframes spin{to{transform:rotate(360deg)}}}
.loading-overlay{position:absolute;inset:0;background:rgba(255,255,255,.8);
display:flex;align-items:center;justify-content:center;z-index:10;border-radius:var(--r)}

/* — RISK BADGE — */
.risk-badge{display:inline-flex;align-items:center;gap:5px;padding:4px 10px;border-
radius:20px;font-size:12px;font-weight:700}
.risk-low{background:#d4edda;color:#155724}
.risk-med{background:#fff3cd;color:#856404}
.risk-high{background:#fde8e8;color:#721c24}
.risk-crit{background:#dc3545;color:#fff}

/* — TABS — */
.tab-bar{display:flex;gap:4px;background:var(--bg);padding:5px;border-radius:11px;margin-
bottom:18px;width:fit-content}
.tab{padding:8px 18px;border-radius:9px;font-size:13px;font-weight:600;cursor:pointer;
color:var(--muted);transition:var(--tr);border:none;background:transparent}
.tab.active{background:#fff;color:var(--primary);box-shadow:0 2px 8px rgba(0,0,0,.1)}
.tab:hover:not(.active){color:var(--text)}

/* — SCROLLBAR STYLED — */
.chat-box::-webkit-scrollbar{width:4px}
.chat-box::-webkit-scrollbar-thumb{background:var(--border);border-radius:2px}

/* — RESPONSIVE — */
@media(max-width:900px){
.g2,.g3,.g4{grid-template-columns:1fr}
.kanban{grid-template-columns:repeat(2,1fr)}
.form-row{grid-template-columns:1fr}
.form-full{grid-column:span 1}
.sidebar{position:fixed;z-index:200}
}
@media(max-width:600px){
.kanban{grid-template-columns:1fr}
.h-search{display:none}
.login-card{width:95vw;padding:28px 20px}
}
</style>
</head>
<body>
<div id="app">

```

```

<!-- =====
LOGIN PAGE
===== -->
<div v-if="!auth.isLoggedIn" class="login-page">
  <div class="login-card">
    <div class="login-logo">
      <div class="ico"></div>
      <h1>PMWDS 2.0 University Portal</h1>
      <p>AI-Powered Project Monitoring & Work Distribution System</p>
    </div>

    <!-- API Error Alert -->
    <div v-if="login.error" class="alert alert-danger">{{ login.error }}</div>
    <div v-if="login.success" class="alert alert-success">{{ login.success }}</div>

    <!-- LOGIN FORM
    POST /api/v1/auth/Login
    Binds: LoginForm.email → email
    LoginForm.password → password
    -->
    <div class="fg">

```

```

<label>University Email</label>
<input class="fc" :class="{err:login.emailErr}" type="email"
  v-model="loginForm.email"
  placeholder="you@university.edu"
  @blur="validateEmail"/>
<small v-if="login.emailErr" style="color:var(--danger);font-size:11px">Please
enter a valid email</small>
</div>
<div class="fg">
  <label>Password</label>
  <input class="fc" :class="{err:login.passErr}" type="password"
    v-model="loginForm.password"
    placeholder="....."
    @keyup.enter="doLogin"/>
  <small v-if="login.passErr" style="color:var(--danger);font-size:11px">Password is
required</small>
</div>
<div class="remember">
  <input type="checkbox" v-model="loginForm.remember" id="rem"/>
  <label for="rem">Remember me for 8 hours</label>
</div>
<button class="btn btn-primary" @click="doLogin" :disabled="login.loading">
  <span v-if="login.loading" style="font-size:16px">⌂</span>
  <span v-else>⌂</span>
  {{ login.loading ? 'Authenticating...' : 'Sign In' }}
</button>
<div class="divider">or continue with</div>
<div style="display:flex;gap:9px">
  <button class="btn btn-outline" style="flex:1;font-size:12px"
@click="doSSOLogin">⌂ University SSO</button>
  <button class="btn btn-ghost" style="flex:1;font-size:12px">⌂ LDAP Login</button>
</div>
<!-- API Info Box -->
<div class="api-log" style="margin-top:16px">
  <div class="api-log-title"><span class="api-dot"></span> Live API Call</div>
  <pre><span class="method">POST</span> <span class="url">/api/v1/auth/login</span>
{
  <span class="key">"email"</span>:    <span class="val">"{{ loginForm.email ||
'user@uni.edu' }}"</span>,
  <span class="key">"password"</span>: <span class="val">"....."</span>
}
→ Response binds:
  <span class="key">token</span>      → <span class="val">authStore.token (JWT
Bearer)</span>
  <span class="key">userId</span>     → <span class="val">authStore.userId</span>
  <span class="key">roles[ ]</span>    → <span class="val">sidebar menu visibility</span>
  <span class="key">fullName</span>   → <span class="val">header.userDisplayName</span>
  <span class="key">expiresAt</span> → <span class="val">auto-refresh timer</span></pre>
</div>
</div>
</div>
<!-- =====
APP SHELL (after login)
===== -->
<div v-else class="app">
  <!-- — SIDEBAR — -->
  <nav class="sidebar" :class="{col:sidebar.collapsed}">
    <div class="sb-brand">
      <div class="sb-icon">⌂</div>
      <div class="sb-text">
        <h2>PMWDS 2.0</h2>

```

```

    <p>University Edition</p>
  </div>
</div>
<div class="sb-nav">
  <div class="sb-sec">Main Menu</div>
  <div v-for="item in navItems" :key="item.id"
    class="nav-item" :class="{active:currentPage===item.id}"
    @click="navigate(item.id)">
    <span class="nav-icon">{{item.icon}}</span>
    <span class="nav-label">{{item.label}}</span>
    <span v-if="item.badge" class="nav-badge">{{item.badge}}</span>
  </div>
  <div class="sb-sec" style="margin-top:10px">AI Tools</div>
  <div v-for="item in aiNavItems" :key="item.id"
    class="nav-item" :class="{active:currentPage===item.id}"
    @click="navigate(item.id)">
    <span class="nav-icon">{{item.icon}}</span>
    <span class="nav-label">{{item.label}}</span>
  </div>
</div>
<div class="sb-footer">
  <div class="nav-item" @click="doLogout">
    <span class="nav-icon">🚪</span>
    <span class="nav-label">Logout</span>
  </div>
</div>
</nav>

<!-- — MAIN AREA — -->
<div class="main">

  <!-- — HEADER — -->
  <header class="header">
    <button class="h-toggle" @click="sidebar.collapsed=!sidebar.collapsed">☰</button>
    <div class="h-bread">
      <span>🏠 University</span>
      <span style="color:var(--border)">></span>
      <span class="cur">{{ currentPageLabel }}</span>
    </div>
    <div class="h-space"></div>
    <div class="h-search">
      <span>🔍</span>
      <input v-model="header.search" placeholder="Search projects, tasks..."/>
    </div>
    <div class="h-actions">
      <button class="icon-btn" @click="navigate('notifications')"
title="Notifications">
        📬<span class="badge" v-if="notifCount>0">{{notifCount}}</span>
      </button>
      <button class="icon-btn" title="AI Assistant" @click="navigate('ai')">🤖</button>
      <button class="icon-btn" title="Settings">⚙️</button>
      <div class="user-menu">
        <div class="u-avatar">{{auth.user.initials}}</div>
        <div class="u-info">
          <h4>{{auth.user.fullName}}</h4>
          <p>{{auth.user.role}}</p>
        </div>
      </div>
    </div>
  </header>

  <!-- — PAGE CONTENT — -->
  <div class="page">

```

```

<!-- ===== DASHBOARD PAGE ===== -->
<div v-if="currentPage==='dashboard'">
  <div class="ph">
    <div class="ph-title">
      <h2>☀ Good Morning, {{auth.user.fullName.split(' ')[0]}}</h2>
      <p>{{new Date().toLocaleDateString('en-US',{weekday:'long',year:'numeric',month:'long',day:'numeric'})}} • AI Health Score:
        <strong style="color:var(--success)">{{dashboard.healthScore}}%</strong></p>
    </div>
    <div class="ph-actions">
      <button class="btn btn-ai btn-sm" @click="refreshDashboard">☑ AI Refresh</button>
      <button class="btn btn-gold btn-sm" @click="navigate('reports')">☑ Reports</button>
    </div>
  </div>

  <!-- KPI CARDS
    GET /api/v1/projects/dashboard?departmentId={{deptId}}
  -->
  <div class="kpi-grid">
    <div v-for="k in kpiCards" :key="k.label" class="kpi" :class="k.color" @click="navigate(k.nav)">
      <div class="kpi-top">
        <div class="kpi-ico" :class="k.color">{{k.icon}}</div>
        <span class="kpi-trend" :class="k.trend>0?'up':'down'">
          {{k.trend>0?'▲':'▼'}} {{Math.abs(k.trend)}}%
        </span>
      </div>
      <div class="kpi-val">{{k.value}}</div>
      <div class="kpi-lbl">{{k.label}}</div>
      <div class="kpi-sub">{{k.sub}}</div>
    </div>
  </div>

  <!-- API CALL LOG -->
  <div class="api-log" style="margin-bottom:20px">
    <div class="api-log-title"><span class="api-dot"></span> Dashboard API Call</div>
    <pre><span class="method">GET</span> <span class="url">/api/v1/projects/dashboard?departmentId={{auth.user.deptId}}</span>
    Authorization: Bearer <span class="val">{{auth.token.substring(0,22)}}...</span>
    → Bound: <span class="key">totalProjects</span>→<span class="val">KPI Card</span> | <span class="key">activeProjects</span>→<span class="val">KPI Card</span>
    <span class="key">budgetVariance</span>→<span class="val">Budget Card color</span> | <span class="key">overallHealthScore</span>→<span class="val">Donut Chart</span>
    <span class="key">highRiskProjects[]</span>→<span class="val">At-Risk table</span> | <span class="key">workloadDistribution[]</span>→<span class="val">Bar chart</span></pre>
  </div>

  <div class="g2">
    <!-- TASK TREND CHART -->
    <div class="card">
      <div class="card-header">
        <h3>☑ Task Completion Trend (30 Days)</h3>
        <span class="badge badge-success">● Live</span>
      </div>
      <div class="card-body">
        <canvas id="trendChart" height="180"></canvas>
      </div>
    </div>
  </div>

```

```

    </div>
  </div>

  <!-- AI HEALTH DONUT -->
  <div class="card">
    <div class="card-header">
      <h3>📊 Project Health Overview</h3>
      <button class="btn btn-ai btn-sm" @click="fetchHealth">🔍 Analyse</button>
    </div>
    <div class="card-body">
      <div class="donut-wrap" style="margin-bottom:18px">
        <canvas id="healthDonut" width="140" height="140"></canvas>
        <div class="donut-center">
          <div class="donut-val">{{dashboard.healthScore}}%</div>
          <div class="donut-lbl">Health</div>
        </div>
      </div>
      <div v-for="h in healthBars" :key="h.label" class="health-item">
        <div class="health-top">
          <span>{{h.label}}</span>
          <span :style="{color: h.val>=70?'var(--success)':h.val>=50?'var(--warning)':'var(--danger)'}">{{h.val}}%</span>
        </div>
        <div class="pb-wrap">
          <div class="pb" :class="h.val>=70?'green':h.val>=50?'yellow':'red'"
:style="{width:h.val+'%'}"></div>
        </div>
      </div>
    </div>
  </div>

  <div class="g2">
    <!-- AT-RISK PROJECTS -->
    <div class="card">
      <div class="card-header">
        <h3>📌 At-Risk Projects</h3>
        <span class="badge badge-danger">{{atRiskProjects.length}} Critical</span>
      </div>
      <div class="tbl-wrap">
        <table>

<thead><tr><th>Code</th><th>Project</th><th>Progress</th><th>Risk</th><th>Action</th></tr>
</thead>

        <tbody>
          <tr v-for="p in atRiskProjects" :key="p.id">
            <td><span class="badge badge-primary">{{p.code}}</span></td>
            <td><strong>{{p.name}}</strong></td>
            <td style="min-width:100px">
              <div class="pb-wrap"><div class="pb"
:class="p.progress>=70?'green':p.progress>=40?'yellow':'red'"
:style="{width:p.progress+'%'}"></div></div>
              <span style="font-size:11px;color:var(--muted)">{{p.progress}}%</span>
            </td>
            <td>
              <span class="risk-badge" :class="p.risk>80?'risk-crit':p.risk>60?'risk-high':p.risk>40?'risk-med':'risk-low'"
              >{{p.risk}}%
            </span>
            </td>
            <td><button class="btn btn-sm btn-danger"
@click="escalateProject(p)">🚨 Escalate</button></td>
          </tr>
        </tbody>
      </div>
    </div>
  </div>

```

```

        </tr>
      </tbody>
    </table>
  </div>
</div>

<!-- WORKLOAD DISTRIBUTION -->
<div class="card">
  <div class="card-header">
    <h3>📊 Team Workload Distribution</h3>
    <button class="btn btn-sm btn-outline" @click="fetchWorkload">🔄
Refresh</button>
  </div>
  <div class="card-body">
    <div v-for="w in workload" :key="w.name" style="margin-bottom:14px">
      <div style="display:flex;justify-content:space-between;margin-
bottom:5px;font-size:13px">
        <div style="display:flex;align-items:center;gap:8px">
          <div class="mini-av">{{w.name[0]}}</div>
          <span style="font-weight:600">{{w.name}}</span>
          <span class="badge badge-gray" style="font-
size:10px">{{w.role}}</span>
        </div>
        <span :style="{color:w.load>80?'var(--danger)':w.load>60?'var(--
warning)':'var(--success)',fontWeight:'700'}">
          {{w.load}}%
        </span>
      </div>
      <div class="pb-wrap">
        <div class="pb" :class="w.load>80?'red':w.load>60?'yellow':'green'"
:style="{width:w.load+'%'}"></div>
      </div>
      <div style="font-size:11px;color:var(--muted);margin-top:3px">{{w.tasks}}
active tasks • Burnout risk: {{w.burnout}}%</div>
    </div>
  </div>
</div>
</div>
</div>
</div>
</div>

<!-- ===== PROJECTS PAGE ===== -->
<div v-if="currentPage==='projects'">
  <div class="ph">
    <div class="ph-title">
      <h2>📁 Research Projects & Courses</h2>
      <p>Manage all departmental projects and research initiatives</p>
    </div>
    <div class="ph-actions">
      <button class="btn btn-primary" @click="openProjectModal">+ New
Project</button>
      <button class="btn btn-gold btn-sm" @click="navigate('reports')">📄
Export</button>
    </div>
  </div>
</div>

<!-- Filters
GET
/api/v1/projects?departmentId={{deptId}}&status={{filter.status}}&priority={{filter.priorit
y}}
-->
<div class="filter-bar">
  <div class="s-input">
    <span>🔍</span>

```

```

        <input v-model="projectFilter.search" placeholder="Search projects..."/>
    </div>
    <select class="fsel" v-model="projectFilter.status" @change="fetchProjects">
        <option value="">All Status</option>
        <option value="InProgress">🔄 Active</option>
        <option value="Delayed">🕒 Delayed</option>
        <option value="OnHold">🛑 On Hold</option>
        <option value="Completed">✅ Completed</option>
    </select>
    <select class="fsel" v-model="projectFilter.priority" @change="fetchProjects">
        <option value="">All Priority</option>
        <option value="Critical">🔴 Critical</option>
        <option value="High">🟠 High</option>
        <option value="Medium">🟡 Medium</option>
        <option value="Low">🟢 Low</option>
    </select>
    <select class="fsel" v-model="projectFilter.dept">
        <option value="">All Departments</option>
        <option value="se">Software Engineering</option>
        <option value="ai">Artificial Intelligence</option>
        <option value="cs">Computer Science</option>
    </select>
</div>

<!-- API LOG -->
<div class="api-log" style="margin-bottom:18px">
    <div class="api-log-title"><span class="api-dot"></span> Projects API</div>
    <pre><span class="method">GET</span> <span
class="url">/api/v1/projects?departmentId={{auth.user.deptId}}&status={{projectFilter.statu
s|'all'}}</span>
→ <span class="key">[].name</span><span class="val">Name column</span> | <span
class="key">[].status</span><span class="val">badge color</span>
    <span class="key">[].progressPercentage</span><span class="val">progress bar</span> |
<span class="key">[].aiHealthScore</span><span class="val">health icon</span>
    <span class="key">[].aiDelayRiskScore</span><span class="val">risk badge</span></pre>
</div>

<div class="card">
    <div class="tbl-wrap">
        <table>
            <thead>
                <tr>
                    <th>Code</th><th>Project / Course</th><th>Department</th>
                    <th>Status</th><th>Progress</th><th>Health</th><th>Risk</th>
                    <th>Due Date</th><th>Actions</th>
                </tr>
            </thead>
            <tbody>
                <tr v-for="p in filteredProjects" :key="p.id">
                    <td><span class="badge badge-primary">{{p.code}}</span></td>
                    <td>
                        <strong>{{p.name}}</strong>
                        <div style="font-size:11px;color:var(--muted)">PM:
{{p.manager}}</div>
                    </td>
                    <td><span class="badge badge-info">{{p.dept}}</span></td>
                    <td>
                        <span class="badge"
: class="p.status==='Active'? 'badge-
success':p.status==='Delayed'? 'badge-danger':p.status==='OnHold'? 'badge-info': 'badge-
gray'">

```

```

{{p.status==='Active'?'\u25a1':p.status==='Delayed'?'\u25a2':p.status==='OnHold'?'\u25a3':'\u25a4'}}
{{p.status}}
    </span>
  </td>
  <td style="min-width:120px">
    <div class="pb-wrap"><div class="pb"
: class="p.progress">=70?'green':p.progress>=40?'yellow': 'red'"
: style="{width:p.progress+'%'}"></div></div>
    <span style="font-size:11px">{{p.progress}}%</span>
  </td>
  <td>
    <span>{{p.health}>=70?''\u25a1':p.health>=50?''\u25a2':'\u25a3'}}</span>
    <span style="font-size:12px;font-weight:700;margin-
left:3px">{{p.health}}</span>
  </td>
  <td>
    <span class="risk-badge" :class="p.risk">70?'risk-
high':p.risk>40?'risk-med': 'risk-low'">{{p.risk}}%</span>
  </td>
  <td style="font-size:12px" :style="{color:new Date(p.dueDate)<new
Date()?'var(--danger)': 'var(--text)'}">{{p.dueDate}}</td>
  <td>
    <div style="display:flex;gap:5px">
      <button class="btn btn-sm btn-outline" @click="viewProject(p)"
title="View">\u25b6</button>
      <button class="btn btn-sm btn-gold" @click="editProject(p)"
title="Edit">\u25b6</button>
      <button class="btn btn-sm btn-ai" @click="viewAIHealth(p)"
title="AI Health">\u25b6</button>
    </div>
  </td>
</tr>
</tbody>
</table>
</div>
</div>
<!-- Pagination -->
<div style="display:flex;align-items:center;justify-content:space-
between;padding:12px 0">
  <span style="font-size:13px;color:var(--muted)">Showing
{{filteredProjects.length}} of {{projects.length}} projects</span>
  <div style="display:flex;gap:6px">
    <button class="btn btn-ghost btn-sm">\u25c4 Prev</button>
    <button class="btn btn-primary btn-sm"
style="width:32px;padding:0">1</button>
    <button class="btn btn-ghost btn-sm" style="width:32px;padding:0">2</button>
    <button class="btn btn-ghost btn-sm">Next \u25c6</button>
  </div>
</div>
</div>
</div>

<!-- ===== TASKS / KANBAN PAGE ===== -->
<div v-if="currentPage==='tasks'">
  <div class="ph">
    <div class="ph-title">
      <h2>\u25aa Task Management \u2013 Kanban Board</h2>
      <p>PROJ-001: AI Research Initiative \u2022 Drag tasks to update status</p>
    </div>
    <div class="ph-actions">
      <button class="btn btn-primary" @click="openTaskModal">+ New Task</button>
      <button class="btn btn-ai btn-sm" @click="aiAssignAll">\u25aa AI Auto-
Assign</button>
    </div>
  </div>
</div>

```

```

</div>
</div>

<!-- VIEW TOGGLE -->
<div class="tab-bar">
  <button class="tab" :class="{active:taskView==='kanban'}"
@click="taskView='kanban'"> Kanban</button>
  <button class="tab" :class="{active:taskView==='list'}"
@click="taskView='list'"> List</button>
  <button class="tab" :class="{active:taskView==='gantt'}"
@click="taskView='gantt'"> Gantt</button>
</div>

<!-- API LOG -->
<div class="api-log" style="margin-bottom:18px">
  <div class="api-log-title"><span class="api-dot"></span> Tasks API</div>
  <pre><span class="method">GET</span> <span class="url">/api/v1/tasks/by-
project/{{selectedProject.id}}</span>
→ <span class="key">[].status</span>→<span class="val">Kanban column</span> | <span
class="key">[].priority</span>→<span class="val">card border color</span>
  <span class="key">[].assignedToUserId</span>→<span class="val">avatar</span> | <span
class="key">[].aiDelayProbability</span>→<span class="val">risk badge</span>
  <span class="key">[].isEscalated</span>→<span class="val"> icon</span>

<span class="method">PATCH</span> <span
class="url">/api/v1/tasks/{{ '{' }}{taskId}}/progress</span>
{ <span class="key">"progressPercentage"</span>: <span class="val">slider.value</span>,
<span class="key">"notes"</span>: <span class="val">textarea.value</span> }</pre>
</div>

<!-- KANBAN BOARD -->
<div v-if="taskView==='kanban'" class="kanban">
  <div v-for="col in kanbanCols" :key="col.status" class="k-col">
    <div class="k-col-hdr">
      <div class="k-col-title">
        <span>{{col.icon}}</span>
        <span>{{col.label}}</span>
      </div>
      <span class="k-cnt">{{tasksInCol(col.status).length}}</span>
    </div>
    <div v-for="t in tasksInCol(col.status)" :key="t.id"
      class="task-card" :class="t.priority.toLowerCase()"
      @click="openTaskDetail(t)">
      <div class="tc-hdr">
        <span class="tc-title">{{t.title}}</span>
        <span class="badge"
: class="t.priority==='Critical'? 'badge-
danger':t.priority==='High'? 'badge-warning':t.priority==='Medium'? 'badge-primary': 'badge-
success'"
style="font-size:10px">{{t.priority}}</span>
      </div>
      <div class="pb-wrap"><div class="pb blue"
:style="{width:t.progress+'%'}"></div></div>
      <div class="tc-meta">
        <div class="tc-assignee">
          <div class="mini-av">{{(t.assignee||'')[0]}}</div>
          <span>{{t.assignee||'Unassigned'}}</span>
        </div>
        <span class="tc-due" :class="{late:isLate(t.dueDate)}">
{{t.dueDate}}</span>
      </div>
      <div style="margin-top:8px;display:flex;align-items:center;gap:6px;flex-
wrap:wrap">

```

```

        <span class="risk-badge" :class="t.aiRisk>70?'risk-high':t.aiRisk>40?'risk-med': 'risk-low'" style="font-size:10px">
            Risk: {{t.aiRisk}}%
        </span>
        <span v-if="t.escalated" class="badge badge-danger" style="font-size

```

Claude Sonnet 4.6

Let me read the uploaded file first, then continue the complete HTML wireframe.

Now I have the complete picture from the SRS and code. Here is the **full, complete, self-contained HTML wireframe** — save as `pmwds-university.html` and open in any browser:

```

html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"/>
<meta name="viewport" content="width=device-width,initial-scale=1.0"/>
<title> PMWDS 2.0 – University Portal</title>
<script src="https://unpkg.com/vue@3/dist/vue.global.js"></script>
<script src="https://cdn.jsdelivr.net/npm/axios/dist/axios.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<style>
:root{
  --primary:#1B3A6B;--pl:#2a52a0;--pd:#122848;
  --gold:#C8A951;--gl:#e8c97a;
  --ok:#28a745;--warn:#ffc107;--err:#dc3545;--inf:#17a2b8;
  --ai:#6f42c1;--ail:#9b6ee8;
  --bg:#f0f2f7;--card:#fff;--bdr:#dee2e6;
  --tx:#212529;--mx:#6c757d;
  --sw:260px;--hh:64px;--r:12px;
  --sh:0 4px 16px rgba(27,58,107,.12);
  --shh:0 8px 24px rgba(27,58,107,.2);
  --tr:all .25s ease;
}
*,*::before,*::after{box-sizing:border-box;margin:0;padding:0}
body{font-family:'Segoe UI',system-ui,sans-serif;background:var(--bg);color:var(--tx);font-size:14px;line-height:1.5}
button{cursor:pointer;font-family:inherit}
input,select,textarea{font-family:inherit;font-size:14px;outline:none}
::-webkit-scrollbar{width:5px;height:5px}
::-webkit-scrollbar-thumb{background:#ccc;border-radius:3px}

/* LOGIN */
.lp{min-height:100vh;display:flex;align-items:center;justify-content:center;
  background:linear-gradient(135deg,var(--pd),var(--primary),var(--pl));position:relative;overflow:hidden}
.lp::before{content:'';position:absolute;top:-50%;right:-20%;width:600px;height:600px;
  background:rgba(200,169,81,.08);border-radius:50%}
.lc{background:#fff;border-radius:20px;padding:48px 44px;width:440px;
  box-shadow:0 24px 64px rgba(0,0,0,.2);position:relative;z-index:1}
.ll{text-align:center;margin-bottom:30px}
.ll .ico{width:70px;height:70px;background:linear-gradient(135deg,var(--primary),var(--pl));
  border-radius:18px;display:inline-flex;align-items:center;justify-content:center;font-size:30px;
  margin-bottom:12px;box-shadow:0 8px 20px rgba(27,58,107,.3)}
.ll h1{font-size:19px;font-weight:700;color:var(--primary)}
.ll p{color:var(--mx);font-size:12px;margin-top:3px}
.fg{margin-bottom:16px}
.fg label{display:block;font-weight:600;margin-bottom:6px;font-size:13px}
.fc{width:100%;padding:11px 14px;border:2px solid var(--bdr);border-radius:10px;
  font-size:14px;transition:var(--tr);background:#fafafa}

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.fc:focus{border-color:var(--primary);background:#fff;box-shadow:0 0 0 3px
rgba(27,58,107,.1)}
.fc.err{border-color:var(--err)}
.rem{display:flex;align-items:center;gap:8px;margin-bottom:18px;font-size:13px;color:var(--
mx)}
.rem input{width:15px;height:15px;accent-color:var(--primary)}

/* BUTTONS */
.btn{display:inline-flex;align-items:center;justify-content:center;gap:6px;
padding:10px 20px;border:none;border-radius:10px;font-weight:600;font-
size:13px;transition:var(--tr);cursor:pointer}
.btn-p{background:linear-gradient(135deg,var(--primary),var(--
pl));color:#fff;width:100%;padding:13px}
.btn-p:hover{transform:translateY(-1px);box-shadow:0 6px 16px rgba(27,58,107,.35)}
.btn-o{background:transparent;border:2px solid var(--primary);color:var(--primary)}
.btn-o:hover{background:var(--primary);color:#fff}
.btn-s{padding:6px 12px;font-size:12px;border-radius:8px}
.btn-ok{background:var(--ok);color:#fff}.btn-ok:hover{background:#218838}
.btn-err{background:var(--err);color:#fff}.btn-err:hover{background:#c82333}
.btn-warn{background:var(--warn);color:#000}
.btn-inf{background:var(--inf);color:#fff}
.btn-gold{background:linear-gradient(135deg,var(--gold),var(--gl));color:var(--pd);font-
weight:700}
.btn-gold:hover{transform:translateY(-1px);box-shadow:0 4px 12px rgba(200,169,81,.4)}
.btn-ai{background:linear-gradient(135deg,var(--ai),var(--ail));color:#fff}
.btn-ai:hover{transform:translateY(-1px);box-shadow:0 4px 12px rgba(111,66,193,.3)}
.btn-gh{background:transparent;color:var(--mx);border:1.5px solid var(--bdr)}
.btn-gh:hover{background:var(--bg)}

/* ALERTS */
.alert{padding:11px 14px;border-radius:10px;margin-bottom:14px;font-
size:13px;display:flex;align-items:center;gap:8px}
.a-err{background:#fde8ea;border:1px solid #f5c6cb;color:#721c24}
.a-ok{background:#d4edda;border:1px solid #c3e6cb;color:#155724}
.a-warn{background:#fff3cd;border:1px solid #ffeeba;color:#856404}
.a-inf{background:#d1ecf1;border:1px solid #bee5eb;color:#0c5460}
.div{text-align:center;margin:14px 0;color:var(--mx);font-size:12px;position:relative}
.div::before,.div::after{content:'';position:absolute;top:50%;width:42%;height:1px;backgrou
nd:var(--bdr)}
.div::before{left:0}.div::after{right:0}

/* SHELL */
.app{display:flex;height:100vh;overflow:hidden}

/* SIDEBAR */
.sb{width:var(--sw);background:linear-gradient(180deg,var(--pd),var(--primary));
height:100vh;display:flex;flex-direction:column;flex-shrink:0;transition:var(--tr);z-
index:100}
.sb.col{width:68px}
.sb-brand{padding:16px 14px;border-bottom:1px solid
rgba(255,255,255,.1);display:flex;align-items:center;gap:10px}
.sb-ico{width:40px;height:40px;background:rgba(200,169,81,.2);border:2px solid var(--gold);
border-radius:10px;display:flex;align-items:center;justify-content:center;font-
size:19px;flex-shrink:0}
.sb-txt{overflow:hidden;transition:var(--tr)}
.sb-txt h2{color:#fff;font-size:14px;font-weight:700;white-space:nowrap}
.sb-txt p{color:var(--gold);font-size:11px;white-space:nowrap}
.sb.col .sb-txt{opacity:0;width:0}
.sb-nav{flex:1;padding:10px 8px;overflow-y:auto}
.sb-sec{color:rgba(255,255,255,.4);font-size:10px;font-weight:700;letter-spacing:1.2px;
text-transform:uppercase;padding:10px 8px 4px;white-
space:nowrap;overflow:hidden;transition:var(--tr)}
.sb.col .sb-sec{opacity:0}
.ni{display:flex;align-items:center;gap:10px;padding:10px 11px;border-radius:10px;
color:rgba(255,255,255,.75);cursor:pointer;transition:var(--tr);margin-bottom:2px;white-
space:nowrap}

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.ni:hover{background:rgba(255,255,255,.1);color:#fff}
.ni.act{background:var(--gold);color:var(--pd);font-weight:600;box-shadow:0 4px 12px
rgba(200,169,81,.3)}
.n-ico{font-size:16px;flex-shrink:0;width:21px;text-align:center}
.n-lbl{font-size:13px;overflow:hidden;transition:var(--tr)}
.sb.col .n-lbl{opacity:0;width:0}
.nb{margin-left:auto;background:var(--err);color:#fff;font-size:10px;font-
weight:700;padding:2px 6px;border-radius:10px;flex-shrink:0}
.sb.col .nb{display:none}
.sb-foot{padding:10px 8px;border-top:1px solid rgba(255,255,255,.1)}

/* MAIN */
.main{flex:1;display:flex;flex-direction:column;overflow:hidden}

/* HEADER */
.hdr{height:var(--hh);background:#fff;border-bottom:1px solid var(--bdr);
display:flex;align-items:center;padding:0 20px;gap:12px;flex-shrink:0;
box-shadow:0 2px 8px rgba(0,0,0,.06);z-index:50}
.h-tog{background:none;border:none;font-size:21px;color:var(--mx);padding:6px;border-
radius:8px;cursor:pointer}
.h-tog:hover{background:var(--bg)}
.h-br{display:flex;align-items:center;gap:7px;color:var(--mx);font-size:13px}
.h-br .cur{color:var(--primary);font-weight:600}
.h-sp{flex:1}
.h-srch{display:flex;align-items:center;background:var(--bg);border:1.5px solid var(--bdr);
border-radius:10px;padding:7px 12px;gap:7px;width:220px}
.h-srch input{background:none;border:none;flex:1;font-size:13px;color:var(--tx)}
.h-acts{display:flex;align-items:center;gap:7px}
.ib{width:36px;height:36px;border:none;border-radius:10px;background:var(--bg);
color:var(--mx);font-size:16px;display:flex;align-items:center;justify-content:center;
cursor:pointer;position:relative;transition:var(--tr)}
.ib:hover{background:var(--bdr);color:var(--tx)}
.ib .bdg{position:absolute;top:4px;right:4px;background:var(--err);color:#fff;
font-size:9px;font-weight:700;width:15px;height:15px;border-radius:50%;
display:flex;align-items:center;justify-content:center}
.um{display:flex;align-items:center;gap:8px;padding:5px 10px;border-
radius:10px;cursor:pointer;transition:var(--tr)}
.um:hover{background:var(--bg)}
.uav{width:35px;height:35px;border-radius:50%;background:linear-gradient(135deg,var(--
primary),var(--pl));
color:#fff;font-weight:700;font-size:13px;display:flex;align-items:center;justify-
content:center;flex-shrink:0}
.ui h4{font-size:13px;font-weight:600}
.ui p{font-size:11px;color:var(--mx)}

/* PAGE */
.pg{flex:1;overflow-y:auto;padding:20px}
.ph{display:flex;align-items:center;justify-content:space-between;margin-bottom:20px;flex-
wrap:wrap;gap:12px}
.ph-t h2{font-size:20px;font-weight:700;color:var(--primary)}
.ph-t p{color:var(--mx);font-size:13px;margin-top:2px}
.ph-a{display:flex;gap:8px;flex-wrap:wrap}

/* KPI */
.kgrid{display:grid;grid-template-columns:repeat(auto-
fill,minmax(185px,1fr));gap:14px;margin-bottom:20px}
.kpi{background:var(--card);border-radius:var(--r);padding:17px;box-shadow:var(--sh);
border-left:4px solid transparent;transition:var(--tr);cursor:pointer}
.kpi:hover{transform:translateY(-2px);box-shadow:var(--shh)}
.kpi.bl{border-color:var(--primary)}.kpi.gr{border-color:var(--ok)}
.kpi.rd{border-color:var(--err)}.kpi.gd{border-color:var(--gold)}
.kpi.pu{border-color:var(--ai)}.kpi.or{border-color:#fd7e14}
.kpi-top{display:flex;align-items:center;justify-content:space-between;margin-bottom:10px}
.kico{width:42px;height:42px;border-radius:10px;display:flex;align-items:center;justify-
content:center;font-size:19px}
.kico.bl{background:rgba(27,58,107,.1)}.kico.gr{background:rgba(40,167,69,.1)}

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.kico.rd{background:rgba(220,53,69,.1)}.kico.gd{background:rgba(200,169,81,.15)}
.kico.pu{background:rgba(111,66,193,.1)}.kico.or{background:rgba(253,126,20,.1)}
.ktr{font-size:11px;font-weight:600;padding:2px 7px;border-radius:20px}
.ktr.up{background:#d4edda;color:#155724}.ktr.dn{background:#fde8ea;color:#721c24}
.kval{font-size:26px;font-weight:800;line-height:1}
.klbl{font-size:12px;color:var(--mx);margin-top:3px;font-weight:500}
.ksub{font-size:11px;color:var(--mx);margin-top:6px}

/* CARD */
.card{background:var(--card);border-radius:var(--r);box-shadow:var(--sh);overflow:hidden;margin-bottom:18px}
.ch{padding:14px 18px;border-bottom:1px solid var(--bdr);display:flex;align-items:center;justify-content:space-between;gap:10px}
.ch h3{font-size:14px;font-weight:700;color:var(--primary)}
.cb{padding:16px}

/* GRIDS */
.g2{display:grid;grid-template-columns:1fr 1fr;gap:16px}
.g3{display:grid;grid-template-columns:repeat(3,1fr);gap:16px}
.g4{display:grid;grid-template-columns:repeat(4,1fr);gap:16px}
.ga{display:grid;grid-template-columns:repeat(auto-fill,minmax(280px,1fr));gap:16px}
.s2{grid-column:span 2}.s3{grid-column:span 3}

/* TABLE */
.tw{overflow-x:auto}
table{width:100%;border-collapse:collapse;font-size:13px}
thead tr{background:#f8f9ff}
th{padding:10px 14px;text-align:left;font-weight:700;color:var(--primary);font-size:11px;letter-spacing:.5px;text-transform:uppercase;white-space:nowrap}
td{padding:10px 14px;border-bottom:1px solid #f0f0f0;vertical-align:middle}
tr:hover td{background:#f8f9ff}
tr:last-child td{border-bottom:none}

/* BADGES */
.bdg{display:inline-flex;align-items:center;gap:3px;padding:3px 9px;border-radius:20px;font-size:11px;font-weight:600;white-space:nowrap}
.b-p{background:rgba(27,58,107,.1);color:var(--primary)}
.b-ok{background:rgba(40,167,69,.12);color:#1a7431}
.b-warn{background:rgba(255,193,7,.15);color:#856404}
.b-err{background:rgba(220,53,69,.12);color:#721c24}
.b-inf{background:rgba(23,162,184,.12);color:#0c5460}
.b-ai{background:rgba(111,66,193,.12);color:#5a2d8a}
.b-gd{background:rgba(200,169,81,.15);color:#7a5800}
.b-gr{background:#eee;color:#555}

/* PROGRESS */
.pw{background:#eee;border-radius:10px;height:7px;overflow:hidden}
.pb{height:100%;border-radius:10px;transition:width .6s ease}
.pb.g{background:linear-gradient(90deg,var(--ok),#5cb85c)}
.pb.y{background:linear-gradient(90deg,var(--warn),#ffd740)}
.pb.r{background:linear-gradient(90deg,var(--err),#ff6b6b)}
.pb.b{background:linear-gradient(90deg,var(--primary),var(--pl))}
.pb.p{background:linear-gradient(90deg,var(--ai),var(--ail))}
.pb.gd{background:linear-gradient(90deg,var(--gold),var(--gl))}

/* FILTERS */
.fb{display:flex;align-items:center;gap:8px;margin-bottom:16px;flex-wrap:wrap}
.fsel{padding:8px 28px 8px 10px;border:1.5px solid var(--bdr);border-radius:10px;background:#fff;appearance:none;cursor:pointer;font-size:13px;color:var(--tx);min-width:120px}
.fsel:focus{border-color:var(--primary)}
.si{display:flex;align-items:center;background:#fff;border:1.5px solid var(--bdr);border-radius:10px;padding:8px 10px;gap:6px;flex:1;min-width:160px}
.si input{background:none;border:none;flex:1;color:var(--tx)}

/* KANBAN */

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.kb{display:grid;grid-template-columns:repeat(4,1fr);gap:14px;min-height:320px}
.kc{background:#f8f9ff;border-radius:var(--r);padding:10px;min-height:220px}
.kch{display:flex;align-items:center;justify-content:space-between;margin-bottom:10px}
.kct{display:flex;align-items:center;gap:6px;font-weight:700;font-size:13px}
.kcn{background:rgba(0,0,0,.08);color:#555;font-size:11px;font-weight:700;padding:2px
6px;border-radius:10px}
.tc{background:#fff;border-radius:10px;padding:12px;margin-bottom:8px;
  box-shadow:0 2px 8px rgba(0,0,0,.06);border-left:3px solid
transparent;cursor:pointer;transition:var(--tr)}
.tc:hover{transform:translateY(-2px);box-shadow:0 4px 14px rgba(0,0,0,.1)}
.tc.crit{border-color:var(--err)}.tc.high{border-color:var(--warn)}
.tc.med{border-color:var(--primary)}.tc.low{border-color:var(--ok)}
.tch{display:flex;align-items:flex-start;justify-content:space-between;margin-bottom:7px}
.tct{font-weight:600;font-size:13px;flex:1}
.tcm{display:flex;align-items:center;justify-content:space-between;margin-top:9px}
.tca{display:flex;align-items:center;gap:5px;font-size:11px;color:var(--mx)}
.av{width:22px;height:22px;border-radius:50%;background:linear-gradient(135deg,var(--
primary),var(--pl));
  color:#fff;font-size:9px;font-weight:700;display:flex;align-items:center;justify-
content:center}
.tcd{font-size:11px;color:var(--mx)}.tcd.late{color:var(--err);font-weight:600}

/* CHAT */
.chat{height:300px;overflow-y:auto;padding:12px;background:#f8f9ff;border-
radius:10px;margin-bottom:10px}
.msg{margin-bottom:10px;display:flex;flex-direction:column}
.msg.ai{align-items:flex-start}.msg.u{align-items:flex-end}
.mb{max-width:80%;padding:9px 13px;border-radius:12px;font-size:13px;line-height:1.5}
.msg.ai .mb{background:#fff;border:1px solid var(--bdr);color:var(--tx)}
.msg.u .mb{background:var(--primary);color:#fff}
.mt{font-size:10px;color:var(--mx);margin-top:2px}
.cir{display:flex;gap:8px}
.ciw{flex:1;display:flex;align-items:center;background:#fff;border:1.5px solid var(--
bdr);border-radius:10px;padding:8px 12px;gap:7px}
.ciw input{background:none;border:none;flex:1;font-size:13px}
.chips{display:flex;gap:6px;flex-wrap:wrap;margin-top:9px}
.chip{padding:4px 10px;background:rgba(111,66,193,.1);color:var(--ai);
  border-radius:20px;font-size:12px;font-weight:600;cursor:pointer;border:1px solid
rgba(111,66,193,.2);transition:var(--tr)}
.chip:hover{background:var(--ai);color:#fff}

/* DONUT */
.dw{position:relative;width:130px;height:130px;margin:0 auto}
.dc{position:absolute;top:50%;left:50%;transform:translate(-50%,-50%);text-
align:center;pointer-events:none}
.dv{font-size:21px;font-weight:800;color:var(--primary)}
.dl{font-size:11px;color:var(--mx)}

/* HEALTH BARS */
.hi{margin-bottom:13px}
.ht{display:flex;justify-content:space-between;margin-bottom:4px;font-size:13px}
.ht span:first-child{font-weight:600}
.ht span:last-child{font-weight:700}

/* NOTIFICATIONS */
.nl{max-height:400px;overflow-y:auto}
.nitem{padding:12px 14px;border-bottom:1px solid #f0f0f0;cursor:pointer;
  transition:var(--tr);display:flex;gap:10px;align-items:flex-start}
.nitem:hover{background:#f8f9ff}
.nitem.unr{background:#f0f4ff}
.nitem:last-child{border-bottom:none}
.nico{width:37px;height:37px;border-radius:10px;display:flex;align-items:center;justify-
content:center;font-size:16px;flex-shrink:0}
.nb2{flex:1}
.nt{font-weight:600;font-size:13px;margin-bottom:2px}
.nm{font-size:12px;color:var(--mx);margin-bottom:3px}

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.ntime{font-size:11px;color:var(--mx)}
.ndot{width:8px;height:8px;border-radius:50%;background:var(--primary);flex-shrink:0;margin-top:5px}

/* MODAL */
.ov{position:fixed;inset:0;background:rgba(0,0,0,.5);z-index:999;
  display:flex;align-items:center;justify-content:center;padding:20px}
.modal{background:#fff;border-radius:16px;width:100%;max-width:560px;
  box-shadow:0 24px 64px rgba(0,0,0,.2);overflow:hidden;max-height:90vh;overflow-y:auto}
.mh{padding:16px 20px;border-bottom:1px solid var(--bdr);display:flex;align-items:center;justify-content:space-between}
.mh h3{font-size:15px;font-weight:700;color:var(--primary)}
.mx-btn{background:none;border:none;font-size:20px;color:var(--mx);cursor:pointer;
  width:32px;height:32px;border-radius:8px;display:flex;align-items:center;justify-content:center}
.mx-btn:hover{background:var(--bg)}
.mb2{padding:20px}
.mf{padding:14px 20px;border-top:1px solid var(--bdr);display:flex;gap:8px;justify-content:flex-end}

/* FORM */
.fr{display:grid;grid-template-columns:1fr 1fr;gap:12px}
.fc2{display:flex;flex-direction:column;gap:4px}
.fc2 label{font-size:12px;font-weight:600;color:var(--tx)}
.fc2 input,.fc2 select,.fc2 textarea{
  padding:8px 11px;border:1.5px solid var(--bdr);border-radius:9px;font-size:13px;transition:var(--tr)}
.fc2 input:focus,.fc2 select:focus,.fc2 textarea:focus{
  border-color:var(--primary);box-shadow:0 0 3px rgba(27,58,107,.1)}
.fc2 textarea{resize:vertical;min-height:75px}
.ff{grid-column:span 2}

/* API LOG */
.alog{background:#0d1117;border-radius:10px;padding:12px;margin-top:12px;max-height:170px;overflow-y:auto}
.alog pre{color:#58a6ff;font-size:11px;font-family:'Courier New',monospace;white-space:pre-wrap}
.alog .m{color:#ffa657}.alog .u{color:#79c0ff}
.alog .k{color:#ff7b72}.alog .v{color:#a5d6ff}
.alt{font-size:11px;color:#8b949e;margin-bottom:6px;display:flex;align-items:center;gap:5px}
.adot{width:8px;height:8px;border-radius:50%;background:var(--ok);animation:pulse 1.5s infinite}
@keyframes pulse{0%,100%{opacity:1}50%{opacity:.4}}

/* SPINNER */
.spin{width:34px;height:34px;border:3px solid rgba(27,58,107,.15);
  border-top-color:var(--primary);border-radius:50%;animation:sp .8s linear infinite;margin:0 auto}
@keyframes sp{to{transform:rotate(360deg)}}

/* RISK */
.rb{display:inline-flex;align-items:center;gap:4px;padding:3px 9px;border-radius:20px;font-size:12px;font-weight:700}
.rl{background:#d4edda;color:#155724}.rm{background:#fff3cd;color:#856404}
.rh{background:#fde8e8;color:#721c24}.rc{background:#dc3545;color:#fff}

/* TABS */
.tbr{display:flex;gap:4px;background:var(--bg);padding:4px;border-radius:10px;margin-bottom:16px;width:fit-content}
.tab{padding:7px 16px;border-radius:8px;font-size:13px;font-weight:600;cursor:pointer;
  color:var(--mx);transition:var(--tr);border:none;background:transparent}
.tab.act{background:#fff;color:var(--primary);box-shadow:0 2px 8px rgba(0,0,0,.1)}

/* GANTT */
.gantt{overflow-x:auto}

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.gt{min-width:700px}
.gr2{display:flex;align-items:center;padding:8px 0;border-bottom:1px solid #f0f0f0}
.gl2{width:200px;flex-shrink:0;font-size:13px;font-weight:600;padding-right:10px}
.gbar-wrap{flex:1;background:#f0f0f0;height:20px;border-radius:4px;position:relative;overflow:hidden}
.gbar{height:100%;border-radius:4px;position:absolute;display:flex;align-items:center;padding:0 8px;font-size:11px;color:#fff;font-weight:600}

/* AI SCORE RING */
.ring-wrap{display:flex;gap:16px;flex-wrap:wrap;justify-content:center;padding:8px 0}
.ring-item{text-align:center}
.ring{width:70px;height:70px;border-radius:50%;display:flex;align-items:center;justify-content:center;font-size:16px;font-weight:800;color:#fff;margin:0 auto 6px}
.ring-lbl{font-size:12px;color:var(--mx);font-weight:600}

/* REPORTS */
.rep-grid{display:grid;grid-template-columns:1fr 2fr;gap:18px}
.rep-type{display:flex;flex-direction:column;gap:6px}
.rep-opt{padding:11px 14px;border-radius:10px;border:2px solid var(--bdr);cursor:pointer;transition:var(--tr);font-size:13px;font-weight:600;display:flex;align-items:center;gap:8px}
.rep-opt:hover{border-color:var(--primary);background:#f0f4ff}
.rep-opt.sel{border-color:var(--primary);background:#f0f4ff;color:var(--primary)}

/* SKILL CHIPS */
.skill-chip{display:inline-flex;align-items:center;gap:4px;padding:3px 9px;background:rgba(27,58,107,.08);color:var(--primary);border-radius:20px;font-size:11px;font-weight:600;margin:2px}

/* ESCALATION BANNER */
.esc-ban{background:linear-gradient(135deg,#fde8ea,#fff3cd);border:1px solid #f5c6cb;border-radius:10px;padding:12px 16px;margin-bottom:16px;display:flex;align-items:center;gap:10px}

/* USER CARD */
.ucard{background:var(--card);border-radius:var(--r);padding:18px;box-shadow:var(--sh);transition:var(--tr);cursor:pointer}
.ucard:hover{transform:translateY(-2px);box-shadow:var(--shh)}
.ucav{width:56px;height:56px;border-radius:50%;background:linear-gradient(135deg,var(--primary),var(--pl));color:#fff;font-weight:700;font-size:22px;display:flex;align-items:center;justify-content:center;margin:0 auto 12px}
.uc-name{text-align:center;font-weight:700;font-size:14px;margin-bottom:3px}
.uc-role{text-align:center;font-size:12px;color:var(--mx);margin-bottom:12px}
.uc-stats{display:flex;gap:12px;justify-content:center;margin-bottom:12px}
.uc-stat{text-align:center}
.uc-sv{font-size:18px;font-weight:800;color:var(--primary)}
.uc-sl{font-size:11px;color:var(--mx)}

@media(max-width:900px){
  .g2,.g3,.g4,.rep-grid{grid-template-columns:1fr}
  .kb{grid-template-columns:repeat(2,1fr)}
  .fr{grid-template-columns:1fr}.ff{grid-column:span 1}
}
@media(max-width:600px){
  .kb{grid-template-columns:1fr}
  .h-srch{display:none}
  .lc{width:95vw;padding:28px 18px}
  .kgrid{grid-template-columns:repeat(2,1fr)}
}
</style>
</head>
<body>
<div id="app">

```

```

<!-- LOGIN PAGE -->
<div v-if="!auth.isLoggedInIn" class="lp">
  <div class="lc">
    <div class="ll">
      <div class="ico"></div>
      <h1>PMWDS 2.0 University Portal</h1>
      <p>AI-Powered Project Monitoring & Work Distribution System</p>
    </div>

    <div v-if="login.error" class="alert a-err">{{login.error}}</div>
    <div v-if="login.success" class="alert a-ok">{{login.success}}</div>

    <!-- POST /api/v1/auth/login -->
    <div class="fg">
      <label>University Email</label>
      <input class="fc" :class="{err:login.eErr}" type="email"
        v-model="loginForm.email" placeholder="you@university.edu"
        @blur="valEmail"/>
      <small v-if="login.eErr" style="color:var(--err);font-size:11px">Enter a valid
email</small>
    </div>
    <div class="fg">
      <label>Password</label>
      <input class="fc" :class="{err:login.pErr}" type="password"
        v-model="loginForm.password" placeholder="....." @keyup.enter="doLogin"/>
      <small v-if="login.pErr" style="color:var(--err);font-size:11px">Password
required</small>
    </div>
    <div class="rem">
      <input type="checkbox" v-model="loginForm.remember" id="rem"/>
      <label for="rem">Remember me for 8 hours</label>
    </div>

    <button class="btn btn-p" @click="doLogin" :disabled="login.loading">
      <span>{{login.loading?'':'🔄'}}</span>
      {{login.loading?'Authenticating...':'Sign In'}}
    </button>
    <div class="div">or continue with</div>
    <div style="display:flex;gap:8px">
      <button class="btn btn-o" style="flex:1;font-size:12px" @click="ssoLogin">
University SSO</button>
      <button class="btn btn-gh" style="flex:1;font-size:12px">LDAP Login</button>
    </div>

    <!-- Live API call preview -->
    <div class="alog" style="margin-top:14px">
      <div class="alt"><span class="adot"></span>Live API Call</div>
      <pre><span class="m">POST</span> <span class="u">/api/v1/auth/login</span>
{
  <span class="k">"email"</span>: <span
class="v">"{{loginForm.email||'user@uni.edu'}}"</span>,
  <span class="k">"password"</span>: <span class="v">"....."</span>
}
→ Binds: <span class="k">token</span>→<span class="v">authStore.token</span> | <span
class="k">roles[]</span>→<span class="v">sidebar visibility</span>
      <span class="k">fullName</span>→<span class="v">header display</span> | <span
class="k">expiresAt</span>→<span class="v">auto-refresh timer</span></pre>
    </div>
  </div>
</div>

```

```

<!-- APP SHELL -->
<div v-else class="app">

  <!-- SIDEBAR -->
  <nav class="sb" :class="{col:sb.col}">
    <div class="sb-brand">
      <div class="sb-ico"></div>
      <div class="sb-txt"><h2>PMWDS 2.0</h2><p>University Edition</p></div>
    </div>
    <div class="sb-nav">
      <div class="sb-sec">Main Menu</div>
      <div v-for="item in nav" :key="item.id" class="ni" :class="{act:pg===item.id}"
@click="go(item.id)">
        <span class="n-ico">{{item.icon}}</span>
        <span class="n-lbl">{{item.label}}</span>
        <span v-if="item.badge" class="nb">{{item.badge}}</span>
      </div>
      <div class="sb-sec" style="margin-top:8px">AI Tools</div>
      <div v-for="item in aiNav" :key="item.id" class="ni" :class="{act:pg===item.id}"
@click="go(item.id)">
        <span class="n-ico">{{item.icon}}</span>
        <span class="n-lbl">{{item.label}}</span>
      </div>
    </div>
    <div class="sb-foot">
      <div class="ni" @click="logout">
        <span class="n-ico"></span>
        <span class="n-lbl">Logout</span>
      </div>
    </div>
  </nav>

  <!-- MAIN -->
  <div class="main">

    <!-- HEADER -->
    <header class="hdr">
      <button class="h-tog" @click="sb.col=!sb.col"></button>
      <div class="h-br">
        <span></span> University</span>
        <span style="color:var(--bdr)"></span>
        <span class="cur">{{pgLabel}}</span>
      </div>
      <div class="h-sp"></div>
      <div class="h-srch">
        <span></span>
        <input v-model="hdr.q" placeholder="Search projects, tasks..."/>
      </div>
      <div class="h-acts">
        <button class="ib" @click="go('notifications')" title="Notifications">
          <span class="bdg" v-if="notifCount>0">{{notifCount}}</span>
        </button>
        <button class="ib" @click="go('ai')" title="AI Assistant"></button>
        <button class="ib" title="Settings"></button>
        <div class="um">
          <div class="uav">{{auth.user.ini}}</div>
          <div class="ui">
            <h4>{{auth.user.fullName}}</h4>
            <p>{{auth.user.role}}</p>
          </div>
        </div>
      </div>
    </div>
  </div>

```

```

    </div>
</header>

<!-- PAGE CONTENT -->
<div class="pg">

    <!-- =====
    ? DASHBOARD
    ===== -->

    <div v-if="pg==='dashboard'">
        <div class="ph">
            <div class="ph-t">
                <h2>? Good Morning, {{auth.user.fullName.split(' ')[0]}}</h2>
                <p>{{today}} · AI Health Score: <strong style="color:var(--
ok)">{{dash.health}}%</strong></p>
            </div>
            <div class="ph-a">
                <button class="btn btn-ai btn-s" @click="refreshDash">? AI Refresh</button>
                <button class="btn btn-gold btn-s" @click="go('reports')">? Reports</button>
                <button class="btn btn-p btn-s" @click="go('projects')">+ New
Project</button>
            </div>
        </div>

        <!-- ESCALATION BANNER -->
        <div class="esc-ban" v-if="dash.escalated>0">
            ? <strong>{{dash.escalated}} task(s) escalated</strong> – require immediate
attention.
            <button class="btn btn-s btn-err" style="margin-left:auto"
@click="go('tasks')">View Tasks</button>
        </div>

        <!-- KPI CARDS – GET /api/v1/projects/dashboard -->
        <div class="kgrid">
            <div v-for="k in kpis" :key="k.label" class="kpi" :class="k.col"
@click="go(k.nav)">
                <div class="kpi-top">
                    <div class="kico" :class="k.col">{{k.icon}}</div>
                    <span class="ktr" :class="k.up?'up':'dn'">{{k.up?'▲':'▼'}}
{{Math.abs(k.trend)}}%</span>
                </div>
                <div class="kval">{{k.val}}</div>
                <div class="klbl">{{k.label}}</div>
                <div class="ksub">{{k.sub}}</div>
            </div>
        </div>

        <!-- API LOG -->
        <div class="alog" style="margin-bottom:18px">
            <div class="alt"><span class="adot"></span>Dashboard API</div>
            <pre><span class="m">GET</span> <span
class="u">/api/v1/projects/dashboard?departmentId={{auth.user.deptId}}</span>
Authorization: Bearer <span class="v">{{auth.token.substr(0,24)}}...</span>
→ <span class="k">totalProjects</span><span class="v">KPI</span> | <span
class="k">activeProjects</span><span class="v">KPI</span> | <span
class="k">budgetVariance</span><span class="v">Budget KPI color</span>
<span class="k">overallHealthScore</span><span class="v">Donut</span> | <span
class="k">highRiskProjects[]</span><span class="v">At-Risk table</span>
<span class="k">workloadDistribution[]</span><span class="v">Workload bars</span> |
<span class="k">taskCompletionTrend[]</span><span class="v">Line chart</span></pre>
        </div>

        <div class="g2">

```

```

<!-- TREND CHART -->
<div class="card">
  <div class="ch"><h3>📊 Task Completion Trend (30 Days)</h3><span class="bdg b-
ok">● Live</span></div>
  <div class="cb"><canvas id="trendChart" height="190"></canvas></div>
</div>
<!-- HEALTH DONUT -->
<div class="card">
  <div class="ch">
    <h3>🏥 Project Health</h3>
    <button class="btn btn-ai btn-s" @click="fetchHealth">🔍 Analyse</button>
  </div>
  <div class="cb">
    <div class="dw" style="margin-bottom:16px">
      <canvas id="donutChart" width="130" height="130"></canvas>
      <div class="dc"><div class="dv">{{dash.health}}%</div><div
class="dl">Health</div></div>
    </div>
    <div v-for="h in healthBars" :key="h.lbl" class="hi">
      <div class="ht">
        <span>{{h.lbl}}</span>
        <span :style="{color:h.v>=70?'var(--ok)':h.v>=50?'var(--warn)':'var(--
err)'}">{{h.v}}%</span>
      </div>
      <div class="pw"><div class="pb" :class="h.v>=70?'g':h.v>=50?'y':'r'"
:style="{width:h.v+'%'}"></div></div>
    </div>
  </div>
</div>

<div class="g2">
  <!-- AT-RISK -->
  <div class="card">
    <div class="ch"><h3>🚨 At-Risk Projects</h3><span class="bdg b-
err">{{atRisk.length}} Critical</span></div>
    <div class="tw">
      <table>

<thead><tr><th>Code</th><th>Project</th><th>Progress</th><th>Risk</th><th>Action</th></tr>
</thead>

      <tbody>
        <tr v-for="p in atRisk" :key="p.id">
          <td><span class="bdg b-p">{{p.code}}</span></td>
          <td><strong>{{p.name}}</strong></td>
          <td style="min-width:90px">
            <div class="pw"><div class="pb"
:~class="p.prog>=70?'g':p.prog>=40?'y':'r'" :style="{width:p.prog+'%'}"></div></div>
            <small style="color:var(--mx)">{{p.prog}}%</small>
          </td>
          <td><span class="rb"
:~class="p.risk>80?'rc':p.risk>60?'rh':p.risk>40?'rm':'rl'">{{p.risk}}%</span></td>
          <td><button class="btn btn-s btn-err" @click="escalate(p)">🔊
Escalate</button></td>
        </tr>
      </tbody>
    </table>
  </div>
</div>
<!-- WORKLOAD -->
<div class="card">
  <div class="ch"><h3>👥 Team Workload</h3><button class="btn btn-s btn-o"
@click="fetchWL">🔄</button></div>

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        <div class="cb">
            <div v-for="w in workload" :key="w.name" style="margin-bottom:13px">
                <div style="display:flex;justify-content:space-between;margin-
bottom:4px">
                    <div style="display:flex;align-items:center;gap:7px">
                        <div class="av">{{w.name[0]}}</div>
                        <span style="font-weight:600;font-size:13px">{{w.name}}</span>
                        <span class="bdg b-gr" style="font-size:10px">{{w.role}}</span>
                    </div>
                    <span :style="{color:w.load>80?'var(--err)':w.load>60?'var(--
warn)':'var(--ok)',fontWeight:'700',fontSize:'13px'}">{{w.load}}%</span>
                </div>
                <div class="pw"><div class="pb" :class="w.load>80?'r':w.load>60?'y':'g'"
:style="{width:w.load+'%'}"></div></div>
                <div style="font-size:11px;color:var(--mx);margin-top:2px">{{w.tasks}}
tasks • Burnout: {{w.burnout}}%</div>
            </div>
        </div>
    </div>
</div>

<!-- BUDGET OVERVIEW -->
<div class="card">
    <div class="ch"><h3>📊 Budget Overview</h3><span class="bdg b-gd">FY
2025</span></div>
    <div class="cb">
        <canvas id="budgetChart" height="120"></canvas>
    </div>
</div>
</div>

<!-- =====
📁 PROJECTS
===== -->

<div v-if="pg==='projects'">
    <div class="ph">
        <div class="ph-t">
            <h2>📁 Research Projects & Courses</h2>
            <p>Manage all departmental projects and research initiatives</p>
        </div>
        <div class="ph-a">
            <button class="btn btn-p" @click="showProjModal=true">+ New Project</button>
            <button class="btn btn-ai btn-s" @click="aiOptimize">🤖 AI Optimize</button>
            <button class="btn btn-gold btn-s">📄 Export</button>
        </div>
    </div>

    <!-- GET /api/v1/projects -->
    <div class="alog" style="margin-bottom:16px">
        <div class="alt"><span class="adot"></span>Projects API</div>
        <pre><span class="m">GET</span> <span
class="u">/api/v1/projects?departmentId={{auth.user.deptId}}&status={{pf.status||'all'}}&pr
iority={{pf.priority||'all'}}</span>
→ <span class="k">[].name</span>→<span class="v">Name col</span> | <span
class="k">[].status</span>→<span class="v">badge color</span> | <span
class="k">[].progressPercentage</span>→<span class="v">progress bar</span>
    <span class="k">[].aiHealthScore</span>→<span class="v">health icon</span> | <span
class="k">[].aiDelayRiskScore</span>→<span class="v">risk badge</span>
    <span class="k">[].plannedEndDate</span>→<span class="v">due date (red if
overdue)</span></pre>
        </div>

    <div class="fb">

```

```

<div class="si"><span>🔍</span><input v-model="pf.q" placeholder="Search
projects..."></div>
<select class="fsel" v-model="pf.status" @change="fetchProjects">
  <option value="">All Status</option>
  <option value="InProgress">🔄 Active</option>
  <option value="Delayed">🔴 Delayed</option>
  <option value="OnHold">🟡 On Hold</option>
  <option value="Completed">✅ Completed</option>
</select>
<select class="fsel" v-model="pf.priority">
  <option value="">All Priority</option>
  <option value="Critical">🔴 Critical</option>
  <option value="High">🟠 High</option>
  <option value="Medium">🟡 Medium</option>
  <option value="Low">🟢 Low</option>
</select>
<select class="fsel" v-model="pf.dept">
  <option value="">All Departments</option>
  <option value="SE">Software Eng.</option>
  <option value="AI">Artificial Intelligence</option>
  <option value="CS">Computer Science</option>
</select>
</div>

<div class="card">
  <div class="tw">
    <table>
      <thead>
        <tr><th>Code</th><th>Project / Course</th><th>Dept</th><th>Status</th>
        <th>Progress</th><th>Health</th><th>Risk</th><th>Due
Date</th><th>Budget</th><th>Actions</th></tr>
      </thead>
      <tbody>
        <tr v-for="p in filteredProjs" :key="p.id">
          <td><span class="bdg b-p">{{p.code}}</span></td>
          <td>
            <strong>{{p.name}}</strong>
            <div style="font-size:11px;color:var(--mx)">PM: {{p.manager}}</div>
          </td>
          <td><span class="bdg b-inf">{{p.dept}}</span></td>
          <td>
            <span class="bdg" :class="p.status==='Active'? 'b-
ok':p.status==='Delayed'? 'b-err':p.status==='OnHold'? 'b-inf': 'b-gr'">
            {{p.status==='Active'? '🔄':p.status==='Delayed'? '🔴':p.status==='OnHold'? '🟡': '✅'}}
            {{p.status}}
          </span>
          </td>
          <td style="min-width:110px">
            <div class="pw"><div class="pb"
: class="p.prog>=70?'g':p.prog>=40?'y': 'r'" :style="{width:p.prog+'%}"></div></div>
            <small>{{p.prog}}%</small>
          </td>
          <td>
            {{p.health>=70?'🟢':p.health>=50?'🟡': '🔴'}}
            <strong style="font-size:12px">{{p.health}}</strong>
          </td>
          <td><span class="rb"
: class="p.risk>70?'rh':p.risk>40?'rm': 'r1'">{{p.risk}}%</span></td>
          <td style="font-size:12px" :style="{color:isLate(p.due)? 'var(--
err)': 'var(--tx)'">{{p.due}}</td>
          <td style="font-size:12px">

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        <div style="font-size:11px;color:var(--mx)">${{p.budget}}K /
${{p.actual}}K</div>
        <div class="pw" style="margin-top:2px">
            <div class="pb" :class="p.actual>p.budget?'r':'g'"
:style="{width:Math.min(100,(p.actual/p.budget*100))+'%'}"></div>
        </div>
    </td>
    <td>
        <div style="display:flex;gap:4px">
            <button class="btn btn-s btn-o" @click="viewProj(p)"></button>
            <button class="btn btn-s btn-gold" @click="editProj(p)"></button>
            <button class="btn btn-s btn-ai" @click="aiHealth(p)"></button>
        </div>
    </td>
</tr>
</tbody>
</table>
</div>
</div>
<!-- Pagination -->
<div style="display:flex;align-items:center;justify-content:space-
between;padding:10px 0">
    <span style="font-size:13px;color:var(--mx)">Showing {{filteredProjs.length}}
of {{projects.length}} projects</span>
    <div style="display:flex;gap:5px">
        <button class="btn btn-gh btn-s">← Prev</button>
        <button class="btn btn-p btn-s" style="width:30px;padding:0">1</button>
        <button class="btn btn-gh btn-s" style="width:30px;padding:0">2</button>
        <button class="btn btn-gh btn-s">Next →</button>
    </div>
</div>
</div>
<!-- =====>
    TASKS / KANBAN
    <!-- =====>
    <div v-if="pg==='tasks'">
        <div class="ph">
            <div class="ph-t">
                <h2>Task Management</h2>
                <p>PROJ-001: AI Research Initiative · Sprint 3</p>
            </div>
            <div class="ph-a">
                <button class="btn btn-p" @click="showTaskModal=true">+ New Task</button>
                <button class="btn btn-ai btn-s" @click="aiAssignAll">AI Auto-
Assign</button>
                <button class="btn btn-gold btn-s" @click="go('reports')">Export</button>
            </div>
        </div>
    </div>
    <!-- API LOG -->
    <div class="alog" style="margin-bottom:14px">
        <div class="alt"><span class="adot"></span>Tasks API</div>
        <pre><span class="m">GET</span>    <span class="u">/api/v1/tasks/by-
project/{{selProj.id}}</span> → <span class="k">[.status</span>→<span class="v">Kanban
col</span>
<span class="m">PATCH</span> <span class="u">/api/v1/tasks/{taskId}/progress</span> →
<span class="k">progressPercentage</span>,<span class="k">notes</span>
<span class="m">PATCH</span> <span class="u">/api/v1/tasks/{taskId}/assign</span> →
<span class="k">assigneeId</span>
<span class="m">POST</span> <span class="u">/api/v1/tasks/{taskId}/escalate</span> →
escalation trigger

```



```

<thead>
  <tr><th>ID</th><th>Title</th><th>Status</th><th>Priority</th>
  <th>Assignee</th><th>Progress</th><th>AI
Risk</th><th>Due</th><th>Actions</th></tr>
</thead>
<tbody>
  <tr v-for="t in tasks" :key="t.id">
    <td><span class="bdg b-p" style="font-size:10px">TSK-
    {{t.id}}</span></td>
    <td>
      <strong>{{t.title}}</strong>
      <span v-if="t.escalated" class="bdg b-err" style="font-
      size:10px;margin-left:5px">⚠</span>
    </td>
    <td>
      <span class="bdg"
      :class="t.status==='Done'? 'b-ok':t.status==='InProgress'? 'b-
      p':t.status==='InReview'? 'b-warn': 'b-gr'">
        {{t.status}}
      </span>
    </td>
    <td>
      <span class="bdg"
      :class="t.pri==='Critical'? 'b-err':t.pri==='High'? 'b-
      warn':t.pri==='Medium'? 'b-p': 'b-ok'">
        {{t.pri}}
      </span>
    </td>
    <td>
      <div style="display:flex;align-items:center;gap:5px">
        <div class="av">{{(t.assignee||'')[0]}}</div>
        <span style="font-size:12px">{{t.assignee||'Unassigned'}}</span>
      </div>
    </td>
    <td style="min-width:100px">
      <div class="pw"><div class="pb b"
      :style="{width:t.prog+'%'}"></div></div>
      <small>{{t.prog}}%</small>
    </td>
    <td>
      <span class="rb"
      :class="t.aiRisk>70?'rh':t.aiRisk>40?'rm':'rl'">{{t.aiRisk}}%</span>
    </td>
    <td style="font-size:12px :style="{color:isLate(t.due)? 'var(--
    err)': 'var(--tx)}'">
      {{t.due}}
    </td>
    <td>
      <div style="display:flex;gap:4px">
        <button class="btn btn-s btn-o" @click="openTask(t)">⏏</button>
        <button class="btn btn-s btn-gold">⏏</button>
        <button class="btn btn-s btn-err" @click="escalateTask(t)"
        title="Escalate">⚠</button>
      </div>
    </td>
  </tr>
</tbody>
</table>
</div>
</div>
<!-- GANTT VIEW -->
<div v-if="tv==='gantt'" class="card">

```

```

<div class="ch"><h3>📅 Gantt Chart – Sprint 3</h3>
  <span class="bdg b-inf">Feb 1 → Mar 31, 2025</span>
</div>
<div class="cb">
  <div class="alog" style="margin-bottom:12px">
    <div class="alt"><span class="adot"></span>Gantt API</div>
    <pre><span class="m">GET</span> <span class="u">/api/v1/tasks/by-
project/{{selProj.id}}?includeGantt=true</span>
→ <span class="k">[].startDate</span><span class="v">bar left offset</span> | <span
class="k">[].dueDate</span><span class="v">bar width</span>
  <span class="k">[].progressPercentage</span><span class="v">fill overlay</span></pre>
    </div>
    <div class="gantt">
      <div class="gt">
        <!-- Header -->
        <div style="display:flex;padding:0 0 8px 200px;gap:0;border-bottom:2px
solid var(--bdr);margin-bottom:6px">
          <div v-for="m in ganttMonths" :key="m" style="flex:1;text-
align:center;font-size:12px;font-weight:700;color:var(--primary)">{{m}}</div>
        </div>
        <!-- Rows -->
        <div v-for="t in tasks" :key="t.id" class="gr2">
          <div class="gl2">
            <span class="bdg b-p" style="font-size:10px;margin-right:4px">TSK-
{{t.id}}</span>
            {{t.title}}
          </div>
          <div class="gbar-wrap">
            <div class="gbar"
              :style="{
                left: t.ganttLeft+'%',
                width: t.ganttWidth+'%',
                background: t.pri==='Critical'?var(--err):t.pri==='High'?var(-
-warn):t.pri==='Medium'?var(--primary):var(--ok)'
              }">
              {{t.prog}}%
            </div>
          </div>
        </div>
      </div>
    </div>
  </div>
</div>
<!-- =====>


📅 TEAM / USERS PAGE
  <div v-if="pg==='team'">
    <div class="ph">
      <div class="ph-t">
        <h2>📅 Team Members & Workload</h2>
        <p>Manage team availability, skills, and workload distribution</p>
      </div>
      <div class="ph-a">
        <button class="btn btn-p" @click="showUserModal=true">+ Add Member</button>
        <button class="btn btn-ai btn-s" @click="aiBurnout">📅 Burnout
Analysis</button>
        <button class="btn btn-gold btn-s" @click="go('reports')">📅 Workload
Report</button>
      </div>
    </div>
  </div>


```

```

<!-- API LOG -->
<div class="alog" style="margin-bottom:16px">
  <div class="alt"><span class="adot"></span>Users API</div>
  <pre><span class="m">GET</span> <span
class="u">/api/v1/users?departmentId={{auth.user.deptId}}</span>
<span class="m">GET</span> <span
class="u">/api/v1/users/workload?departmentId={{auth.user.deptId}}</span>
<span class="m">GET</span> <span class="u">/api/v1/users/available</span>
→ <span class="k">[.availabilityPercentage</span>→<span class="v">workload bar</span>
  <span class="k">[.aiBurnoutRiskScore</span>→<span class="v">burnout badge</span>
  <span class="k">[.aiWorkloadScore</span>→<span class="v">overload warning</span></pre>
</div>

<!-- WORKLOAD SUMMARY KPIS -->
<div class="kgrid" style="margin-bottom:18px">
  <div class="kpi bl">
    <div class="kpi-top"><div class="kico bl">📊</div><span class="ktr up">▲
2%</span></div>
    <div class="kval">{{teamMembers.length}}</div>
    <div class="klbl">Total Members</div>
  </div>
  <div class="kpi gr">
    <div class="kpi-top"><div class="kico gr">📊</div><span class="ktr up">▲
5%</span></div>
    <div
class="kval">{{teamMembers.filter(m=>m.status==='Available').length}}</div>
    <div class="klbl">Available Now</div>
  </div>
  <div class="kpi rd">
    <div class="kpi-top"><div class="kico rd">📊</div><span class="ktr dn">▼
1%</span></div>
    <div class="kval">{{teamMembers.filter(m=>m.load>80).length}}</div>
    <div class="klbl">Overloaded</div>
  </div>
  <div class="kpi or">
    <div class="kpi-top"><div class="kico or">📊</div><span class="ktr dn">▼
3%</span></div>
    <div class="kval">{{teamMembers.filter(m=>m.burnout>60).length}}</div>
    <div class="klbl">Burnout Risk</div>
  </div>
</div>

<!-- TEAM GRID -->
<div class="ga">
  <div v-for="m in teamMembers" :key="m.id" class="ucard">
    <div class="ucav">{{m.name[0]}}</div>
    <div class="uc-name">{{m.name}}</div>
    <div class="uc-role">
      <span class="bdg b-p" style="font-size:11px">{{m.role}}</span>
      <span class="bdg" :class="m.status==='Available'?'b-ok':m.status==='Busy'?'b-warn':'b-gr'"
style="font-size:11px;margin-left:4px">{{m.status}}</span>
    </div>
    <div class="uc-stats">
      <div class="uc-stat">
        <div class="uc-sv">{{m.tasks}}</div>
        <div class="uc-sl">Tasks</div>
      </div>
      <div class="uc-stat">
        <div class="uc-sv">{{m.done}}</div>
        <div class="uc-sl">Done/Mo</div>
      </div>
    </div>
  </div>

```

```

        <div class="uc-sv" :style="{color:m.perf>=80?'var(--
ok)':'m.perf>=60?'var(--warn)':'var(--err)'}">{{m.perf}}</div>
        <div class="uc-sl">Score</div>
    </div>
</div>
<!-- Workload bar -->
<div style="margin-bottom:8px">
    <div style="display:flex;justify-content:space-between;font-
size:12px;margin-bottom:3px">
        <span style="font-weight:600">Workload</span>
        <span :style="{color:m.load>80?'var(--err)':'m.load>60?'var(--
warn)':'var(--ok)',fontWeight:'700'}">{{m.load}}%</span>
    </div>
    <div class="pw"><div class="pb" :class="m.load>80?'r':m.load>60?'y':'g'"
:style="{width:m.load+'%'}"></div></div>
</div>
<!-- Burnout -->
<div style="margin-bottom:10px">
    <div style="display:flex;justify-content:space-between;font-
size:12px;margin-bottom:3px">
        <span style="font-weight:600"> Burnout Risk</span>
        <span :style="{color:m.burnout>60?'var(--err)':'m.burnout>40?'var(--
warn)':'var(--ok)',fontWeight:'700'}">{{m.burnout}}%</span>
    </div>
    <div class="pw"><div class="pb"
:class="m.burnout>60?'r':m.burnout>40?'y':'g'" :style="{width:m.burnout+'%'}"></div></div>
</div>
<!-- Skills -->
<div style="margin-bottom:12px">
    <div style="font-size:11px;font-weight:600;margin-bottom:5px;color:var(--
mx)">SKILLS</div>
    <div>
        <span v-for="sk in m.skills" :key="sk" class="skill-chip">{{sk}}</span>
    </div>
</div>
<div style="display:flex;gap:6px">
    <button class="btn btn-s btn-o" style="flex:1" @click="viewMember(m)">
View</button>
    <button class="btn btn-s btn-ai" @click="aiAssignTo(m)"> Assign</button>
</div>
</div>
</div>
</div>

<!-- =====
NOTIFICATIONS PAGE
===== -->
<div v-if="pg==='notifications'">
    <div class="ph">
        <div class="ph-t">
            <h2> Notifications</h2>
            <p>{{notifCount}} unread · System alerts, escalations & AI insights</p>
        </div>
        <div class="ph-a">
            <button class="btn btn-o btn-s" @click="markAllRead"> Mark All Read</button>
            <button class="btn btn-err btn-s"> Clear All</button>
        </div>
    </div>
</div>

<!-- API LOG -->
<div class="alog" style="margin-bottom:16px">
    <div class="alt"><span class="adot"></span>Notifications API</div>

```

```

        <pre><span class="m">GET</span>    <span
class="u">/api/v1/notifications?unreadOnly=false&page=1&pageSize=20</span>
<span class="m">GET</span>    <span class="u">/api/v1/notifications/unread-count</span> →
<span class="k">count</span><span class="v">header badge</span>
<span class="m">PATCH</span> <span class="u">/api/v1/notifications/{id}/read</span> →
mark single read
<span class="m">PATCH</span> <span class="u">/api/v1/notifications/read-all</span> →
mark all read
<span class="m">POST</span> <span class="u">/api/v1/notifications/broadcast</span> →
admin broadcast (Manager+)
→ SignalR hub: <span class="k">NotificationHub</span> → real-time push to connected
clients</pre>
    </div>

    <div class="card">
        <div class="ch">
            <h3>All Notifications</h3>
            <div style="display:flex;gap:7px">
                <select class="fsel" style="min-width:100px">
                    <option>All Types</option>
                    <option>🔔 Escalations</option>
                    <option>🔔 AI Insights</option>
                    <option>🔔 Deadlines</option>
                    <option>🔔 Tasks</option>
                </select>
            </div>
        </div>
        <div class="nl">
            <div v-for="n in notifications" :key="n.id"
                class="nitem" :class="{unr:!n.read}" @click="readNotif(n)">
                <div class="nico" :style="{background:n.bg}">{{n.icon}}</div>
                <div class="nb2">
                    <div class="nt">{{n.title}}</div>
                    <div class="nm">{{n.msg}}</div>
                    <div style="display:flex;align-items:center;gap:8px">
                        <span class="ntime">{{n.time}}</span>
                        <span v-if="n.ai" class="bdg b-ai" style="font-size:10px">🔔 AI
Generated</span>
                        <span class="bdg" :class="n.type==='Escalation'? 'b-
err':n.type==='Deadline'? 'b-warn': 'b-p'" style="font-size:10px">
                            {{n.type}}
                        </span>
                    </div>
                </div>
            </div>
            <div v-if="!n.read" class="ndot"></div>
        </div>
    </div>
</div>

<!-- =====
🔔 AI ASSISTANT PAGE
===== -->

<div v-if="pg==='ai'">
    <div class="ph">
        <div class="ph-t">
            <h2>🔔 AI Assistant & Analytics</h2>
            <p>Powered by GPT-4o • ML.NET Delay Prediction • Smart Task Allocation</p>
        </div>
        <div class="ph-a">
            <button class="btn btn-ai btn-s" @click="trainModel">🔔 Retrain
Models</button>

```

```

        <button class="btn btn-gold btn-s" @click="go('reports')"> AI
Report</button>
    </div>
</div>

<!-- API LOG -->
<div class="alog" style="margin-bottom:16px">
    <div class="alt"><span class="adot"></span>AI APIs</div>
    <pre><span class="m">POST</span> <span class="u">/api/v1/ai/chat</span>
→ GPT-4o chat
<span class="m">GET</span> <span class="u">/api/v1/tasks/{id}/ai/recommend-
assignee</span> → ML allocation
<span class="m">GET</span> <span class="u">/api/v1/tasks/{id}/ai/delay-prediction</span>
→ ML.NET FastTree
<span class="m">GET</span> <span class="u">/api/v1/projects/{id}/ai/health</span>
→ health scoring
<span class="m">GET</span> <span class="u">/api/v1/projects/{id}/ai/insights</span>
→ NLG insights
<span class="m">GET</span> <span class="u">/api/v1/users/{id}/ai/burnout-risk</span>
→ burnout score
<span class="m">POST</span> <span class="u">/api/v1/ai/train</span>
→ trigger retraining</pre>
    </div>

    <div class="g2">
        <!-- AI CHAT -->
        <div class="card">
            <div class="ch">
                <h3> AI Chat Assistant</h3>
                <span class="bdg b-ok">● Online · GPT-4o</span>
            </div>
            <div class="cb">
                <div class="chat" ref="chatBox">
                    <div v-for="(m,i) in chatMsgs" :key="i" class="msg" :class="m.role">
                        <div class="mb">{{m.text}}</div>
                        <div class="mt">{{m.time}}</div>
                    </div>
                    <div v-if="ai.typing" class="msg ai">
                        <div class="mb" style="color:var(--mx)"> Thinking<span
class="dots">...</span></div>
                    </div>
                </div>
                <div class="cir">
                    <div class="ciw">
                        <span></span>
                        <input v-model="ai.input" placeholder="Ask about projects, risks,
workload..."
                        @keyup.enter="sendChat"/>
                    </div>
                    <button class="btn btn-ai" @click="sendChat">Send</button>
                </div>
            </div>
            <div class="chips">
                <span class="chip" @click="quickChat('Which tasks are at highest risk of
delay?')"> Delay Risks</span>
                <span class="chip" @click="quickChat('Who is overloaded this week?')">
Overload</span>
                <span class="chip" @click="quickChat('Summarize project health')">
Health</span>
                <span class="chip" @click="quickChat('Recommend task assignments')">
Assign</span>
                <span class="chip" @click="quickChat('Show burnout risk report')">
Burnout</span>
            </div>

```

```

        </div>
    </div>

    <!-- AI SCORES -->
    <div style="display:flex;flex-direction:column;gap:16px">
        <!-- Model Performance -->
        <div class="card">
            <div class="ch"><h3>📊 AI Model Performance</h3><span class="bdg b-ai">v2.4.1</span></div>
            <div class="cb">
                <div class="ring-wrap">
                    <div v-for="r in aiMetrics" :key="r.label" class="ring-item">
                        <div class="ring" :style="{background:r.val>=80?'var(--ok)':'r.val>=60?'var(--warn)':'var(--err)'}">{{r.val}}%</div>
                        <div class="ring-lbl">{{r.label}}</div>
                    </div>
                </div>
                <div class="alog" style="margin-top:10px">
                    <div class="alt"><span class="adot"></span>Model Stats</div>
                    <pre><span class="k">Task Allocation</span> → <span class="v">ML.NET
+ Rule Engine</span>
<span class="k">Delay Prediction</span> → <span class="v">FastTree Regression</span>
<span class="k">Health Analysis</span> → <span class="v">Weighted Scoring</span>
<span class="k">NLP Chat</span> → <span class="v">GPT-4o via OpenAI</span>
<span class="k">Last Trained</span> → <span class="v">{{new
Date().toLocaleDateString()}}</span>
<span class="k">Next Training</span> → <span class="v">Daily 2:00 AM
(Hangfire)</span></pre>
                </div>
            </div>
        </div>

        <!-- AI Insights Feed -->
        <div class="card">
            <div class="ch"><h3>📢 Live AI Insights</h3><button class="btn btn-s btn-ai"
@click="refreshInsights">🔄 Refresh</button></div>
            <div class="cb" style="padding:10px">
                <div v-for="ins in aiInsights" :key="ins.id"
                    style="padding:10px;border-radius:10px;margin-bottom:8px;border-
left:3px solid"
                    :style="{borderColor:ins.sev==='High'? 'var(--err)':'ins.sev==='Medium'? 'var(--warn)':'var(--ok)',background:ins.sev==='High'? '#fde8ea':ins.sev==='Medium'? '#fff3cd':'#d4edda'}">
                    <div style="font-weight:700;font-size:13px;margin-
bottom:3px">{{ins.icon}} {{ins.title}}</div>
                    <div style="font-size:12px;color:var(--mx)">{{ins.msg}}</div>
                    <div style="display:flex;justify-content:space-between;margin-top:5px">
                        <span class="rb"
:~:class="ins.sev==='High'? 'rh':ins.sev==='Medium'? 'rm':'rl'" style="font-
size:10px">{{ins.sev}}</span>
                        <span style="font-size:11px;color:var(--mx)">{{ins.time}}</span>
                    </div>
                </div>
            </div>
        </div>

        <!-- AI PREDICTION TABLE -->
        <div class="card">
            <div class="ch">
                <h3>📊 Delay Prediction – All Active Tasks</h3>
                <span class="bdg b-ai">ML.NET FastTree Model</span>
            </div>
        </div>
    </div>

```

```

    </div>
    <div class="tw">
      <table>
        <thead>
          <tr><th>Task</th><th>Project</th><th>Assignee</th><th>Due Date</th>
            <th>Delay Prob.</th><th>Expected Delay</th><th>Risk
Level</th><th>Mitigations</th></tr>
        </thead>
        <tbody>
          <tr v-for="p in aiPredictions" :key="p.id">
            <td><strong>{{p.task}}</strong></td>
            <td><span class="bdg b-p" style="font-size:10px">{{p.proj}}</span></td>
            <td>
              <div style="display:flex;align-items:center;gap:5px">
                <div class="av">{{p.assignee[0]}}</div>
                <span>{{p.assignee}}</span>
              </div>
            </td>
            <td style="font-size:12px" :style="{color:isLate(p.due)?'var(--err)':'var(--tx)'}">{{p.due}}</td>
            <td>
              <div class="pw" style="margin-bottom:2px"><div class="pb"
: class="p.prob>70?'r':p.prob>40?'y':'g'" :style="{width:p.prob+'%'}"></div></div>
              <strong :style="{color:p.prob>70?'var(--err)':p.prob>40?'var(--warn)':'var(--ok)'}">{{p.prob}}%</strong>
            </td>
            <td style="font-size:12px">{{p.delayDays > 0 ? '+'+p.delayDays+ ' days'
: 'On track'}}</td>
            <td><span class="rb"
: class="p.risk==='Critical'? 'rc':p.risk==='High'? 'rh':p.risk==='Medium'? 'rm':'rl'">{{p.risk}}</span></td>
            <td>
              <ul style="font-size:11px;color:var(--mx);padding-left:14px;margin:0">
                <li v-for="m in p.mitigations.slice(0,2)" :key="m">{{m}}</li>
              </ul>
            </td>
          </tr>
        </tbody>
      </table>
    </div>
  </div>
</div>

<!-- =====
REPORTS PAGE
===== -->
<div v-if="pg==='reports'">
  <div class="ph">
    <div class="ph-t">
      <h2>📊 Reports & Analytics</h2>
      <p>Generate PDF/Excel reports · AI-narrated insights</p>
    </div>
    <div class="ph-a">
      <button class="btn btn-p btn-s" @click="generateReport">📄 Generate
Report</button>
      <button class="btn btn-ai btn-s" @click="aiNarrate">🗣️ AI Narrate</button>
    </div>
  </div>
</div>

<!-- API LOG -->
<div class="alog" style="margin-bottom:16px">
  <div class="alt"><span class="adot"></span>Reports API</div>

```

```

    <pre><span class="m">GET</span> <span class="u">/api/v1/reports/project-
status/{id}?format={{rep.format}}</span>
<span class="m">POST</span> <span class="u">/api/v1/reports/task-
completion?format={{rep.format}}</span>
<span class="m">POST</span> <span class="u">/api/v1/reports/department-
workload?format={{rep.format}}</span>
<span class="m">GET</span> <span class="u">/api/v1/reports/budget-
variance/{id}?format={{rep.format}}</span>
<span class="m">POST</span> <span class="u">/api/v1/reports/delay-
analysis?format={{rep.format}}</span>
<span class="m">GET</span> <span class="u">/api/v1/reports/ai-
insights/{id}?format={{rep.format}}</span>
→ Returns: <span class="v">byte[]</span> with Content-Type: <span
class="k">application/pdf</span> or <span
class="k">application/vnd.openxmlformats</span></pre>
</div>

```

```

<div class="rep-grid">
  <!-- Report Type Selector -->
  <div class="card">
    <div class="ch"><h3> Report Type</h3></div>
    <div class="cb">
      <div class="rep-type">
        <div v-for="r in reportTypes" :key="r.id"
          class="rep-opt" :class="{sel:rep.type===r.id}"
          @click="rep.type=r.id">
            {{r.icon}} {{r.label}}
          </div>
        </div>
      </div>
      <div style="margin-top:16px">
        <div class="fc2" style="margin-bottom:10px">
          <label>Format</label>
          <select class="fsel" v-model="rep.format" style="width:100%">
            <option value="pdf"> PDF</option>
            <option value="excel"> Excel</option>
            <option value="csv"> CSV</option>
          </select>
        </div>
        <div class="fc2" style="margin-bottom:10px">
          <label>Department</label>
          <select class="fsel" v-model="rep.dept" style="width:100%">
            <option value="">All Departments</option>
            <option value="SE">Software Engineering</option>
            <option value="AI">AI Research</option>
            <option value="CS">Computer Science</option>
          </select>
        </div>
        <div class="fr">
          <div class="fc2">
            <label>From</label>
            <input type="date" v-model="rep.from"/>
          </div>
          <div class="fc2">
            <label>To</label>
            <input type="date" v-model="rep.to"/>
          </div>
        </div>
        <button class="btn btn-p" style="width:100%;margin-top:14px"
@click="generateReport">
          Generate {{rep.format.toUpperCase()}}
        </button>
        <button class="btn btn-ai" style="width:100%;margin-top:8px"
@click="aiNarrate">

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```

         AI Narrated Report
      </button>
    </div>
  </div>
</div>

<!-- Report Preview -->
<div class="card">
  <div class="ch">
    <h3><img alt="Report Preview icon" data-bbox="315 205 328 215"/> Report Preview –
    {{reportTypes.find(r=>r.id===rep.type)?.label}}</h3>
    <span class="bdg b-ok">{{new Date().toLocaleDateString()}}</span>
  </div>
  <div class="cb">
    <!-- Summary stats row -->
    <div style="display:flex;gap:20px;margin-bottom:18px;flex-wrap:wrap">
      <div style="text-align:center">
        <div style="font-size:28px;font-weight:800;color:var(--primary)">{{projects.length}}</div>
        <div style="font-size:12px;color:var(--mx)">Total Projects</div>
      </div>
      <div style="text-align:center">
        <div style="font-size:28px;font-weight:800;color:var(--ok)">{{tasks.filter(t=>t.status==='Done').length}}</div>
        <div style="font-size:12px;color:var(--mx)">Completed Tasks</div>
      </div>
      <div style="text-align:center">
        <div style="font-size:28px;font-weight:800;color:var(--warn)">{{tasks.filter(t=>t.aiRisk>60).length}}</div>
        <div style="font-size:12px;color:var(--mx)">High Risk Tasks</div>
      </div>
      <div style="text-align:center">
        <div style="font-size:28px;font-weight:800;color:var(--ai)">{{dash.health}}%</div>
        <div style="font-size:12px;color:var(--mx)">Avg Health Score</div>
      </div>
    </div>
    <canvas id="reportChart" height="200"></canvas>
    <!-- AI Narration -->
    <div v-if="rep.narration" class="alert a-inf" style="margin-top:14px">
      <img alt="AI Narration icon" data-bbox="275 610 288 620"/> <strong>AI Narration:</strong> {{rep.narration}}
    </div>
  </div>
</div>
</div>
</div>
</div>

<!-- =====>
<div>
  <img alt="DEPARTMENTS PAGE icon" data-bbox="228 715 241 725"/> DEPARTMENTS PAGE
  <!-- ===== -->
  <div v-if="pg==='departments'">
    <div class="ph">
      <div class="ph-t">
        <h2><img alt="DEPARTMENTS PAGE icon" data-bbox="228 715 241 725"/> Departments</h2>
        <p>Manage university departments and their project portfolios</p>
      </div>
      <div class="ph-a">
        <button class="btn btn-p" @click="showDeptModal=true">+ New
        Department</button>
      </div>
    </div>
  </div>
  <div class="alog" style="margin-bottom:16px">

```

```

        <div class="alt"><span class="adot"></span>Departments API</div>
        <pre><span class="m">GET</span>    <span class="u">/api/v1/departments</span>
<span class="m">GET</span>    <span class="u">/api/v1/departments/{id}/dashboard</span>
<span class="m">POST</span><span>    <span class="u">/api/v1/departments</span>    { <span
class="k">name</span>, <span class="k">code</span>, <span class="k">description</span>,
<span class="k">budget</span> }
<span class="m">PUT</span>    <span class="u">/api/v1/departments/{id}</span>
<span class="m">DELETE</span><span>    <span class="u">/api/v1/departments/{id}</span></span></pre>
        </div>

        <div class="ga">
            <div v-for="d in departments" :key="d.id" class="card"
style="cursor:pointer;transition:var(--tr)" @click="viewDept(d)">
                <div class="ch">
                    <div>
                        <h3>{{d.icon}} {{d.name}}</h3>
                        <span class="bdg b-p" style="margin-top:3px;display:inline-
block">{{d.code}}</span>
                    </div>
                    <span class="bdg" :class="d.health>=70?'b-ok':d.health>=50?'b-warn':'b-
err'">Health: {{d.health}}%</span>
                </div>
                <div class="cb">
                    <div style="display:flex;justify-content:space-around;margin-bottom:14px">
                        <div style="text-align:center">
                            <div style="font-size:22px;font-weight:800;color:var(--
primary)">{{d.projects}}</div>
                            <div style="font-size:11px;color:var(--mx)">Projects</div>
                        </div>
                        <div style="text-align:center">
                            <div style="font-size:22px;font-weight:800;color:var(--
ok)">{{d.members}}</div>
                            <div style="font-size:11px;color:var(--mx)">Members</div>
                        </div>
                        <div style="text-align:center">
                            <div style="font-size:22px;font-weight:800;color:var(--
gold)">{{d.budget}}K</div>
                            <div style="font-size:11px;color:var(--mx)">Budget ($)</div>
                        </div>
                    </div>
                    <div style="margin-bottom:8px">
                        <div style="display:flex;justify-content:space-between;font-
size:12px;margin-bottom:3px">
                            <span>Budget Used</span>
                            <span :style="{color:d.budgetUsed>90?'var(--
err)':d.budgetUsed>70?'var(--warn)':'var(--ok)',fontWeight:'700'}">{{d.budgetUsed}}%</span>
                        </div>
                        <div class="pw"><div class="pb"
:class="d.budgetUsed>90?'r':d.budgetUsed>70?'y':'g'"
:style="{width:d.budgetUsed+'%'}"></div></div>
                    </div>
                    <div style="display:flex;gap:6px">
                        <button class="btn btn-s btn-o" style="flex:1">🏠 Dashboard</button>
                        <button class="btn btn-s btn-gold">🔧 Edit</button>
                    </div>
                </div>
            </div>
        </div>
    </div><!-- end .pg -->
</div><!-- end .main -->
</div><!-- end .app -->

```

```
<!-- =====  
MODALS  
===== -->
```

```
<!-- NEW PROJECT MODAL - POST /api/v1/projects -->  
<div v-if="showProjModal" class="ov" @click.self="showProjModal=false">  
  <div class="modal">  
    <div class="mh">  
      <h3> Create New Project</h3>  
      <button class="mx-btn" @click="showProjModal=false">X</button>  
    </div>  
    <div class="mb2">  
      <div class="alog" style="margin-bottom:14px">  
        <div class="alt"><span class="adot"></span>API Call</div>  
        <pre><span class="m">POST</span> <span class="u">/api/v1/projects</span>  
{ <span class="k">"projectCode"</span>:<span class="v">"{{pForm.code}}"</span>, <span  
class="k">"name"</span>:<span class="v">"{{pForm.name}}"</span>,  
  <span class="k">"plannedStartDate"</span>:<span class="v">"{{pForm.start}}"</span>,  
<span class="k">"plannedEndDate"</span>:<span class="v">"{{pForm.end}}"</span>,  
  <span class="k">"plannedBudget"</span>:<span class="v">"{{pForm.budget||0}}"</span>, <span  
class="k">"priority"</span>:<span class="v">"{{pForm.priority}}"</span>,  
  <span class="k">"departmentId"</span>:<span class="v">"{{auth.user.deptId}}"</span>  
</pre>  
      </div>  
      <div class="fr">  
        <div class="fc2">  
          <label>Project Code *</label>  
          <input v-model="pForm.code" placeholder="PROJ-002"/>  
        </div>  
        <div class="fc2">  
          <label>Priority *</label>  
          <select v-model="pForm.priority">  
  
<option>Critical</option><option>High</option><option>Medium</option><option>Low</option>  
          </select>  
        </div>  
        <div class="fc2 ff">  
          <label>Project Name *</label>  
          <input v-model="pForm.name" placeholder="e.g. AI Research Initiative Phase 2"/>  
        </div>  
        <div class="fc2 ff">  
          <label>Description</label>  
          <textarea v-model="pForm.desc" placeholder="Project scope and  
objectives..."></textarea>  
        </div>  
        <div class="fc2">  
          <label>Category</label>  
          <select v-model="pForm.cat">  
            <option>Research</option><option>Software Development</option>  
            <option>Infrastructure</option><option>Education</option>  
          </select>  
        </div>  
        <div class="fc2">  
          <label>Department</label>  
          <select v-model="pForm.dept">  
            <option value="SE">Software Engineering</option>  
            <option value="AI">Artificial Intelligence</option>  
            <option value="CS">Computer Science</option>  
          </select>  
        </div>  
        <div class="fc2">  
          <label>Start Date *</label>
```

```

        <input type="date" v-model="pForm.start"/>
      </div>
      <div class="fc2">
        <label>End Date *</label>
        <input type="date" v-model="pForm.end"/>
      </div>
      <div class="fc2">
        <label>Planned Budget ($)</label>
        <input type="number" v-model="pForm.budget" placeholder="250000"/>
      </div>
      <div class="fc2">
        <label>Project Manager</label>
        <select v-model="pForm.manager">
          <option v-for="m in teamMembers" :key="m.id"
:value="m.id">{{m.name}}</option>
        </select>
      </div>
    </div>
  </div>
  <div class="mf">
    <button class="btn btn-gh" @click="showProjModal=false">Cancel</button>
    <button class="btn btn-ai" @click="aiValidate">AI Validate</button>
    <button class="btn btn-p" @click="createProject">Create Project</button>
  </div>
</div>
</div>

<!-- NEW TASK MODAL - POST /api/v1/tasks -->
<div v-if="showTaskModal" class="ov" @click.self="showTaskModal=false">
  <div class="modal">
    <div class="mh">
      <h3>Create New Task</h3>
      <button class="mx-btn" @click="showTaskModal=false">X</button>
    </div>
    <div class="mb2">
      <div class="alog" style="margin-bottom:14px">
        <div class="alt"><span class="adot"></span>API Call</div>
        <pre><span class="m">POST</span> <span class="u">/api/v1/tasks</span>
{ <span class="k">"title"</span>:<span class="v">{{tForm.title}}</span>, <span
class="k">"priority"</span>:<span class="v">{{tForm.priority}}</span>,
  <span class="k">"projectId"</span>:<span class="v">{{selProj.id}}</span>, <span
class="k">"assignedToUserId"</span>:<span class="v">{{tForm.assignee}}</span>,
  <span class="k">"dueDate"</span>:<span class="v">{{tForm.due}}</span>, <span
class="k">"estimatedHours"</span>:<span class="v">{{tForm.hours||0}}</span> }</pre>
      </div>
      <div class="fr">
        <div class="fc2 ff">
          <label>Task Title *</label>
          <input v-model="tForm.title" placeholder="e.g. Design database schema"/>
        </div>
        <div class="fc2 ff">
          <label>Description</label>
          <textarea v-model="tForm.desc" placeholder="Task details..."></textarea>
        </div>
        <div class="fc2">
          <label>Priority *</label>
          <select v-model="tForm.priority">
            <option>Critical</option><option>High</option><option
selected>Medium</option><option>Low</option>
          </select>
        </div>
        <div class="fc2">
          <label>Estimated Hours</label>

```

```

        <input type="number" v-model="tForm.hours" placeholder="40"/>
      </div>
      <div class="fc2">
        <label>Start Date</label>
        <input type="date" v-model="tForm.start"/>
      </div>
      <div class="fc2">
        <label>Due Date *</label>
        <input type="date" v-model="tForm.due"/>
      </div>
      <div class="fc2">
        <label>Assignee</label>
        <select v-model="tForm.assignee">
          <option value="">-- AI Auto-Assign --</option>
          <option v-for="m in teamMembers" :key="m.id" :value="m.id">{{m.name}}
({{m.load}}% load)</option>
        </select>
      </div>
      <div class="fc2">
        <label>Milestone</label>
        <select v-model="tForm.milestone">
          <option value="">No Milestone</option>
          <option>Phase 1 – Architecture</option>
          <option>Phase 2 – Development</option>
          <option>Phase 3 – Testing</option>
        </select>
      </div>
    </div>
  </div>
  <div class="mf">
    <button class="btn btn-gh" @click="showTaskModal=false">Cancel</button>
    <button class="btn btn-ai" @click="aiRecommendAssignee">👤 AI Recommend</button>
    <button class="btn btn-p" @click="createTask">👤 Create Task</button>
  </div>
</div>
</div>

<!-- TASK DETAIL MODAL -->
<div v-if="showTaskDetail && selTask" class="ov" @click.self="showTaskDetail=false">
  <div class="modal" style="max-width:640px">
    <div class="mh">
      <h3>TSK-{{selTask.id}} · {{selTask.title}}</h3>
      <button class="mx-btn" @click="showTaskDetail=false">X</button>
    </div>
    <div class="mb2">
      <!-- Status & badges row -->
      <div style="display:flex;gap:8px;flex-wrap:wrap;margin-bottom:14px">
        <span class="bdg" :class="selTask.status==='Done'?'b-ok':selTask.status==='InProgress'?'b-p':'b-gr'">{{selTask.status}}</span>
        <span class="bdg" :class="selTask.pri==='Critical'?'b-err':selTask.pri==='High'?'b-warn':'b-p'">{{selTask.pri}}</span>
        <span v-if="selTask.escalated" class="bdg b-err">🔪 Escalated</span>
        <span class="bdg b-ai">🔪 Risk: {{selTask.aiRisk}}%</span>
      </div>
      <!-- Progress update -->
      <div style="margin-bottom:16px">
        <label style="font-weight:600;font-size:13px;display:block;margin-bottom:6px">
          Progress: {{taskProg}}%
        </label>
        <input type="range" v-model="taskProg" min="0" max="100"
style="width:100%;accent-color:var(--primary)"/>
        <div class="pw" style="margin-top:6px"><div class="pb b"
:style="{width:taskProg+'%'}"></div></div>

```

```

    </div>
    <div class="fc2" style="margin-bottom:14px">
      <label>Progress Notes</label>
      <textarea v-model="taskNote" placeholder="Describe what was
completed..."></textarea>
    </div>
    <!-- AI Delay Prediction -->
    <div class="alert a-warn" style="margin-bottom:14px">
      <strong>AI Delay Prediction:</strong>
      {{selTask.aiRisk}}% delay probability ·
      <span v-if="selTask.aiRisk>60">Expected +{{Math.round(selTask.aiRisk/10)}} days
delay · <strong>Escalation recommended</strong></span>
      <span v-else>On track</span>
    </div>
    <!-- API call preview -->
    <div class="alog">
      <div class="alt"><span class="adot"></span>Progress Update API</div>
      <pre><span class="m">PATCH</span> <span
class="u">/api/v1/tasks/{{selTask.id}}/progress</span>
{ <span class="k">"progressPercentage"</span>: <span class="v">{{taskProg}}</span>, <span
class="k">"notes"</span>: <span
class="v">"{{taskNote||'...'||(taskNote.substring?taskNote.substring(0,30):'...')}}"</span>
}</pre>
    </div>
  </div>
  <div class="mf">
    <button class="btn btn-gh" @click="showTaskDetail=false">Close</button>
    <button class="btn btn-err btn-s" @click="escalateTask(selTask)">⚠
Escalate</button>
    <button class="btn btn-ai" @click="aiPredictTask(selTask)">🧠 AI Predict</button>
    <button class="btn btn-p" @click="updateProgress">💾 Save Progress</button>
  </div>
</div>
</div>

<!-- end #app -->

<script>
const { createApp, ref, computed, onMounted, nextTick } = Vue;

createApp({
  setup() {

    // — AUTH STATE —
    const auth = ref({
      isLoggedIn: false,
      token: 'eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.demo.token_2025',
      user: {
        fullName: 'Dr. Sarah Johnson',
        ini: 'SJ',
        role: 'Department Head',
        deptId: 'dept-se-001',
        email: 'sarah.johnson@university.edu'
      }
    });

    const loginForm = ref({ email:'', password:'', remember:false });
    const login = ref({ error:'', success:'', loading:false, eErr:false, pErr:false });

    function valEmail() {
      login.value.eErr = !loginForm.value.email.includes('@');
    }

    async function doLogin() {

```

```

login.value.eErr = !loginForm.value.email.includes('@');
login.value.pErr = !loginForm.value.password;
if (login.value.eErr || login.value.pErr) return;
login.value.loading = true;
login.value.error = '';
// Simulate POST /api/v1/auth/Login
await new Promise(r => setTimeout(r, 1200));
login.value.loading = false;
login.value.success = 'Authenticated successfully! Redirecting...';
await new Promise(r => setTimeout(r, 700));
auth.value.isLoggedIn = true;
initCharts();
}

function ssoLogin() {
  login.value.success = 'Redirecting to University SSO...';
  setTimeout(() => { auth.value.isLoggedIn = true; initCharts(); }, 900);
}

function logout() {
  auth.value.isLoggedIn = false;
  loginForm.value = { email:'', password:'', remember:false };
  login.value = { error:'', success:'', loading:false, eErr:false, pErr:false };
}

// — NAVIGATION —
const sb = ref({ col: false });
const pg = ref('dashboard');
const hdr = ref({ q:'' });

const nav = [
  { id:'dashboard', icon:'🏠', label:'Dashboard' },
  { id:'projects', icon:'📁', label:'Projects', badge:3 },
  { id:'tasks', icon:'📋', label:'Tasks', badge:7 },
  { id:'team', icon:'👥', label:'Team Members' },
  { id:'departments', icon:'🏢', label:'Departments' },
  { id:'notifications', icon:'🔔', label:'Notifications', badge:5 },
  { id:'reports', icon:'📊', label:'Reports' },
];
const aiNav = [
  { id:'ai', icon:'🤖', label:'AI Assistant' },
];

const pgLabel = computed(() => {
  const all = [...nav, ...aiNav];
  return all.find(n => n.id === pg.value)?.label || 'Dashboard';
});

function go(id) { pg.value = id; if (id==='reports') nextTick(initReportChart); }
const today = new Date().toLocaleDateString('en-US',{weekday:'long',year:'numeric',month:'long',day:'numeric'});

// — DASHBOARD DATA —
const dash = ref({ health:78, escalated:2 });

const kpis = [
  { label:'Total Projects', val:24, icon:'📁', col:'bl', up:true, trend:12, sub:'3 new this month', nav:'projects' },
  { label:'Active Projects', val:18, icon:'📁', col:'gr', up:true, trend:8, sub:'2 ahead of schedule', nav:'projects' },
  { label:'Tasks This Sprint', val:142, icon:'📋', col:'bl', up:true, trend:5, sub:'89 completed', nav:'tasks' },

```

```

    { label:'Overdue Tasks',      val:7,   icon:'🔴', col:'rd', up:false, trend:3,
    sub:'Requires attention', nav:'tasks' },
    { label:'Team Members',      val:38,   icon:'👤', col:'pu', up:true, trend:2, sub:'4
on leave', nav:'team' },
    { label:'Budget Used',      val:'67%',icon:'📊',col:'gd', up:false, trend:4,
    sub:'$892K of $1.3M', nav:'reports' },
  ];

  const healthBars = [
    { lbl:'Schedule Adherence', v:82 },
    { lbl:'Budget Compliance', v:67 },
    { lbl:'Team Velocity', v:74 },
    { lbl:'Quality Index', v:91 },
    { lbl:'Risk Management', v:58 },
  ];

  const atRisk = [
    { id:1, code:'PROJ-003', name:'Cloud Migration Initiative', prog:31, risk:87 },
    { id:2, code:'PROJ-007', name:'Mobile App Development', prog:45, risk:73 },
    { id:3, code:'PROJ-011', name:'Data Analytics Platform', prog:22, risk:68 },
  ];

  const workload = [
    { name:'Ahmad Hassan', role:'Dev Lead', load:94, tasks:12, burnout:78 },
    { name:'Sarah Lee', role:'ML Engineer', load:72, tasks:8, burnout:45 },
    { name:'John Williams', role:'Designer', load:55, tasks:6, burnout:30 },
    { name:'Fatima Al-Zahra', role:'DevOps', load:88, tasks:10, burnout:62 },
    { name:'Carlos Marin', role:'Analyst', load:41, tasks:5, burnout:22 },
  ];

  function refreshDash() { dash.value.health = Math.floor(Math.random()*20)+70; }
  function fetchHealth() {}
  function fetchWL() {}
  function escalate(p) { alert(`🚨 Escalating: ${p.name}\nPOST /api/v1/tasks/escalate`); }

  // — PROJECTS DATA —
  const projects = ref([
    { id:1, code:'PROJ-001', name:'PMWDS Portal Development', dept:'SE', manager:'Dr. Sarah Johnson',
      status:'Active', prog:67, health:82, risk:28, due:'2025-08-31', budget:250, actual:168 },
    { id:2, code:'PROJ-002', name:'AI Research Initiative', dept:'AI', manager:'Prof. Ahmed Ali',
      status:'Active', prog:45, health:71, risk:42, due:'2025-06-15', budget:180, actual:82 },
    { id:3, code:'PROJ-003', name:'Cloud Migration', dept:'CS', manager:'Dr. James Liu',
      status:'Delayed', prog:31, health:44, risk:87, due:'2025-03-31', budget:320, actual:290 },
    { id:4, code:'PROJ-004', name:'Mobile App v2.0', dept:'SE', manager:'Dr. Sarah Johnson',
      status:'OnHold', prog:55, health:61, risk:55, due:'2025-09-30', budget:140, actual:78 },
    { id:5, code:'PROJ-005', name:'Data Analytics Platform', dept:'AI', manager:'Prof. Ahmed Ali',
      status:'Active', prog:22, health:38, risk:68, due:'2025-04-30', budget:210, actual:47 },
  ]);
  const pf = ref({ q:'', status:'', priority:'', dept:'' });

  const filteredProjs = computed(() =>
    projects.value.filter(p =>
      (!pf.value.q || p.name.toLowerCase().includes(pf.value.q.toLowerCase())) &&

```

```

        (!pf.value.status || p.status === pf.value.status) &&
        (!pf.value.dept || p.dept === pf.value.dept)
    )
);
function fetchProjects() {}
function viewProj(p) { alert(`  Viewing: ${p.name}\nGET /api/v1/projects/${p.id}`); }
function editProj(p) { showProjModal.value = true; }
function aiHealth(p) { alert(`  AI Health Analysis\nGET
/api/v1/projects/${p.id}/ai/health\n\nHealth Score: ${p.health}%\nRisk Score: ${p.risk}%`);
}
function aiOptimize() { alert(`  AI Resource Optimization\nPOST
/api/v1/projects/{id}/ai/optimize-resources`); }

// — TASKS / KANBAN DATA —
const tv = ref('kanban');
const selProj = ref({ id: 'proj-001' });

const kanbanCols = [
  { status: 'Todo', icon: '📌', label: 'To Do' },

```

I now have full context from the SRS and code document. Here is the **exact continuation** from where the code was cut off — pick up at the Kanban cols array definition and continue all the way through to the complete closing `</script>` and `</html>` tags:

```

javascript
    { status: 'InProgress', icon: '🔄', label: 'In Progress' },
    { status: 'InReview', icon: '🔍', label: 'In Review' },
    { status: 'Done', icon: '✅', label: 'Done' },
  ];

  const tasks = ref([
    { id: 1, title: 'Design System Architecture', status: 'Done', pri: 'High',
    assignee: 'Ahmad Hassan', prog: 100, due: '2025-02-15', aiRisk: 12, escalated: false,
    ganttLeft: 0, ganttWidth: 20 },
    { id: 2, title: 'Setup CI/CD Pipeline', status: 'Done', pri: 'Medium',
    assignee: 'Fatima Al-Zahra', prog: 100, due: '2025-02-20', aiRisk: 18, escalated: false,
    ganttLeft: 5, ganttWidth: 18 },
    { id: 3, title: 'Develop Auth Module', status: 'InProgress',
    pri: 'Critical', assignee: 'Ahmad Hassan', prog: 72, due: '2025-03-10', aiRisk: 65,
    escalated: false, ganttLeft: 15, ganttWidth: 25 },
    { id: 4, title: 'AI Delay Prediction Engine', status: 'InProgress', pri: 'High',
    assignee: 'Sarah Lee', prog: 48, due: '2025-03-20', aiRisk: 72, escalated: true,
    ganttLeft: 20, ganttWidth: 30 },
    { id: 5, title: 'Dashboard UI Components', status: 'InProgress', pri: 'Medium',
    assignee: 'John Williams', prog: 55, due: '2025-03-25', aiRisk: 38, escalated: false,
    ganttLeft: 22, ganttWidth: 28 },
    { id: 6, title: 'Reports & PDF Export', status: 'InReview', pri: 'High',
    assignee: 'Carlos Marin', prog: 90, due: '2025-03-15', aiRisk: 22, escalated: false,
    ganttLeft: 18, ganttWidth: 20 },
    { id: 7, title: 'SignalR Real-time Notifications', status: 'InReview', pri: 'Medium',
    assignee: 'Fatima Al-Zahra', prog: 88, due: '2025-03-18', aiRisk: 30, escalated: false,
    ganttLeft: 25, ganttWidth: 18 },
    { id: 8, title: 'Mobile App API Integration', status: 'Todo', pri: 'Low',
    assignee: null, prog: 0, due: '2025-04-10', aiRisk: 55, escalated: false,
    ganttLeft: 40, ganttWidth: 25 },
    { id: 9, title: 'Burnout Risk ML Model', status: 'Todo', pri: 'High',
    assignee: null, prog: 0, due: '2025-04-20', aiRisk: 80, escalated: true,
    ganttLeft: 42, ganttWidth: 22 },

```

```

    { id:10, title:'User Acceptance Testing', status:'Todo',
    pri:'Critical', assignee:null, prog:0, due:'2025-05-01', aiRisk:45,
    escalated:false, ganttLeft:55, ganttWidth:20 },
  ]);

  const ganttMonths = ['Feb 2025', 'Mar 2025', 'Apr 2025', 'May 2025', 'Jun 2025'];

  function tasksIn(status) { return tasks.value.filter(t => t.status === status); }

  const showTaskModal = ref(false);
  const showTaskDetail = ref(false);
  const selTask = ref(null);
  const taskProg = ref(0);
  const taskNote = ref('');

  const tForm = ref({ title:'', desc:'', priority:'Medium', hours:'', start:'', due:'',
  assignee:'', milestone:'' });

  function openTask(t) {
    selTask.value = t;
    taskProg.value = t.prog;
    taskNote.value = '';
    showTaskDetail.value = true;
  }

  function updateProgress() {
    if (selTask.value) {
      selTask.value.prog = Number(taskProg.value);
      if (taskProg.value >= 100) selTask.value.status = 'Done';
      alert(`📌 Progress saved!\nPATCH /api/v1/tasks/${selTask.value.id}/progress\n{
progressPercentage: ${taskProg.value}, notes: "${taskNote.value} | 'Updated'" }`);
      showTaskDetail.value = false;
    }
  }

  function escalateTask(t) {
    t.escalated = true;
    dash.value.escalated++;
    alert(`📌 Task escalated!\nPOST /api/v1/tasks/${t.id}/escalate\n→ Notifications sent
to Project Manager & Department Head\n→ SignalR push to all connected clients`);
  }

  function createTask() {
    if (!tForm.value.title) { alert('Task title is required'); return; }
    const newTask = {
      id: tasks.value.length + 1,
      title: tForm.value.title,
      status: 'Todo',
      pri: tForm.value.priority,
      assignee: teamMembers.value.find(m => m.id === tForm.value.assignee)?.name ||
null,
      prog: 0,
      due: tForm.value.due || '2025-06-01',
      aiRisk: Math.floor(Math.random() * 50) + 10,
      escalated: false,
      ganttLeft: 60,
      ganttWidth: 15
    };
    tasks.value.push(newTask);
    showTaskModal.value = false;
    tForm.value = { title:'', desc:'', priority:'Medium', hours:'', start:'', due:'',
    assignee:'', milestone:'' };
    alert(`📌 Task created!\nPOST /api/v1/tasks → 201 Created\nID: TSK-${newTask.id}`);
  }

```

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    }

    function aiAssignAll() {
        tasks.value.filter(t => !t.assignee).forEach(t => {
            const avail = teamMembers.value.filter(m => m.load < 80);
            if (avail.length) {
                t.assignee = avail[Math.floor(Math.random() * avail.length)].name;
                t.status = 'InProgress';
            }
        });
        alert(`🔔 AI Auto-Assignment complete!\nGET /api/v1/tasks/{id}/ai/recommend-assignee\n→ ML.NET FastTree model scored all candidates\n→ 3 tasks assigned based on availability, skills & burnout score`);
    }

    function aiPredictTask(t) {
        alert(`🔔 AI Delay Prediction for TSK-${t.id}\nGET /api/v1/tasks/${t.id}/ai/delay-prediction\n\nDelay Probability: ${t.aiRisk}%\nExpected Delay: +${Math.round(t.aiRisk/10)} days\nRisk Level: ${t.aiRisk>70?'High':'Medium'}\n\nModel: ML.NET FastTree Regression`);
    }

    // — TEAM DATA —
    const teamMembers = ref([
        { id:'u1', name:'Ahmad Hassan', role:'Dev Lead', status:'Busy', tasks:12, done:28, perf:88, load:94, burnout:78, skills:['C#','ASP.NET','SQL','Azure'] },
        { id:'u2', name:'Sarah Lee', role:'ML Engineer', status:'Available', tasks:8, done:21, perf:91, load:72, burnout:45, skills:['Python','ML.NET','TensorFlow'] },
        { id:'u3', name:'John Williams', role:'UI Designer', status:'Available', tasks:6, done:18, perf:76, load:55, burnout:30, skills:['Vue.js','CSS','Figma','UX'] },
        { id:'u4', name:'Fatima Al-Zahra', role:'DevOps Eng.', status:'Busy', tasks:10, done:32, perf:85, load:88, burnout:62, skills:['Docker','Kubernetes','CI/CD'] },
        { id:'u5', name:'Carlos Marin', role:'Data Analyst', status:'Available', tasks:5, done:15, perf:79, load:41, burnout:22, skills:['Power BI','SQL','Python','Tableau'] },
        { id:'u6', name:'Layla Nasser', role:'QA Engineer', status:'OnLeave', tasks:3, done:12, perf:83, load:25, burnout:18, skills:['Selenium','Postman','JIRA'] },
    ]);

    function viewMember(m) { alert(`🔍 Viewing profile: ${m.name}\nGET /api/v1/users/${m.id}\n\nWorkload: ${m.load}% | Burnout: ${m.burnout}%\nActive Tasks: ${m.tasks} | Done/Month: ${m.done}`); }

    function aiAssignTo(m) { alert(`🔔 AI Task Assignment for ${m.name}\nGET /api/v1/users/${m.id}/ai/burnout-risk\n\nBurnout Risk: ${m.burnout}%\n\nRecommended Max Tasks: ${m.load>80?'No new tasks':'m.load>60?'1 task':'Multiple tasks'}`); }

    function aiBurnout() { alert(`🔔 Burnout Analysis complete!\nGET /api/v1/users/workload?departmentId=...\n\n🔔 Ahmad Hassan – 78% burnout risk (critical)\n🔔 Fatima Al-Zahra – 62% burnout risk (high)\n🔔 Carlos Marin – 22% burnout risk (low)`); }

    function showUserModal() {}

    // — NOTIFICATIONS DATA —
    const notifications = ref([
        { id:1, icon:'🔔', bg:'#fde8ea', title:'Task Escalated – AI Delay Risk', msg:'TSK-4: AI Delay Prediction Engine – 72% delay probability detected by ML model', time:'2 min ago', type:'Escalation', ai:true, read:false },
        { id:2, icon:'🔔', bg:'#fff3cd', title:'Deadline Alert – PROJ-003', msg:'Cloud Migration is 69 days overdue. Immediate intervention required.', time:'15 min ago', type:'Deadline', ai:false, read:false },
    ])

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    { id:3, icon:'🔍', bg:'#e8d5ff', title:'AI Insight – Burnout Risk Detected',
msg:'Ahmad Hassan shows 78% burnout risk. Recommend redistributing 2 tasks.',
time:'1 hr ago', type:'AI Insight', ai:true, read:false },
    { id:4, icon:'✅', bg:'#d4edda', title:'Task Completed',
msg:'TSK-2: Setup CI/CD Pipeline marked complete by Fatima Al-Zahra',
time:'2 hrs ago', type:'Task', ai:false, read:false },
    { id:5, icon:'📌', bg:'#d1ecf1', title:'New Project Assigned',
msg:'You have been assigned as Project Manager for PROJ-005: Data Analytics Platform',
time:'3 hrs ago', type:'Project', ai:false, read:false },
    { id:6, icon:'⚠️', bg:'#fff3cd', title:'Budget Warning – PROJ-003',
msg:'Cloud Migration has consumed 90.6% of planned budget ($290K of $320K)',
time:'5 hrs ago', type:'Budget', ai:true, read:true },
    { id:7, icon:'🔍', bg:'#e8d5ff', title:'AI Model Retrained',
msg:'Delay Prediction model retrained with 847 new data points. Accuracy: 87.3%',
time:'8 hrs ago', type:'AI Insight', ai:true, read:true },
    { id:8, icon:'👤', bg:'#d1ecf1', title:'New Team Member Added',
msg:'Layla Nasser (QA Engineer) has been added to the Software Engineering team',
time:'1 day ago', type:'Team', ai:false, read:true },
  ]);

  const notifCount = computed(() => notifications.value.filter(n => !n.read).length);

  function readNotif(n) { n.read = true; }
  function markAllRead() { notifications.value.forEach(n => n.read = true); }

  // — AI PAGE DATA —
  const ai = ref({ input:'', typing:false });
  const chatMsgs = ref([
    { role:'ai', text:'👋 Hello Dr. Johnson! I\'m your PMWDS AI Assistant powered by GPT-4o. I can help you analyse project health, predict task delays, recommend assignments, and generate insights. What would you like to know?', time:'09:00 AM' },
    { role:'u', text:'Which tasks are at highest risk of delay this week?', time:'09:01 AM' },
    { role:'ai', text:'📊 Based on my ML analysis, the top 3 high-risk tasks are:\n\n1. 📌 TSK-9: Burnout Risk ML Model (80% delay risk) – unassigned, deadline April 20\n2. 📌 TSK-4: AI Delay Prediction Engine (72% delay risk) – Ahmad Hassan at 94% workload\n3. 📌 TSK-3: Auth Module Development (65% delay risk) – team lead overloaded\n\nRecommendation: Reassign TSK-9 to Sarah Lee (72% workload, ML expertise match).', time:'09:01 AM' },
  ]);

  async function sendChat() {
    if (!ai.value.input.trim()) return;
    const userMsg = ai.value.input.trim();
    chatMsgs.value.push({ role:'u', text:userMsg, time: new Date().toLocaleTimeString([], {hour:'2-digit',minute:'2-digit'}) });
    ai.value.input = '';
    ai.value.typing = true;
    await new Promise(r => setTimeout(r, 1400));
    ai.value.typing = false;
    const responses = {
      default: '🔄 Processing your query via POST /api/v1/ai/chat → GPT-4o...\n\nBased on current data: I\'ve analysed all 142 active tasks across 18 projects. Your overall portfolio health is 78%. Key concern: 3 projects are at critical risk. Recommend scheduling a review meeting.',
      overload: '📊 Workload Analysis:\n\n👤 Ahmad Hassan – 94% load (CRITICAL)\n👤 Fatima Al-Zahra – 88% load (HIGH)\n👤 Carlos Marin – 41% load (available for reallocation)\n\nAI Recommendation: Move 2 tasks from Ahmad to Carlos to reduce burnout risk from 78% → 45%.',
      health: '📈 Project Health Summary:\n\n📌 PROJ-001 PMWDS Portal – 82% (Good)\n📌 PROJ-002 AI Research – 71% (Fair)\n📌 PROJ-003 Cloud Migration – 44% (Critical)\n📌 PROJ-005 Data Analytics – 38% (Critical)\n\nOverall Score: 78%. 2 projects need immediate intervention.',
      assign: '📌 Task Assignment Recommendations:\n\nTSK-8 Mobile API → Carlos Marin (41% load, API skills match: 92%)\nTSK-9 ML Model → Sarah Lee (72% load, ML expertise'
    };
  }

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match: 97%)\nTSK-10 UAT → Layla Nasser (25% load, QA skills match: 95%)\n\nEstimated
efficiency gain: +23% after reallocation.',
    burnout: '🔴 Burnout Risk Report:\n\n🔴 Ahmad Hassan – 78% (Immediate action
required)\n🔴 Fatima Al-Zahra – 62% (Monitor closely)\n🔴 Sarah Lee – 45% (Acceptable)\n🔴
John Williams – 30% (Good)\n🔴 Carlos Marin – 22% (Excellent)\n\nRecommendation: Reduce
Ahmad\'s task load by 3 tasks immediately.'
    });
    const lower = userMsg.toLowerCase();
    const reply = lower.includes('overload') || lower.includes('workload') ?
responses.overload
    : lower.includes('health') ? responses.health
    : lower.includes('assign') ? responses.assign
    : lower.includes('burnout') ? responses.burnout
    : responses.default;
    chatMsgs.value.push({ role:'ai', text:reply, time: new Date().toLocaleTimeString([],
{hour:'2-digit',minute:'2-digit'}) });
}

function quickChat(msg) { ai.value.input = msg; sendChat(); }

function trainModel() {
    alert('🔔 Retraining triggered!\nPOST /api/v1/ai/train\n→ Hangfire background job
queued\n→ ML.NET pipeline: data load → feature engineering → FastTree training\n→ Expected
completion: ~4 minutes\n→ Next auto-train: Tomorrow 2:00 AM');
}

function refreshInsights() {
    alert('🔔 Refreshing AI Insights...\nGET /api/v1/projects/{id}/ai/insights\n→ NLG
engine generating natural language insights\n→ 6 new insights generated');
}

const aiMetrics = [
    { label:'Task Allocation', val:87 },
    { label:'Delay Prediction', val:83 },
    { label:'Health Analysis', val:91 },
    { label:'Burnout Detection', val:79 },
];

const aiInsights = ref([
    { id:1, icon:'🔴', title:'Critical Workload Imbalance', msg:'Ahmad Hassan
is at 94% capacity. Recommend immediate reallocation of 2-3 tasks to available team
members.', sev:'High', time:'Just now' },
    { id:2, icon:'🟡', title:'PROJ-003 Budget Overrun Predicted', msg:'Cloud
Migration is projected to exceed budget by $45K (14%) based on current spending velocity.',
sev:'High', time:'5 min ago' },
    { id:3, icon:'🟢', title:'Sprint Velocity Improving', msg:'Team
completed 89 of 142 tasks (63%) with 8 days remaining. Predicted sprint success rate:
78%.', sev:'Medium', time:'1 hr ago' },
    { id:4, icon:'🟢', title:'PROJ-001 On Track for Delivery', msg:'PMWDS Portal
is 67% complete with 82% health score. No intervention needed. Keep current pace.',
sev:'Low', time:'2 hrs ago' },
    { id:5, icon:'🔴', title:'Delay Risk Spike Detected – TSK-9', msg:'Burnout Risk
ML Model task has 80% delay probability due to no assignee and approaching deadline (Apr
20).', sev:'High', time:'3 hrs ago' },
]);

const aiPredictions = [
    { id:1, task:'AI Delay Prediction Engine', proj:'PROJ-002', assignee:'Sarah Lee',
due:'2025-03-20', prob:72, delayDays:8, risk:'High', mitigations:['Reassign 1 sub-
task','Extend deadline by 1 week'] },
    { id:2, task:'Burnout Risk ML Model', proj:'PROJ-002', assignee:'Unassigned',
due:'2025-04-20', prob:80, delayDays:14, risk:'Critical', mitigations:['Assign
immediately','Split into 2 sub-tasks'] },

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    { id:3, task:'Auth Module Development', proj:'PROJ-001', assignee:'Ahmad Hassan', due:'2025-03-10', prob:65, delayDays:5, risk:'High', mitigations:['Reduce other tasks','Add pair programmer'] },
    { id:4, task:'Dashboard UI Components', proj:'PROJ-001', assignee:'John Williams', due:'2025-03-25', prob:38, delayDays:0, risk:'Medium', mitigations:['Monitor weekly','Review milestone progress'] },
    { id:5, task:'Reports & PDF Export', proj:'PROJ-001', assignee:'Carlos Marin', due:'2025-03-15', prob:22, delayDays:0, risk:'Low', mitigations:['On track - no action needed'] },
  ];

  // — REPORTS DATA —
  const rep = ref({ type:'project-status', format:'pdf', dept:'', from:'2025-01-01', to:'2025-12-31', narration:'' });

  const reportTypes = [
    { id:'project-status', icon:'📄', label:'Project Status Report' },
    { id:'task-completion', icon:'📄', label:'Task Completion Report' },
    { id:'department-workload', icon:'📄', label:'Department Workload Report' },
    { id:'budget-variance', icon:'📄', label:'Budget Variance Report' },
    { id:'delay-analysis', icon:'📄', label:'Delay Analysis Report' },
    { id:'ai-insights', icon:'📄', label:'AI Insights Report' }
  ];

  function generateReport() {
    const ep = {
      'project-status': `GET /api/v1/reports/project-status/{id}?format=${rep.value.format}`,
      'task-completion': `POST /api/v1/reports/task-completion?format=${rep.value.format}`,
      'department-workload': `POST /api/v1/reports/department-workload?format=${rep.value.format}`,
      'budget-variance': `GET /api/v1/reports/budget-variance/{id}?format=${rep.value.format}`,
      'delay-analysis': `POST /api/v1/reports/delay-analysis?format=${rep.value.format}`,
      'ai-insights': `GET /api/v1/reports/ai-insights/{id}?format=${rep.value.format}`,
    };
    alert(`📄 Generating ${rep.value.format.toUpperCase()} Report...\n${ep[rep.value.type]}\n→ QuestPDF engine rendering...\n→ File ready for download: ${rep.value.type}.${rep.value.format} (Est. ~2.4 MB)`);
  }

  function aiNarrate() {
    rep.value.narration = 'Overall portfolio health is 78% – 2 of 24 projects are at critical risk requiring immediate escalation. Budget variance across all projects is +7.2% over plan, primarily driven by PROJ-003 Cloud Migration (+14%). Team velocity has improved 8% MoM. AI recommends priority reallocation of 3 overloaded team members.';
  }

  // — DEPARTMENTS DATA —
  const departments = ref([
    { id:'d1', icon:'🏢', name:'Software Engineering', code:'SE', health:79, projects:8, members:14, budget:750, budgetUsed:68 },
    { id:'d2', icon:'🏢', name:'Artificial Intelligence', code:'AI', health:65, projects:6, members:9, budget:520, budgetUsed:82 },
    { id:'d3', icon:'🏢', name:'Computer Science', code:'CS', health:44, projects:5, members:11, budget:480, budgetUsed:91 },
    { id:'d4', icon:'🏢', name:'Data Science', code:'DS', health:88, projects:3, members:7, budget:310, budgetUsed:54 },
    { id:'d5', icon:'🏢', name:'Cybersecurity', code:'CY', health:92, projects:2, members:5, budget:220, budgetUsed:42 },
  ]

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    });

    const showDeptModal = ref(false);
    function viewDept(d) { alert(`📄 ${d.name} Department Dashboard\nGET
/api/v1/departments/${d.id}/dashboard\n\nProjects: ${d.projects} | Members:
${d.members}\nBudget Used: ${d.budgetUsed}% | Health: ${d.health}%`); }

    // — MODALS —
    const showProjModal = ref(false);
    const pForm = ref({ code:'', name:'', desc:'', cat:'Research', dept:'SE', start:'',
end:'', budget:'', manager:'', priority:'High' });

    function createProject() {
        if (!pForm.value.name || !pForm.value.code) { alert('Project code and name are
required'); return; }
        const np = {
            id: projects.value.length + 1,
            code: pForm.value.code,
            name: pForm.value.name,
            dept: pForm.value.dept,
            manager: 'Dr. Sarah Johnson',
            status: 'Active',
            prog: 0,
            health: 100,
            risk: 5,
            due: pForm.value.end || '2025-12-31',
            budget: Number(pForm.value.budget)/1000 || 100,
            actual: 0
        };
        projects.value.unshift(np);
        showProjModal.value = false;
        pForm.value = { code:'', name:'', desc:'', cat:'Research', dept:'SE', start:'',
end:'', budget:'', manager:'', priority:'High' };
        alert(`📄 Project created!\nPOST /api/v1/projects → 201 Created\nCode: ${np.code} | AI
Health monitoring started in background`);
    }

    function aiValidate() { alert(`📄 AI Validation complete!\n📄 Project code is
unique\n📄 Timeline is realistic (234 days)\n📄 Budget seems low for scope – recommend
+15%\n📄 Department has available capacity`); }
    function aiRecommendAssignee() { alert(`📄 AI Recommends:\n📄 Sarah Lee (ML
Engineer)\nMatch Score: 94.2%\nReason: Skill match, 72% workload, 45% burnout
risk\n\nAlternative: Carlos Marin (Score: 81.5%)`); }

    // — CHARTS —
    let trendChartInst, donutChartInst, budgetChartInst, reportChartInst;

    function initCharts() {
        nextTick(() => {
            // Trend Chart
            const tc = document.getElementById('trendChart');
            if (tc) {
                if (trendChartInst) trendChartInst.destroy();
                trendChartInst = new Chart(tc, {
                    type: 'line',
                    data: {
                        labels: ['Feb 1', 'Feb 5', 'Feb 10', 'Feb 15', 'Feb 20', 'Feb 25', 'Mar 1', 'Mar
5', 'Mar 10', 'Mar 15'],
                        datasets: [
                            { label: 'Completed', data: [8,15,22,31,40,49,57,68,76,89],
                                borderColor: '#28a745', backgroundColor: 'rgba(40,167,69,.1)',
                                tension:0.4, fill:true, pointRadius:4 },
                            { label: 'Planned', data: [10,20,30,40,50,60,70,80,90,100],

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        borderColor: '#1B3A6B', borderDash: [6, 3], backgroundColor: 'transparent',
        tension: 0.4, pointRadius: 0 },
        { label: 'At Risk', data: [2, 3, 5, 4, 6, 7, 5, 8, 7, 9],
        borderColor: '#dc3545', backgroundColor: 'rgba(220,53,69,.08)',
        tension: 0.4, fill: true, pointRadius: 3 },
    ]
},
options: {
    responsive: true, maintainAspectRatio: false,
    plugins: { legend: { labels: { font: { size: 11 } } } },
    scales: {
        x: { grid: { color: '#f0f0f0' }, ticks: { font: { size: 10 } } },
        y: { grid: { color: '#f0f0f0' }, ticks: { font: { size: 10 } }, beginAtZero: true }
    }
}
});
}
// Donut Chart
const dc = document.getElementById('donutChart');
if (dc) {
    if (donutChartInst) donutChartInst.destroy();
    donutChartInst = new Chart(dc, {
        type: 'doughnut',
        data: {
            labels: ['Healthy', 'At Risk', 'Critical'],
            datasets: [{ data: [14, 6, 4],
            backgroundColor: ['#28a745', '#ffc107', '#dc3545'],
            borderWidth: 0, hoverOffset: 6 }]
        },
        options: {
            responsive: true, maintainAspectRatio: false, cutout: '72%',
            plugins: { legend: { position: 'bottom', labels: { font: { size: 11 }, padding: 10 } } }
        }
    }
});
}
// Budget Chart
const bc = document.getElementById('budgetChart');
if (bc) {
    if (budgetChartInst) budgetChartInst.destroy();
    budgetChartInst = new Chart(bc, {
        type: 'bar',
        data: {
            labels: ['PROJ-001', 'PROJ-002', 'PROJ-003', 'PROJ-004', 'PROJ-005'],
            datasets: [
                { label: 'Planned ($K)', data: [250, 180, 320, 140, 210],
                backgroundColor: 'rgba(27,58,107,.15)', borderColor: '#1B3A6B',
                borderWidth: 1.5, borderRadius: 6 },
                { label: 'Actual ($K)', data: [168, 82, 290, 78, 47],
                backgroundColor: ['rgba(40,167,69,.7)', 'rgba(40,167,69,.7)', 'rgba(220,53,69,.7)', 'rgba(40,167,69,.7)', 'rgba(40,167,69,.7)'],
                borderRadius: 6 },
            ]
        },
        options: {
            responsive: true, maintainAspectRatio: false,
            plugins: { legend: { labels: { font: { size: 11 } } } },
            scales: {
                x: { grid: { display: false }, ticks: { font: { size: 11 } } },
                y: { grid: { color: '#f0f0f0' }, ticks: { font: { size: 11 }, callback: v =>
                '$'+v+'K' } } }
        }
    }
}

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    }
  });
}
});
}

function initReportChart() {
  nextTick(() => {
    const rc = document.getElementById('reportChart');
    if (rc) {
      if (reportChartInst) reportChartInst.destroy();
      reportChartInst = new Chart(rc, {
        type: 'bar',
        data: {
          labels: ['SE Dept', 'AI Dept', 'CS Dept', 'DS Dept', 'CY Dept'],
          datasets: [
            { label: 'Health Score', data: [79, 65, 44, 88, 92],
backgroundcolor: 'rgba(27,58,107,.7)', borderRadius: 6 },
            { label: 'Task Completion', data: [82, 71, 58, 91, 95],
backgroundcolor: 'rgba(40,167,69,.6)', borderRadius: 6 },
            { label: 'Budget Used %', data: [68, 82, 91, 54, 42],
backgroundcolor: 'rgba(200,169,81,.7)', borderRadius: 6 },
          ]
        },
        options: {
          responsive: true, maintainAspectRatio: false,
          plugins: { legend: { labels: { font: { size: 11 } } } },
          scales: {
            x: { grid: { display: false }, ticks: { font: { size: 11 } } },
            y: { grid: { color: '#f0f0f0' }, ticks: { font: { size: 10 }, max: 100, callback:
v => v+'%' } }
          }
        }
      });
    }
  });
}

// — HELPERS —
function isLate(dateStr) {
  if (!dateStr) return false;
  return new Date(dateStr) < new Date();
}

onMounted(() => {
  // Auto-login for demo mode – remove in production
  // auth.value.isLoggedIn = true;
  // initCharts();
});

return {
  // auth & login
  auth, loginForm, login, valEmail, doLogin, ssoLogin, logout,
  // nav
  sb, pg, hdr, nav, aiNav, pgLabel, go, today,
  // dashboard
  dash, kpis, healthBars, atRisk, workload, refreshDash, fetchHealth, fetchWL,
  escalate,
  // projects
  projects, pf, filteredProjs, fetchProjects, viewProj, editProj, aiHealth, aiOptimize,
  showProjModal, pForm, createProject, aiValidate,
  // tasks
  tv, selProj, tasks, tasksIn, kanbanCols, ganttMonths,
  showTaskModal, showTaskDetail, selTask, taskProg, taskNote,

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    tForm, openTask, updateProgress, escalateTask, createTask, aiAssignAll,
aiPredictTask,
    aiRecommendAssignee,
    // team
    teamMembers, viewMember, aiAssignTo, aiBurnout, showUserModal,
    // notifications
    notifications, notifCount, readNotif, markAllRead,
    // ai
    ai, chatMsgs, sendChat, quickChat, trainModel, refreshInsights,
    aiMetrics, aiInsights, aiPredictions,
    // reports
    rep, reportTypes, generateReport, aiNarrate,
    // departments
    departments, showDeptModal, viewDept,
    // helpers
    isLate,
  };
},

mounted() {
  // Auto-boot charts after Login happens via watch
  this.$watch('auth.isLoggedIn', (val) => {
    if (val) this.initCharts();
  });
},

methods: {
  initCharts() {
    // Delegated to setup's initCharts - called via watch above
    const nextTick = Vue.nextTick;
    nextTick(() => {
      // Trigger chart init on the reactive setup function
      this.$options.setup && this.initCharts && this.initCharts?();
    });
  }
},

watch: {
  'auth.isLoggedIn'(val) {
    if (val) {
      Vue.nextTick(() => {
        // Re-initialise charts using the setup function ref
        const initFn = this.$.setupState?.initCharts;
        if (initFn) initFn();
      });
    }
  },
  pg(val) {
    if (val === 'reports') {
      const initFn = this.$.setupState?.initReportChart;
      if (initFn) initFn();
    }
    if (val === 'dashboard') {
      const initFn = this.$.setupState?.initCharts;
      if (initFn) initFn();
    }
  }
}
}).mount('#app');
</script>
</body>
</html>

```

❑ What This Completes

The file is now **100% complete**. Here's every section included across all three parts:

Section	Status
❑ Login page with live API call preview	❑ Complete
❑ App shell — Sidebar + Header	❑ Complete
❑ Dashboard — KPIs, charts, at-risk, workload	❑ Complete
❑ Projects — Table, filters, AI health	❑ Complete
❑ Tasks — Kanban, List, Gantt views	❑ Complete
❑ Team — Member cards, workload, burnout	❑ Complete
❑ Notifications — Real-time feed, SignalR info	❑ Complete
❑ AI Assistant — Chat, predictions, insights	❑ Complete
❑ Reports — Type selector, chart preview, narration	❑ Complete
❑ Departments — Portfolio cards	❑ Complete
❑ New Project Modal + API preview	❑ Complete
❑ New Task Modal + API preview	❑ Complete
❑ Task Detail Modal + progress update	❑ Complete
❑ Chart.js — Trend, Donut, Budget, Report charts	❑ Complete
❑ Vue 3 Composition API — all reactive data & methods	❑ Complete

